Chapter 3

Development, Implementation and Dissemination of AQI

3.1 Indian Air Quality Index (IND-AQI)

Air quality standards are the basic foundation that provides a legal framework for air pollution control. An air quality standard is a description of a level of air quality that is adopted by a regulatory authority as enforceable. The basis of development of standards is to provide a rational for protecting public health from adverse effects of air pollutants, to eliminate or reduce exposure to hazardous air pollutants, and to guide national/local authorities for pollution control decisions. With these objectives, CPCB notified (http://www.cpcb.nic.in) a new set of Indian National Air Quality Standards (INAQS) for 12 parameters [carbon monoxide (CO) nitrogen dioxide (NO_2), sulphur dioxide (SO_2), particulate matter (PM) of less than 2.5 microns size ($PM_{2.5}$), PM of less than 10 microns size (PM_{10}), Ozone (O_3), Lead (Pb), Ammonia (NH_3), Benzo(a)Pyrene (BaP), Benzene (C_6H_6), Arsenic (As), and Nickel (Ni)] . The first eight parameters (Table 3.1) have short-term (1/8/24 hrs) and annual standards (except for CO and O_3) and rest four parameters have only annual standards.

Table 3.1: Indian National Air Quality Standards (units: µg/m^3 unless mentioned otherwise)

Pollutant	**SO_2**	**NO_2**	**$PM_{2.5}$**	**PM_{10}**	**O_3**		**CO (mg/m^3)**		**Pb**	**NH_3**
Averaging time (hr)	24	24	24	24	1	8	1	8	24	24
Standard	80	80	60	100	180	100	4	2	1	400

BaP, C_6H_6, As, and Ni have annual standards

As stated in Chapter 1, an AQI scheme transforms weighted values of individual air pollutant concentrations into a single number or set of numbers. It is important that an AQI system should initially build on the sacrosanct AQS and then embark on pollutant dose-response relationships to describe air quality in simple terms which clearly relate to health impacts. While complexity in building the AQI is inevitable, simplicity in AQI description is essential for general public to understand the air pollution, possibly take actions to protect themselves and for policy makers to take quick and broad decisions to improve air quality.

The objective of an AQI is to quickly disseminate air quality information (almost in real-time) that entails the system to account for pollutants which have short-term impacts. It is equally important that most of these pollutants are measured continuously through an online monitoring network. Consequently, in the AQI system, the following pollutants are considered CO, NO_2, SO_2, $PM_{2.5}$, PM_{10}, O_3, NH_3 and Pb. It is recognized that air concentrations of Pb are not known in real-time and cannot contribute to real-time AQI but its consideration in AQI calculation of past days will help in scrutinizing the status of this important toxic.

It is described in Chapter 2 that the aggregation function, F of combining sub-indices of individual pollutants is delicate as most indices suffer from ambiguity and eclipsing. For the AQI, a maximum operator system has been adopted which is free from ambiguity and eclipsing, as shown below:

$$AQI=Max(I_1,I_2,...,I_n)$$

Figure 3.1 shows the operational scheme of AQI system based of maximum operator function (i.e. maximum sub-index being the overall index). To present status of the air quality and its effects on human health, the following description categories have been adopted for IND-AQI (Table 3.2):

Table 3.2: IND-AQI Category and Range

AQI Category	AQI Range
Good	0 – 50
Satisfactory	51 – 100
Moderate	101 – 200
Poor	201 – 300
Very Poor	301 – 400
Severe	401 - 500

These categories/AQI ranges should map to key references (breakpoints) of concentration of each pollutants through a segmented linear or a nonlinear function.

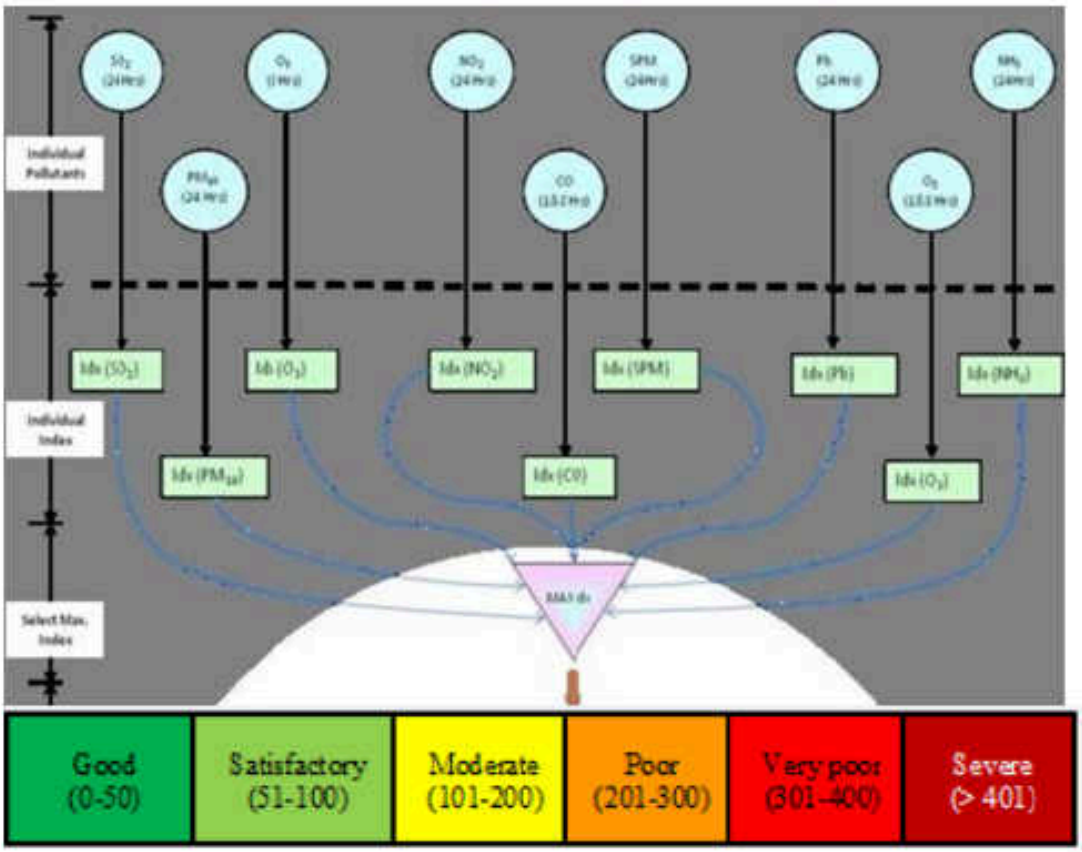


Figure 3.1 Overall AQI System

3.2 Air Quality Monitoring and AQI Considerations

The air quality monitoring network in India can be classified as (i) online and (ii) manual. The pollutant parameters, frequency of measurement and monitoring methodologies for two networks are very different. The AQI system for these networks could be at variance, especially for reporting and completeness in terms of parameters.

(i) ***Online Monitoring Network*:** These are automated air quality monitoring stations which record continuous hourly, monthly or annually averaged data. In India, ~ 40 automatic monitoring stations are operated (e.g. Figure 3.2: online stations in Delhi), where parameters like PM_{10}, $PM_{2.5}$, NO_2, SO_2, CO, O_3, etc. are monitored continuously. Data from these stations are available almost in real-time. Thus such networks are most suitable for computation of AQI sub-indices, as information on AQI can be generated in real time. For AQI to be more useful and effective, there is a need to set up more

online monitoring stations for continuous and easy availability of air quality data for computation of AQI for more Indian cities.



Figure 3.2 Online monitoring station (ITO, New Delhi) (www.cpcb.nic.in)

(ii) ***Manual Monitoring Network:*** The manual stations involve mostly intermittent air quality data collection, thus such stations are not suitable for AQI calculation particularly for its quick dissemination. In India, air quality is being monitored manually at 573 locations under National Air Monitoring Programme (NAMP). In most of these manually operated stations, only three criteria pollutants viz. PM_{10}, sulphur dioxide (SO_2) and nitrogen dioxide (NO_2) are measured, at some stations $PM_{2.5}$ and Pb are also measured. The monitoring frequency is twice a week. Such manual networks are not suitable for computing AQI, as availability of monitored data could have a lag of 1-3 days and sometimes not available at all. However, some efforts are required to use the information in some productive manner. Historical AQIs on weekly basis can be calculated and used for data interpretation and ranking of cities or towns for further prioritization of actions on air pollution control.

3.3 Computation and Basis of Sub-index Breakpoints

Segmented linear functions are used for relating actual air pollution concentration (X_i) (of each pollutant) to a normalized number referred to as sub-index (I_i). While AQI system is not complex in understanding, to arrive at breakpoints which will relate to AQI description is of paramount significance. Consequences of inappropriate adoption of breakpoints could be far reaching; it may lead to incorrect information to general public (on health effects) and decisions taken for pollution control may be incorrect.

The basis for linear functions (for this study) to relate air quality levels to AQI requires careful consideration. Services of practicing doctors and experts in this field (see Appendix I) have proved very useful. In this study, in addition to dose response relationship, the breakpoints adopted by other countries/agencies (USEPA 2014; UK 2013; Malaysia 2013; GVAQI 2013; Ontario 2013) have been examined for using these in IND-AQI.

It is important that an AQI system should build on AQS and pollutant dose-response relationships to describe air quality in simple terms which clearly relates to health impacts. The first step for arriving at breakpoints for each pollutant is to consider attainment of INAQS (Table 3.1). The index category is classified as 'good' for concentration range up to half of INAQS (for example, for SO_2 AQI=0-50 for concentration range of 0-40 $\mu g/m^3$) and as 'satisfactory' up to attainment of INAQS (i.e. SO_2 range 41-80$\mu g/m^3$ linearly maps to AQI=51-100). To arrive at breakpoints for other categories (for each pollutant), we require a thorough research/review of dose response relationships, which is described here.

3.3.1 Carbon Mono-oxide (CO)

Carbon monoxide (CO) is an important criteria pollutant which is ubiquitous in urban environment. CO production mostly occurs from sources having incomplete combustion. Due to its toxicity and appreciable mass in atmosphere, it should be considered as an important pollutant in AQI scheme.

CO rapidly diffuses across alveolar, capillary and placental membranes. Approximately 80-90% of absorbed CO binds with Hb to from Carboxyhaemoglobin (COHb), which is a specific biomarker of exposure in blood. The affinity of Hb for CO is 200-250 times than that of oxygen. In patients with hemolyticanemia, the CO production rate was 2–8 times higher and blood COHb concentration was

3.3.6 Pb and NH_3

It is to be noted that most of the countries have taken only six pollutants (described above) for formulation of AQI. An attempt has been made to propose breakpoints for NH_3 and Pb as these two pollutants also have short-term standards of 24-hr. While NH_3 can be measured on continuous basis and can be included in the list of real time parameters for AQI, such measurements are not possible for Pb. However, Pb levels can be utilized in calculation of AQI of past days to assess impact of lead pollution.

Inhalation of high levels of NH_3 causes irritation to the nose, throat and respiratory tract. Increased inhalation may result in cough and an increased respiratory rate as well as respiratory distress. An association has been reported between exposure to ammonia and cough, phlegm, wheezing, and asthma at high concentration. A study (http://www.hpa.org.uk/webc/hpawebfile/hpaweb_c/1194947398510) has reported that for NH_3 levels at 18 mg/m^3, reduction in FEV1 and FVC% were significant in symptomatic than asymptomatic individuals. For a factor of safety as 10, concentration of 1800 $\mu g/m^3$ should be considered to be severe in ambient air. The other breakpoints for ammonia have been evolved on a linear scale from the level of 1800 $\mu g/m^3$ to the standard concentration of 400 $\mu g/m^3$.

Pb is a toxic metal and its exposure through all routes results in increased blood lead level. At lower concentrations, the blood lead level is proportional to air concentration (after accounting for all resulting exposure routes). For example, 1$\mu g/m^3$ of annual lead level will result in 5$\mu g/dL$ (on an average) of blood lead level (WHO 2000). The effect of blood level above 10$\mu g/dL$ is seen in hematological changes in sensitive population, therefore, at moderate pollution level the break point is at 2$\mu g/m^3$. At 20$\mu g/dL$ blood lead level the effects become more prominent and this corresponds to break point of 4 $\mu g/m^3$but to account for factor of safety, next break point is kept at 3.0 $\mu g/m^3$ (and not at 4 $\mu g/m^3$) and if the lead concentration in air is more than 3.5 $\mu g/m^3$the AQI category will be severe.

In view of the above discussions, Table 3.10 presents the breakpoints for NH_3 and Pb; due consideration has been given to INAQS in deciding breakpoints for categories, Good and Satisfactory.

Table 3.10 AQI Breakpoints for NH_3 and Pb (24-hr)

(Pb from gasoline phased out in 2000)

AQI Category	NH_3 $\mu g/m^3$	Pb $\mu g/m^3$
Good (0-50)	200	0.5
Satisfactory (51-100)	400	1.0
Moderate(101-200)	800	2.0
Poor (201-300)	1200	3.0
Very poor (301-400)	1800	3.5
Severe (401-500)	1800+	3.5+

Sections 3.3.1 to 3.3.6 have presented basis of AQI breakpoints for eight pollutant parameters considered for AQI and these are summarized below in Table 3.11 with colour scheme to represent the AQI bands. Table 3.12 shows health statements for every AQI category for people to understand health effects and protect themselves from these effects.

Table 3.11 Breakpoints for AQI Scale 0-500 (units: $\mu g/m^3$ unless mentioned otherwise)

AQI Category (Range)	PM_{10} 24-hr	$PM_{2.5}$ 24-hr	NO_2 24-hr	O_3 8-hr	CO 8-hr (mg/m^3)	SO_2 24-hr	NH_3 24-hr	Pb 24-hr
Good (0-50)	0-50	0-30	0-40	0-50	0-1.0	0-40	0-200	0-0.5
Satisfactory (51-100)	51-100	31-60	41-80	51-100	1.1-2.0	41-80	201-400	0.6 –1.0
Moderate (101-200)	101-250	61-90	81-180	101-168	2.1- 10	81-380	401-800	1.1-2.0
Poor (201-300)	251-350	91-120	181-280	169-208	10.1-17	381-800	801-1200	2.1-3.0
Very poor (301-400)	351-430	121-250	281-400	209-748*	17.1-34	801-1600	1201-1800	3.1-3.5
Severe (401-500)	430 +	250+	400+	748+*	34+	1600+	1800+	3.5+

**One hourly monitoring (for mathematical calculation only)*

CPCB may consider reviewing the AQI breakpoints every three years after accounting the new research findings on air pollution exposure and health effects.

Table 3.12 Health Statements for AQI Categories

AQI	Associated Health Impacts
Good (0–50)	Minimal Impact
Satisfactory (51–100)	May cause minor breathing discomfort to sensitive people
Moderate (101–200)	May cause breathing discomfort to the people with lung disease such as asthma and discomfort to people with heart disease, children and older adults
Poor (201–300)	May cause breathing discomfort to people on prolonged exposure and discomfort to people with heart disease with short exposure
Very Poor (301–400)	May cause respiratory illness to the people on prolonged exposure. Effect may be more pronounced in people with lung and heart diseases
Severe (401-500)	May cause respiratory effects even on healthy people and serious health impacts on people with lung/heart diseases. The health impacts may be experienced even during light physical activity

3.4 Broad Guidelines for Actions during Very Poor and Severe Categories of AQI

Regulating Agencies

The regulating agencies should establish source-receptor relationships in terms of impact of emissions on air quality. In case AQI category is severe or very poor, necessary steps need to be taken by further regulating the emissions which are causing maximum impact to ambient air quality. Specific actions, for example, may include: (i) strict vigilance and no-tolerance to visible polluting vehicles, industries, open burning, construction activities, etc.; (ii) regulating traffic; and (iii) identifying sources contributing significantly to rising air quality levels and actions for reducing emissions from such sources.

Public

People should maintain vehicles properly (e.g. get PUC checks, replace car air filter, maintain right tires pressure), follow lane discipline and speed limits, avoid prolong idling and turn off

BE it enacted by Parliament in the Fifty-sixth Year of the Republic of India as follows:—

CHAPTER I

PRELIMINARY

Short title, extent and commencement.

1. (*1*) This Act may be called the Commissions for Protection of Child Rights Act, 2005.

(*2*) It extends to the whole of India except the State of Jammu and Kashmir.

(*3*) It shall come into force on such date as the Central Government may, by notification in the Official Gazette, appoint.

Definitions.

2. In this Act, unless the context otherwise requires,—

(*a*) "Chairperson" means the Chairperson of the Commission or of the State Commission, as the case may be;

(*b*) "child rights" includes the children's rights adopted in the United Nations convention on the Rights of the Child on the 20th November, 1989 and ratified by the Government of India on the 11th December, 1992;

(*c*) "Commission" means the National Commission for Protection of Child Rights constituted under section 3;

(*d*) "Member" means a Member of the Commission or of the State Commission, as the case may be, and includes the Chairperson;

(*e*) "notification" means a notification published in the Official Gazette;

(*f*) "prescribed" means prescribed by rules made under this Act;

(*g*) "State Commission" means a State Commission for Protection of Child Rights constituted under section 17.

CHAPTER II

THE NATIONAL COMMISSION FOR PROTECTION OF CHILD RIGHTS

Constitution of National Commission for Protection of Child Rights.

3. (*1*) The Central Government shall, by notification, constitute a body to be known as the National Commission for Protection of Child Rights to exercise the powers conferred on, and to perform the functions assigned to it, under this Act.

(*2*) The Commission shall consist of the following Members, namely :—

(*a*) a Chairperson who is a person of eminence and has done outstanding work for promoting the welfare of children; and

(*b*) six Members, out of which at least two shall be women, from the following fields, to be appointed by the Central Government from amongst persons of eminence, ability, integrity, standing and experience in, —

(*i*) education;

(*ii*) child health, care, welfare or child development;

(*iii*) juvenile justice or care of neglected or marginalized children or children with disabilities;

(*iv*) elimination of child labour or children in distress;

(*v*) child psychology or sociology; and

(*vi*) laws relating to children.

(*3*) The office of the Commission shall be at Delhi.

Appointment of Chairperson and Members.

4. The Central Government shall, by notification, appoint the Chairperson and other Members:

Provided that the Chairperson shall be appointed on the recommendation of a three member Selection Committee constituted by the Central Government under the Chairmanship of the Minister in-charge of the Ministry of Human Resource Development.

Term of office and conditions of service of Chairperson and Members.

5. (*1*) The Chairperson and every Member shall hold office as such for a term of three years from the date on which he assumes office:

Provided that no Chairperson or a Member shall hold the office for more than two terms:

Provided further that no Chairperson or any other Member shall hold office as such after he has attained—

(*a*) in the case of the Chairperson, the age of sixty-five years; and

(*b*) in the case of a Member, the age of sixty years.

(*2*) The Chairperson or a Member may, by writing under his hand addressed to the Central Government, resign his office at any time.

Salary and allowances of Chairperson and Members.

6. The salary and allowances payable to, and other terms and conditions of service of, the Chairperson and Members, shall be such as may be prescribed by the Central Government:

Provided that neither the salary and allowances nor the other terms and conditions of service of the Chairperson or a Member, as the case may be, shall be varied to his disadvantage after his appointment.

Removal from office.

7. (*1*) Subject to the provisions of sub-section (*2*), the Chairperson may be removed from his office by an order of the Central Government on the ground of proved misbehaviour or incapacity.

(*2*) Notwithstanding anything contained in sub-section (*1*), the Central Government may by order remove from office the Chairperson or any other Member, if the Chairperson or, as the case may be, such other Member, —

(*a*) is adjudged an insolvent; or

(*b*) engages during his term of office in any paid employment outside the duties of his office; or

(*c*) refuses to act or becomes incapable of acting; or

(*d*) is of unsound mind and stands so declared by a competent court; or

(*e*) has so abused his office as to render his continuance in office detrimental to the public interest; or

(*f*) is convicted and sentenced to imprisonment for an offence which in the opinion of the Central Government involves moral turpitude; or

(*g*) is, without obtaining leave of absence from the Commission, absent from three consecutive meetings of the Commission.

(*3*) No person shall be removed under this section until that person has been given an opportunity of being heard in the matter.

Vacation of office by Chairperson or Member.

8. (*1*) If the Chairperson or, as the case may be, a Member, —

(*a*) becomes subject to any of the disqualifications mentioned in section 7; or

(*b*) tenders his resignation under sub-section (*2*) of section 5,

his seat shall thereupon become vacant.

(*2*) If a casual vacancy occurs in the office of the Chairperson or a Member, whether by reason of his death, resignation or otherwise, such vacancy shall be filled within a period of ninety days by making afresh appointment in accordance with the provisions of section 4 and the person so appointed shall hold office for the remainder of the term of office for which the Chairperson, or a Member, as the case may be, in whose place he is so appointed would have held that office.

Vacancies, etc., not to invalidate proceedings of Commission.

9. No act or proceeding of the Commission shall be invalid merely by reason of —

(*a*) any vacancy in, or any defect in the constitution of, the Commission; or

(*b*) any defect in the appointment of a person as the Chairperson or a Member; or

(*c*) any irregularity in the procedure of the Commission not affecting the merits of the case.

Procedure for transaction of business.

10. (*1*) The Commission shall meet regularly at its office at such time as the Chairperson thinks fit, but three months shall not intervene between its last and the next meeting.

(*2*) All decisions at a meeting shall be taken by majority:

Provided that in the case of equality of votes, the Chairperson, or in his absence the person presiding, shall have and exercise a second or casting vote.

(*3*) If for any reason, the Chairperson, is unable to attend the meeting of the Commission, any Member chosen by the Members present from amongst themselves at the meeting, shall preside.

(*4*) The Commission shall observe such rules of procedure in the transaction of its business at a meeting, including the quorum at such meeting, as may be prescribed by the Central Government.

(*5*) All orders and decisions of the Commission shall be authenticated by the Member-Secretary or any other officer of the Commission duly authorised by Member-Secretary in this behalf.

Member-Secretary, officers and other employees of Commission.

11. (*1*) The Central Government shall, by notification, appoint an officer not below the rank of the Joint Secretary or the Additional Secretary to the Government of India as a Member-Secretary of the Commission and shall make available to the Commission such other officers and employees as may be necessary for the efficient performance of its functions.

(*2*) The Member-Secretary shall be responsible for the proper administration of the affairs of the Commission and its day-to-day management and shall exercise and discharge such other powers and perform such other duties as may be prescribed by the Central Government.

(*3*) The salary and allowances payable to, and the other terms and conditions of service of the Member-Secretary, other officers and employees, appointed for the purpose of the Commission shall be such as may be prescribed by the Central Government.

Salaries and allowances to be paid out of grants.

12. The salaries and allowances payable to the Chairperson and Members and the administrative expenses, including salaries, allowances and pensions payable to the Member-Secretary, other officers and employees referred to in section 11, shall be paid out of the grants referred to in sub-section (*1*) of section 27.

CHAPTER III

FUNCTIONS AND POWERS OF THE COMMISSION

Functions of Commission.

13. (*1*) The Commission shall perform all or any of the following functions, namely:—

(*a*) examine and review the safeguards provided by or under any law for the time being in force for the protection of child rights and recommend measures for their effective implementation;

(*b*) present to the Central Government, annually and at such other intervals, as the Commission may deem fit, reports upon the working of those safeguards;

(*c*) inquire into violation of child rights and recommend initiation of proceedings in such cases;

(*d*) examine all factors that inhibit the enjoyment of rights of children affected by terrorism, communal violence, riots, natural disaster, domestic violence, HIV/AIDS, trafficking, maltreatment, torture and exploitation, pornography and prostitution and recommend appropriate remedial measures;

(*e*) look into the matters relating to children in need of special care and protection including children in distress, marginalized and disadvantaged children, children in conflict with law, juveniles, children without family and children of prisoners and recommend appropriate remedial measures;

(*f*) study treaties and other international instruments and undertake periodical review of existing policies, programmes and other activities on child rights and make recommendations for their effective implementation in the best interest of children;

(*g*) undertake and promote research in the field of child rights;

(*h*) spread child rights literacy among various sections of the society and promote awareness of the safeguards available for protection of these rights through publications, the media, seminars and other available means;

(*i*) inspect or cause to be inspected any juvenile custodial home, or any other place of residence or institution meant for children, under the control of the Central Government or any State Government or any other authority, including any institution run by a social organisation; where children are detained or lodged for the purpose of treatment, reformation or protection and take up with these authorities for remedial action, if found necessary;

(*j*) inquire into complaints and take *suo motu* notice of matters relating to,—

(*i*) deprivation and violation of child rights;

(*ii*) non-implementation of laws providing for protection and development of children;

(*iii*) non-compliance of policy decisions, guidelines or instructions aimed at mitigating hardships to and ensuring welfare of the children and to provide relief to such children,

or take up the issues arising out of such matters with appropriate authorities; and

(*k*) such other functions as it may consider necessary for the promotion of child rights and any other matter incidental to the above functions.

(*2*) The Commission shall not inquire into any matter which is pending before a State Commission or any other Commission duly constituted under any law for the time being in force.

Powers relating to inquiries.

14. (*1*) The Commission shall, while inquiring into any matter referred to in clause (*j*) of sub-section (*1*) of section 13 have all the powers of a civil court trying a suit under the Code of Civil Procedure, 1908 and, in particular, in respect of the following matters, namely:—

(*a*) summoning and enforcing the attendance of any person and examining him on oath;

(*b*) discovery and production of any document;

(*c*) receiving evidence on affidavits;

(*d*) requisitioning any public record or copy thereof from any court or office; and

(*e*) issuing commissions for the examination of witnesses or documents.

(*2*) The Commission shall have the power to forward any case to a Magistrate having jurisdiction to try the same and the Magistrate to whom any such case is forwarded shall proceed to hear the complaint against the accused as if the case has been forwarded to him under section 346 of the Code of Criminal Procedure, 1973. 2 of 1974.

Steps after inquiry.

15. The Commission may take any of the following steps upon the completion of an inquiry held under this Act, namely :—

(*i*) where the inquiry discloses, the Commission of violation of child rights of a serious nature or contravention of provisions of any law for the time being in force, it may recommend to the concerned Government or authority the initiation of proceedings for prosecution or such other action as the Commission may deem fit against the concerned person or persons;

(*ii*) approach the Supreme Court or the High Court concerned for such directions, orders or writs as that Court may deem necessary;

(*iii*) recommend to the concerned Government or authority for the grant of such interim relief to the victim or the members of his family as the Commission may consider necessary.

Annual and special reports of Commission.

16. (*1*) The Commission shall submit an annual report to the Central Government and to the State Government concerned and may at any time submit special reports on any matter which, in its opinion, is of such urgency or importance that it should not be deferred till submission of the annual report.

(*2*) The Central Government and the State Government concerned, as the case may be, shall cause the annual and special reports of the Commission to be laid before each House of Parliament or the State Legislature respectively, as the case may be, along with a memorandum of action taken or proposed to be taken on the recommendations of the Commission and the reasons for non-acceptance of the recommendations, if any, within a period of one year from the date of receipt of such report.

(*3*) The annual report shall be prepared in such form, manner and contain such details as may be prescribed by the Central Government.

CHAPTER IV

STATE COMMISSIONS FOR PROTECTION OF CHILD RIGHTS

Constitution of State Commission for Protection of Child Rights.

17. (*1*) A State Government may constitute a body to be known as the .....(name of the State) Commission for Protection of Child Rights to exercise the powers conferred upon, and to perform the functions assigned to, a State Commission under this Chapter.

(*2*) The State Commission shall consist of the following Members, namely: —

(*a*) a Chairperson who is a person of eminence and has done outstanding work for promoting the welfare of children; and

(*b*) six Members, out of which at least two shall be women, from the following fields, to be appointed by the State Government from amongst persons of eminence, ability, integrity, standing and experience in,—

(*i*) education;

(*ii*) child health, care, welfare or child development;

(*iii*) juvenile justice or care of neglected or marginalized children or children with disabilities;

(*iv*) elimination of child labour or children in distress;

(*v*) child psychology or sociology; and

(*vi*) laws relating to children.

(*3*) The headquarter of the State Commission shall be at such place as the State Government may, by notification, specify.

Appointment of Chairperson and other Members.

18. The State Government shall, by notification, appoint the Chairperson and other Members:

Provided that the Chairperson shall be appointed on the recommendation of a three Member Selection Committee constituted by the State Government under the Chairmanship of the Minister in-charge of the Department dealing with children.

Term of office and conditions of service of Chairperson and Members.

19. (*1*) The Chairperson and every Member shall hold office as such for a term of three years from the date on which he assumes office:

Provided that no Chairperson or a Member shall hold the office for more than two terms:

Provided further that no Chairperson or any other Member shall hold office as such after he has attained—

(*a*) in the case of Chairperson, the age of sixty-five years; and

(*b*) in the case of a Member, the age of sixty years.

(*2*) The Chairperson or a Member may, by writing under his hand addressed to the State Government, resign his office at any time.

Salary and allowances of Chairperson and Members.

20. The salaries and allowances payable to, and other terms and conditions of service of, the Chairperson and Members shall be such as may be prescribed by the State Government:

Provided that neither the salary and allowances nor the other terms and conditions of service of the Chairperson or a Member, as the case may be, shall be varied to his disadvantage after his appointment.

Secretary, officers and other employees of the State Commission.

21. (*1*) The State Government shall, by notification, appoint an officer not below the rank of the Secretary to the State Government as the Secretary of the State Commission and shall make available to the State Commission such other officers and employees as may be necessary for the efficient performance of its functions.

(*2*) The Secretary shall be responsible for the proper administration of the affairs of the State Commission and its day-to-day management and shall exercise and discharge such other powers and perform such other duties as may be prescribed by the State Government.

(*3*) The salary and allowances payable to, and the other terms and conditions of service of the Secretary, other officers and employees, appointed for the purpose of the State Commission shall be such as may be prescribed by the State Government.

Salaries and allowances to be paid out of grants.

22. The salaries and allowances payable to the Chairperson and Members and the administrative expenses, including salaries, allowances and pensions payable to the Secretary, other officers and employees referred to in section 21, shall be paid out of the grants referred to in sub-section (*1*) of section 28.

Annual and special reports of State Commission.

23. (*1*) The State Commission shall submit an annual report to the State Government and may at any time submit special reports on any matter which, in its opinion, is of such urgency or importance that it should not be deferred till submission of the annual report.

(*2*) The State Government shall cause all the reports referred to in sub-section (*1*) to be laid before each House of State Legislature, where it consists of two Houses, or where such Legislature consists of one House, before that House along with a memorandum explaining the action taken or proposed to be taken on the recommendations relating to the State and the reasons for the non-acceptance, if any, of any of such recommendations.

(*3*) The annual report shall be prepared in such form, manner and contain such details as may be prescribed by the State Government.

Application of certain provisions relating to National Commission for Protection of Child Rights to State Commissions.

24. The provisions of sections 7, 8, 9, 10, sub-section (*1*) of section 13 and sections 14 and 15 shall apply to a State Commission and shall have effect, subject to the following modifications, namely:—

(*a*) references to "Commission" shall be construed as references to "State Commission";

(*b*) references to "Central Government" shall be construed as references to "State Government"; and

(*c*) references to "Member-Secretary" shall be construed as references to "Secretary".

CHAPTER V

CHILDREN'S COURTS

Children's Courts.

25. For the purpose of providing speedy trial of offences against children or of violation of child rights, the State Government may, with the concurrence of the Chief Justice of the High Court, by notification, specify at least a court in the State or specify, for each district, a Court of Session to be a Children's Court to try the said offences:

Provided that nothing in this section shall apply if –

(*a*) a Court of Session is already specified as a special court; or

(*b*) a special court is already constituted,

for such offences under any other law for the time being in force.

Special Public Prosecutor.

26. For every Children's Court, the State Government shall, by notification, specify a Public Prosecutor or appoint an advocate who has been in practice as an advocate for not less than seven years, as a Special Public Prosecutor for the purpose of conducting cases in that Court.

CHAPTER VI

FINANCE, ACCOUNTS AND AUDIT

Grants by Central Government.

27. (*1*) The Central Government shall, after due appropriation made by Parliament by law in this behalf, pay to the Commission by way of grants such sums of money as the Central Government may think fit for being utilised for the purposes of this Act.

(*2*) The Commission may spend such sums of money as it thinks fit for performing the functions under this Act, and such sums shall be treated as expenditure payable out of the grants referred to in sub-section (*1*).

Grants by State Governments.

28. (*1*) The State Government shall, after due appropriation made by Legislature by law in this behalf, pay to the State Commission by way of grants such sums of money as the State Government may think fit for being utilised for the purposes of this Act.

(*2*) The State Commission may spend such sums of money as it thinks fit for performing the functions under Chapter III of this Act, and such sums shall be treated as expenditure payable out of the grants referred to in sub-section (*1*).

Accounts and audit of Commission.

29. (*1*) The Commission shall maintain proper accounts and other relevant records and prepare an annual statement of accounts in such form as may be prescribed by the Central Government in consultation with the Comptroller and Auditor-General of India.

(*2*) The accounts of the Commission shall be audited by the Comptroller and Auditor-General at such intervals as may be specified by him and any expenditure incurred in connection with such audit shall be payable by the Commission to the Comptroller and Auditor-General.

Box-IX.1: Recent technology interventions in PMFBY

- **Digi-Claim-Payment Module** – A new module has been launched to integrate the National Crop Insurance Programme (NCIP) with public financial management system (PFMS) end-to-end. Now, the government will have visibility of the quantum of eligible claims, claims paid by the Insurance Company and actual claims transferred to beneficiary farmers.
- **Yield Estimation Based on Technology (YES-Tech)** is a technology-based yield estimation mechanism developed after two years of rigorous testing and a pilot that runs across 100 country districts. Nine states, namely, Assam, Haryana, Rajasthan, Madhya Pradesh, Maharashtra, Andhra Pradesh, Tamil Nadu & Karnataka & Odisha, are implementing Yes-Tech from the Kharif 2023 season onwards.
- **Weather Information Network & Data System (WINDS)-** is a pioneering initiative to set up a network of Automatic Weather Stations & Rain Gauges at Taluk/ Block and Gram Panchayat (GP) levels for use of all farmer and farming-oriented services. It is proposed that an automatic rain gauge (ARG) would be installed at every GP and an automatic weather station (AWS) at every block covered under PMFBY.
- **Collection of Real-time Observations and Photographs of Crops (CROPIC)-** is an initiative that has been taken up to collect periodic photographs of crops during their life cycle. These photographs will validate sown and insured crops, assess crop damage when any localised and widespread calamity or climatic condition affects the crops, and act as an input for Technology-based yield estimation models.

Agriculture Marketing-Reaping prosperity

9.16 A comprehensive and diversified marketing network enables the farmer to bring the produce to the market most efficiently and timely. It reduces post-harvest losses, enhances competition, and allows price discovery. With the increase in marketable surplus, it has become all the more essential to provide the farming community with better marketing facilities. Delays in farmers accessing formal institutions, including for government procurement at the Agricultural Price Marketing Committee (APMC), enhance reliance on intermediaries[29].

9.17 The average area served by mandis in the country is 434.48 sq. km against the recommendation of the National Commission on Farmers (2006) of the market within a radius of 5 Km (corresponding market area of about 80 sq. km.). The physical constraints in transporting produce to mandis present another layer of difficulty, particularly for small and marginal farmers who may have to travel long distances[30].

9.18 To Promote efficiency in agriculture marketing, and improve price discovery, the government implemented the e-NAM Scheme. Under the e-NAM Scheme, the Government

29 The Standing Committee on Agriculture (2018-19): Agriculture Marketing and role of Weekly Gramin Haats, Sixty Second Report

30 Final Report of Committee of State Ministers, In-charge of Agriculture Marketing to Promote Reforms(2013)

provides free software and assistance of ₹75 Lakh per APMC mandi for related hardware, including quality assaying equipment and the creation of infrastructure for cleaning, grading, sorting, packaging, etc. As of 14th March 2024, more than 1.77 Crore farmers and 2.56 Lakh traders have been registered on the e-NAM portal. The Government of India launched the scheme to form and promote 10,000 FPOs in 2020 with a budget outlay of ₹6.86 Thousand Crore till 2027-28. As of 29 February 2024, 8,195 FPOs have registered under the new FPO scheme, and equity grants of ₹157.4 Crore were released to 3,325 FPOs. Credit guarantee cover worth ₹278.2 Crore was issued to 1,185 FPOs.

9.19 Studies have evaluated the performance and prospects of the e-NAM[31] and concluded that the initiative has generally had a positive impact on farmers by enabling higher price realisation for their crops, thus fulfilling one of its primary objectives. Farmers who participated in e-NAM reported receiving higher prices for their crops post-implementation. About 66 per cent of farmers from the surveyed states found the quality testing procedures transparent[32]. A significant proportion of farmers (82 per cent, 79 per cent, 64 per cent, and 89 per cent) in states like Gujarat, Haryana, Maharashtra, and Telangana, respectively, observed better pricing and reduced transaction costs. Overall, farmers expressed satisfaction with the e-NAM facilities, including cleaning, drying, weighing, assaying, bid management, and e-auction. Similarly, it is seen that 54 per cent of the farmers prefer transactions through the eNAM portal over traditional markets due to the multiple benefits derived through this portal[33]. Implementation challenges of e-NAM still exist, such as limited awareness, lack of trust and infrastructure issues related to setting up assaying facilities.

Box IX. 2: Futures market for agriculture commodities in India

The establishment of a forward market in commodities was driven by the recognition that while the production of agricultural products was largely seasonal and subject to various risks, consumption was not. The forward market serves as a mechanism that brings the prospects of future production and consumption to influence today's price in a logical manner. This process, among other things, establishes a link between present and future production and consumption cycles, thereby facilitating the inter-temporal smoothing of prices. This understanding of the forward market is crucial in comprehending the evolution and current state of the Indian agricultural commodity markets (Bhattacharya, 2007).

31 Nuthalapati, C. S. R. (2020). Institute of Economic Growth. Link to access. https://desagri.gov.in/wp-content/uploads/2024/04/2020-21-Electronic-National-Agricultural-Market-e-NAM-A-Review-of-Performance-and-Prospects.pdf
Shah, B et,al(2023). Electronic National Agriculture Market (e-NAM): A Review of the game changing Marketing Platform. Link to access https://www.researchgate.net/publication/374975907_Electronic_National_Agriculture_Market_e-NAM_A_Review_of_the_game_changing_Marketing_Platform
Performance Evaluation of e-National
Agriculture Market (2020) CCS National Institute of Agricultural Marketing: Link to access:https://ccsniam.gov.in/images/pdfs/Evaluation.pdf

32 Nuthalapati, C. S. R. (2020). Institute of Economic Growth. Link to access.https://desagri.gov.in/wp-content/uploads/2024/04/2020-21-Electronic-National-Agricultural-Market-e-NAM-A-Review-of-Performance-and-Prospects.pdf

33 https://ccsniam.gov.in/images/pdfs/Evaluation.pdf

Evolution: The Indian commodity derivatives market has a long history. India had around 300 commodity exchanges during the 1940s. Until 1952, trading in these exchanges was not regulated by a standard policy or a market regulator. After independence, the GoI formulated the Forward Contracts (Regulation) Act of 1952 and set up the Forward Market Commission (FMC) in 1953 as the regulator. In 1966, a comprehensive ban on futures trading was imposed to control price volatility. At different times, the GoI appointed different committees to look into the feasibility of reintroducing commodity derivatives trading.

Agriculture futures trade in India got a major revival in 2003 with the setting up of national exchanges such as the National Commodity and Derivative Exchange (NCDEX), Multi Commodity Exchange (MCX), and National Multi Commodity Exchange (NMCE). A significant development in the Indian commodity regulatory landscape happened in 2015 when the GoI repealed the Forward Contracts Regulation Act 1952 (FCRA), and Commodity derivatives markets were brought under the Securities Contracts (Regulations) Act (SCRA), 1956 with the Securities and Exchange Board of India (SEBI) taking over from the Forward Market Commission as the commodity market regulator in September 2015. In parallel, e-NAM were introduced by the GoI in 2016 to provide farmers with an electronic online marketplace for agricultural produce. Electronic spot markets for commodities are crucial in integrating localised physical markets, establishing a direct link between the buyer and seller and providing a transparent price discovery mechanism.

Current state of play: The commodity futures market can effectively contribute to price discovery only when many consumers, producers, traders, and aggregators use these markets to hedge their risk. The interplay of these participants, speculators, and arbitrageurs provides liquidity and helps price discovery for longer periods. However, given that most Indian farmers are marginal with fragmented land holdings, they are often left without the necessary wherewithal to engage/participate in these markets effectively, leading to reduced depth in the Indian commodity futures market. Further, it is also observed that the requirement of standardised exchange contracts with specified quality parameters and delivery requirements has also impeded the majority of Indian farmers from effectively engaging in the commodity futures market since the Indian farmers produce different varieties of commodities in widely varying qualities due to varied geographical, weather, and soil conditions. Additionally, periodic bans imposed by GoI on futures trading on agri-commodities as one of the measures to counter food inflation have also had implications on the traded value and price volume in Indian commodity and derivative exchanges.

The way forward: Studies point out that the Indian commodity market can be strengthened through a sequenced diversification of agri-future portfolio. As is highlighted in the 2008 Abhijit Sen Committee Report, "combining prudence with the benefit of the doubt, the best course of action would be to identify those commodities where there is the possibility of futures trading affecting expectations that may influence inflation in essential commodities and insulate these from futures." Sensitive commodities (e.g., common rice, wheat, most pulses, etc.) may be kept outside the ambit of the futures market until the markets are developed and the regulator has a higher degree of comfort in diversifying the portfolio. The agriculture futures market may focus on less sensitive commodities like oilseed complex (oilseeds, meals, and oils), feed (maise), cotton, basmati rice, spices, etc.

As part of recent policy initiatives to broaden the commodity derivatives market, the

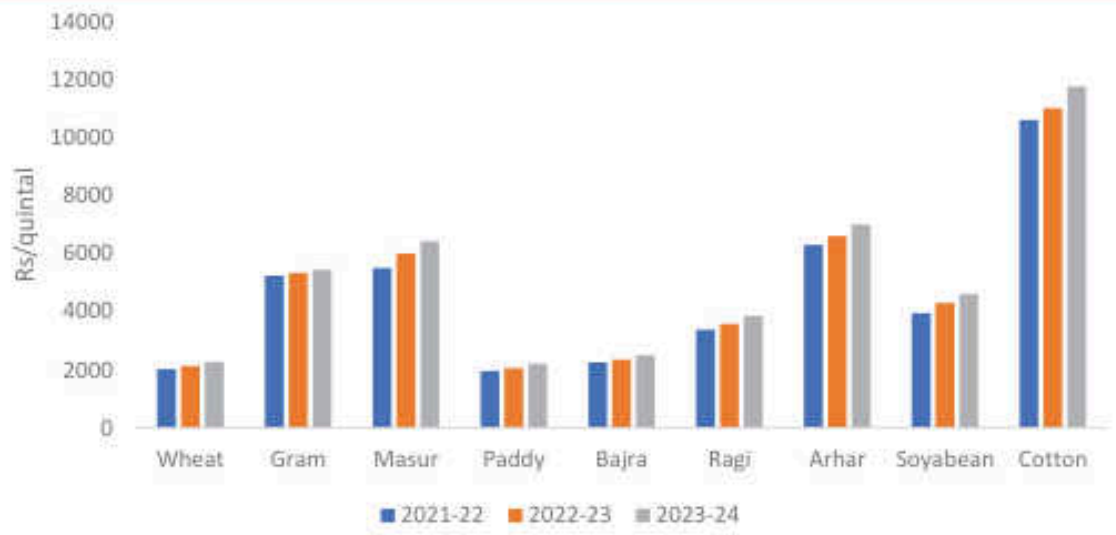
Government of India, on March 1, 2024, expanded the list of commodities eligible for derivatives trading from 91 to 104. The new commodities added to the list include Apples, Cashews, Garlic, Skimmed Milk Powder, White butter, Weather, Processed timber products, Processed bamboo products, etc.

Once the regulators provide clear direction regarding the choice of commodities, they must stay the course by adopting a stable policy with minimal interventions. Farmer Producer Organisations can play a significant role in effectively linking small and dispersed farmers in India and the Commodity markets eco-system. The role of Government, SEBI, and Commodity Exchanges in promoting FPOs in various segments of agri-commodities across the country is pivotal. Skilling and hand-holding the FPOs through financial literacy initiatives can go a long way in encouraging the farmers to benefit from the Agri-derivative markets. As the depth and liquidity in the agri-derivative market increases in the long -run, banning futures trading may no longer be required to stabilise prices unless there is data-backed evidence of futures trading driving up price volatility. The regulator should closely watch the futures market and undertake regular reviews given the fluctuations in domestic production, consumption, and global trade.

Assured Remunerative Prices and Other Income Support Measures

9.20 Agricultural price support assures farmers of remunerative returns and allows the Government to ensure a stable supply of staples at reasonable prices. Minimum Support Price is a factor farmers take into account when making sowing decisions. In India, the government has implemented price policies to encourage the production of food grains, pulses, oilseeds, and nutri-cereals. These policies are designed to provide a safety net to farmers by guaranteeing a minimum price for their produce, thus protecting them from the volatility of market prices. The Union Budget for 2018-19 announced that farmers in India would be given an MSP of at least one and a half times the cost of production. Accordingly, the Government has been increasing the MSP for all 22 Kharif, Rabi and other commercial crops with a margin of at least 50 per cent over the all-India weighted average cost of production since the agricultural year 2018-19.

Chart IX.6: MSP of major crops from 2021-22 to 2023-24



Source: Commission for Agricultural Costs & Prices (CACP)

9.21 Another initiative towards increasing farmers' income is PM-KISAN - a central sector scheme launched on 24 February 2019 to supplement the financial needs of land-holding farmers, subject to exclusions. Under the scheme, an economic benefit of ₹6000/- per year is transferred in three equal four-monthly instalments into the bank accounts of farmers' families across the country through Direct Benefit Transfer (DBT) mode. More than ₹3.24 lakh Crore has been released to more than 11 crore farmers as of 10 July 2024.

9.22 To provide social security to the most vulnerable farmer families, the Government implements Pradhan Mantri Kisan Maandhan Yojna (PMKMY). The scheme offers a monthly pension of ₹3,000 to the enrolled farmers on the attainment of 60 years of age, based on a nominal premium between ₹55 to ₹200 per month paid by the applicant (in the age group 18 to 40 years) subject to exclusion criteria. As per the Ministry of Agriculture, as of 07 July 2024, 23.41 lakh farmers have enrolled under the scheme.

Farm Mechanisation- Powering agriculture

9.23 Agricultural mechanisation covering simple hand tools to more complex machinery has become essential to modern agriculture and contributes towards productivity. Considering that most farmers in India are small and marginal, providing machinery through custom hiring can enhance the adoption of farm mechanisation among these farmers and in regions where mechanisation levels are currently low. The Sub Mission on Agricultural Mechanization (SMAM) provides assistance to the State Government for training and demonstration of agricultural machinery, setting up of Custom Hiring Centres (CHC) and assists farmers in procuring various farm machinery and equipment. The total funds allocated under SMAM from 2014-15 to 2023-24 was ₹7.26 Thousand Crore. In 2023-24, the allocation was ₹859.45 Crore. Further, Farm Machinery Banks promote access to high-tech machinery for small and marginal farm holdings and farm holdings under challenging areas with low levels of mechanisation. During the period 2014-15 to 2023-24, 25527 CHC were established under the scheme, and 607 CHC were set up in 2023-24.

Making Agriculture Sustainable

9.24 A growing challenge in agriculture pertains to sustainability issues like overexploitation, degradation of natural resources, and addressing the impact of climate change. The agricultural methods and inputs used have also had significant implications for sustainable agriculture. For example, increased use of fertiliser and chemicals, overexploitation, and unsustainable use of water resources have affected soil health and fertility. Variability in weather conditions and the relative predominance of rainfed agriculture also impact production and productivity. Sustainability in agriculture is highly relevant to securing the long-term productivity of land holdings, ensuring sufficient farm-based incomes and food security. With 11 of the 17 Sustainable Development Goals (SDGs) directly linked to agriculture, securing improvements in crop yields and ensuring income stability is vital for the country as it strives to achieve the Agenda 2030 goals.

9.25 The climate change impact assessment carried out by the government underscores the need for adaptation in the sector. In the absence of the adoption of adaptation measures, rainfed rice yields in India are projected to drop by 20 per cent in 2050 and 47 per cent in 2080 scenarios, while irrigated rice yields are projected to reduce by 3.5 per cent in 2050 and 5 per

cent in 2080 scenarios. Climate change is projected to reduce wheat yield by 19.3 per cent in 2050 and 40 per cent in 2080 scenarios[34].

9.26 National Mission for Sustainable Agriculture, a part of the National Action Plan on Climate Change (NAPCC), aims to evolve and implement strategies to make Indian agriculture more resilient to the changing climate. One of the most critical interventions in the face of changing climate is to ensure that farmland gets assured irrigation. In this context, Rainfed Area Development (RAD), implemented under the NMSA to enhance productivity and minimise risks associated with climatic variability, is germane. An amount of ₹1.74 Thousand Crore has been released & an area of 7.33 Lakh hectares has been covered under the RAD programme. Pradhan Mantri Krishi Sinchayee Yojana (PMKSY), consisting of two major components, the Accelerated Irrigation Benefit Programme (AIBP) and Har Khet Ko Pani (HKKP), promote the extension of areas under irrigation and water efficiency. Irrigation area coverage increased from 49.3 per cent of gross cropped area (GCA) in 2015-16 to 55 in 2020-21. Similarly, irrigation intensity (ratio of gross irrigated area to net irrigated area) recorded a rise of 10.3 percentage points, from 144.2 per cent in 2015-16 to around 154.5 per cent in 2021-22, while cropping intensity increased by 12.8 percentage points during this period[35]. A Micro Irrigation Fund (MIF) of an initial corpus ₹5 thousand Crore has also been created with NABARD to facilitate the States in mobilising the resources for expanding coverage of micro irrigation. Further, PDMC scheme also supports micro-level water harvesting, storage, management, etc. An area of 90.0 Lakh hectares has been covered under micro irrigation in the country under the PDMC from 2015-16 to 2023-24 as of 6th February 2024[36].

Box-IX.3: Policy led interventions to improve water management-National and International Experience[37]

Automation of Irrigation System: Narayanpur Left Bank Canal System (Karnataka)

The Narayanpur Left Bank Canal (NLBC) system in Karnataka was facing significant challenges such as inadequate water regulation, manual control of gates, and inequitable water distribution. To address these issues, the government implemented an automation system that included over 4,000 automated control and regulating gates, solar-powered integrated gates, and a master VSAT communication system. These interventions have optimised water use efficiency, improved equitable distribution, and enhanced overall agricultural productivity in the region.

Diversion-Based Irrigation System

In the hilly and undulating regions of Barwani and Khargone districts in Madhya Pradesh, the Aga Khan Rural Support Programme (AKRSP) has initiated the development of diversion-

34 https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1909206
35 CACP report on Kharif Price Policy 2024-25
36 ibid
37 Compendium of Best Practices in Water Management NITI Aayog(2023): Link to access; https://www.niti.gov.in/sites/default/files/2023-08/COMPENDIUM-OF-BEST-PRACTICES-IN-WATER-MANAGEMENT-3.0_Water-Resources-Vertical_2_8_23.pdf

based irrigation (DBI) systems. These systems use gravity flow to divert water from streams to agricultural fields. Since 2016, 13 DBI systems have been operationalised, bringing 111 hectares of land under irrigation and benefiting 93 farmers. The cost-effective nature of these systems, requiring approximately ₹300 per running meter, makes them a viable solution for enhancing irrigation coverage in hilly terrains.

Growing Tomatoes Without Soil Using Vertical Farming in Hydroponics

Vertical farming with hydroponics allows the cultivation of tomatoes without soil, offering numerous benefits such as space efficiency, reduced water usage, and year-round production. This method has been implemented in Port Augusta Farm, South Australia, which features a 4.5-hectare greenhouse powered by a 51,500 m^2 concentrated solar power plant. This plant includes 23,000 mirrors that direct sunlight to a 127-meter-high tower weighing 234 tons. The generated heat serves three purposes: maintaining optimal temperatures in 20 hectares of greenhouses, generating electricity via a turbine to power farm systems, and desalinating seawater drawn from the nearby Spencer Gulf. The farm produces one million liters of fresh water daily by desalinating seawater from 3 kilometers away. It grows 7,000 tonnes of tomatoes annually, which accounts for 15 per cent of Australia's total crop, on arid land. In addition, 180,000 tomatoes are grown hydroponically in stacks without soil, saving 2 million liters of diesel and reducing CO_2 emissions by 15,000 tons compared to traditional farming methods.

This method entails growing plants in nutrient-rich water solutions customized to their specific requirements. Vertical hydroponic systems can be employed in urban areas, reducing the environmental impact of transportation and offering locally sourced fresh produce.

Temporary Flood Water Storage in Agricultural Areas in the Middle Tisza River Basin

In the Middle Tisza River Basin, temporary floodwater storage in agricultural areas is being used to reduce flood risks and provide extra irrigation. By storing excess floodwater in specific agricultural fields, the region can better control water levels, decreasing the risk of flooding downstream. This method not only safeguards infrastructure but also improves soil fertility by depositing nutrients from the floodwaters, leading to increased agricultural productivity.

These interventions demonstrate innovative water management and agricultural productivity approaches, addressing specific regional challenges with tailored solutions.

9.27 Another area of intervention is focused on reducing the use of chemical fertilisers in Indian agriculture. Even while the per-hectare usage of agricultural chemicals in India remains significantly lower than in most developed countries, its use of chemical fertilisers has increased over the years[38]. In fact, here it may be noted that the current subsidy structure has contributed to an increased application of urea, which has impacted the nutrient imbalance in the use of major plant nutrients, nitrogen-phosphorus-potassium (NPK), affecting the efficiency of fertiliser use, the quality of soil and output[39]and environment. Revising subsidy policies to support all

38 Chand, R., & Singh, J. (2023). From Green revolution to Amrit Kaal. National Institution for Transforming India. GoI.

39 https://www.epw.in/journal/2023/52/letters/nutrient-imbalance-india.html

major nutrients (N, P, K) can incentivise farmers to use a more balanced approach. While the composition is important, so is the quantity of application. With the view to building knowledge and capacity among farmers, demonstrations to administer precise fertiliser application were carried out with more than 1.79 Lakh drones across several states.

9.28 The PM Programme for Restoration, Awareness Generation, Nourishment, and Amelioration of Mother Earth (PM-PRANAM) initiative incentivises states to reduce chemical fertiliser use. It promotes sustainable methods such as the use of alternative fertilisers, viz. Nano Urea, Nano DAP, and organic fertiliser. Under the said scheme, 50 per cent of the fertiliser subsidy saved by a State/UT in a particular financial year by way of a reduction in consumption of chemical fertilisers (Urea, DAP, NPK, MOP) compared to the previous three years' average consumption, will be passed on to that State/UT as a grant. In addition to these initiatives, the Soil Health Card Scheme was introduced to optimise the usage of nutrients. The introduction of "Urea Gold", which is urea infused with sulphur to address sulphur deficiencies, has been another measure to improve the nutrient balance in the soil.

9.29 Organic and natural farming provides chemical-free fertiliser and pesticide-free food grains and other crops, improving soil health and reducing environmental pollution. About 68.05 Lakh ha was brought under organic farming by 2022-23. Sikkim became the first state in the world to become fully organic, and other states, including Tripura and Uttarakhand, have set similar targets. The Government has also been promoting organic farming by implementing two dedicated schemes, i.e., Paramparagat Krishi Vikas Yojana (PKVY) and Mission Organic Value Chain Development for North Eastern Region (MOVCDNER) since 2015 through cluster/ FPO formation. PKVY Scheme is being implemented in a cluster mode (with a minimum of 20 ha size). Financial assistance of ₹31,500 per ha for three years, out of which ₹15,000 is given as an incentive for organic inputs provided directly through DBT. Under PKVY, as of 2022-23, 48,144 clusters totalling 13.98 Lakh ha area and 24.22 Lakh farmers have been covered.

Box-IX.4: Flexible, farmer-friendly and ecologically sustainable fertiliser subsidy: A suggested way forward

The Lok Sabha Standing Committee on Chemicals & Fertilisers, in their Thirty-Ninth Report, titled 'Nano-Fertilisers for sustainable crop production and maintaining soil health', presented on March 29, 2023, underscored the urgency of the following issue: Fertiliser consumption in India is imbalanced, and Urea accounts for more than 82 per cent of the nitrogenous fertilisers applied to the majority of the crops. As a result, the Nitrogen, Phosphorus and Potassium (NPK) consumption ratio has widened from 4:3.2:1 in 2009-10 to 7:2.8:1 in 2019-20. This imbalance, leading to the deterioration in soil quality and health hazards, necessitates an immediate re-examination of urea subsidy management in agriculture while also considering the sustainability aspects in the long run.

The current design of fertiliser subsidy in India: Based on the recommended dose of nutrients (RDN) calculated by each state[40], the Government of India calculates the recommended dose of fertiliser (RDF) and allocates fertilisers to States for each season.

The States, in turn, sell the fertiliser to the farmers through dealers and primary agriculture

40 The RDN is calculated by each state based on crops grown and soil nutrient status.

cooperative societies using POS devices. Based on the quantity of fertilisers sold to farmers, the Department of Fertilisers pays fertiliser subsidies to the fertiliser companies. However, there are some critical issues with the current design. This includes the following:

- PoS devices at fertiliser outlets are not integrated with land record data
- Any person having an Aadhaar, whether a farmer or not, can buy any quantity of fertilisers
- No limit on the sale of fertiliser to one person or one family
- Adverse financial and ecological impacts such as diversion of subsidised fertiliser for non-agriculture purposes; overuse of fertilisers which adversely affect the soil health; shortage of fertiliser; waste of public resources and health hazards.

Using Agri Stack to improve the targeting of fertiliser subsidy: Agri Stack is the digital foundation set up by the government to make it easier to bring various stakeholders together to improve agriculture in India and enable better outcomes and results for the farmers by using data and digital services. It is now fairly well developed in the major Indian States and can provide the right tool through which the fertiliser subsidy may be better targeted. This will ensure that subsidised fertilisers are sold to only those identified as farmers or authorised by the farmer, and the quantity of subsidised fertiliser is fixed based on parameters such as land ownership and prominent crops of the district (comprising at least 70 per cent of sown area in a season). The parameters may be later refined based on crop grown and soil nutrient status (in convergence with the Soil Health card scheme and provisions may be made to provide top-up entitlement in case of crop damage or calamities caused by volatilities in weather conditions, in convergence with State diaster response fund (SDRF) / National diaster response fund (NDRF).

E-RUPI, a seamless one-time payment mechanism, can be utilised to provide the necessary subsidy to the farmer directly. This system ensures that the Subsidy can only be used through registered PoS devices at authorised fertiliser outlets. Suppose a farmer purchases a quantity of fertilisers that is less than their entitlement. In that case, the remaining subsidy can be used to purchase other agricultural inputs, such as seeds and pesticides, also sold at these outlets. Any unused subsidy at the end of the year can also be converted into a small savings instrument in the farmer's name at a post office. This system not only streamlines the subsidy distribution process but also prevents the misuse of subsidies for non-agricultural purposes. This will give an incentive to the farmer not to use excessive Urea on account of being cheaper than the other NPK fertiliser and may lead to balanced use of fertilisers as per the requirement of the crop & soil.

Some fundamental aspects that would be required to ensure the efficiency of the new mechanism will be as follows:

- Integration of PoS devices with farmer's registry in Agri Stack and the farmer's registry will include the Aadhaar number of each farmer, details of all agricultural lands owned by the farmer as per Record of Rights (RoR), and dynamic updation of land ownership data through mutation module
- Facility to include name, Aadhaar number and other details of family members and any other person authorised to buy subsidised fertilisers
- Facility to update bank details, mobile numbers and other details of farmers, family members, and authorised persons
- Crop sown registry based on digital crop survey to be integrated at a later stage

groundnut were released. Many of the rice varieties that India now exports to the rest of the world came out of research at the International Rice Research Institute. It is a reminder that agricultural research offers one of the highest returns on investment. Hence, reinforcing the agricultural research ecosystem with human and financial resources will continue to pay rich dividends to farmers and to the nation.

FOOD PROCESSING SECTOR (FPI): PROCESSING POTENTIAL

9.39 India is the largest producer of milk and the second largest producer of fruits, vegetables and sugar. The availability of reasonably priced agricultural inputs, the vast labour force, and continuously growing consumer demand provide the essential elements for a robust food processing industry. The sector also plays a vital role in reducing the wastage of perishable agricultural produce, enhancing the shelf life of food products, ensuring value addition to agricultural produce, and incentivises diversification & commercialisation of agriculture. True to that, the food processing industry in India is one of the largest employers in organised manufacturing, with a 12.02 per cent share in the total employment in the organised sector[45]. The value of agri-food exports, including processed food exports during 2022-23, was USD46.44 Billion, accounting for about 11.7 per cent of India's total exports. The share of processed food exports also increased from 14.9 per cent in 2017-18 to 23.4 per cent in 2022-23.

9.40 It is an important industry because it has strong linkages with the agriculture sector and can employ surplus workforce released from the agriculture sector. During the last eight years ending 2022-23, the food processing industries has been growing at an average annual growth rate (AAGR) of around 5.35 per cent at 2011-12 prices. Being labour-intensive, the pandemic adversely affected the sector and it is now recovering. GVA in the food processing sector has increased from ₹1.30 lakh Crore in 2013-14 to ₹1.92 lakh Crore in 2022-23. The sector constituted 7.66 per cent of GVA in Manufacturing in 2022-23 at 2011-12 prices.

Chart IX.9: Share of FPI in manufacturing GVA and growth of FPI in per cent



Source: MoFPI

45 Ministry of Food Processing(MoFPI)

9.41 The Government has taken several initiatives to improve the supply chain management from farm gates to retail outlets. The Production Linked Incentive Scheme for the Food Processing Industry (PLISFPI) supports the creation of global food manufacturing champions, branding and marketing abroad. It is expected to create off-farm employment and provide better prices for farm produce and higher income to farmers. Presently, 173 applications are covered under the PLI Scheme. The beneficiaries of the scheme have invested ₹7.69 Thousand Crore. An incentive amount to the tune of ₹1.07 Thousand Crore was released in FY 2021-22 and FY 2022-23.

9.42 The PM Formalization of Micro Food Processing Enterprises (PMFME) scheme with a total outlay of ₹10 Thousand Crore provides credit-linked subsidy and capacity building, including marketing and branding support. Convergence is being sought with the existing ecosystem to support and complement other schemes such as the National Rural Livelihood Mission, National Urban Livelihood Mission, One District One Product, AIF, and PMKSY implementation. All State/UTs have appointed a state nodal agency, constituted state level approval committee & district level committee for implementation of the scheme. Further, 2 national level technical institutes and 44 state level technical institutes in 36 States/ UTs have also been approved. Against the target of two lakhs, 3,53,608 applications were received, and a loan amount of ₹6.94 Thousand Crore to 86,342 applicants was sanctioned. 522 Master Trainers from 35 States/UTs and 1068 District Level Trainers from 26 States/UTs and 70,936 beneficiaries from 30 States/UTs have been trained.

9.43 The scheme for developing the Tomato, Onion and Potato (TOP) value chain was launched in 2018-19. The coverage of Operation Green has been expanded from 3 crops (tomato, onion & potato) to 22 perishables crops, which include 10 fruits, 11 vegetables (including TOP) and one marine, i.e. shrimp. The scheme's objectives include enhancing farmers' value realisation, reducing post-harvest losses, increasing food processing capacities, and adding value. The scheme has two-pronged strategies: Price Stabilization Measures (short-term measures) and Integrated Value Chain Development Projects (long-term). Under the short-term interventions of the scheme, there is a provision for a 50 per cent subsidy on the cost of transportation and storage for fruits & vegetables for evacuation of surplus production from production centres during the glut situation. For the long-term, grant-in-aid is provided in the range 35 per cent to 50 per cent for setting up food processing project for eligible crops in the identified production clusters in major producing states.

FOOD MANAGEMENT[46]: SOCIAL NET FOR FOOD SECURITY

9.44 The main objectives of food management are the procurement of foodgrains from farmers at remunerative prices, the distribution of foodgrains to consumers, particularly to the vulnerable sections of society, at affordable prices, and the maintenance of food buffer stocks for food security and price stability. The instruments used are procurement at MSP from farmers and Central Issue Price (CIP) for consumers. The nodal agency that undertakes the procurement, distribution, and storage of food grain is the Food Corporation of India (FCI).

46 Food Price Inflation is covered under Chapter 5- Prices and Inflation

For prudent management of foodgrain stock and for ensuring adequate availability of wheat and rice in the central pool, the Central Government implements a decentralized procurement scheme.

9.45 As of 24 May 2024, procurement of wheat during the Rabi Marketing Season(RMS) 2024-25 is going smoothly in the major procuring States nationwide, with 263.33 LMT of wheat already procured for the central pool, surpassing last year's total procurement of 262.02 LMT. A total of 22.42 lakh farmers have been benefitted during RMS 2024-25. Similarly, 489.20 LMT of rice was procured directly from 98.26 lakh farmers during the Kharif Marketing Season (KMS) 2023-24. With the above procurement quantity, the combined stock of wheat and rice in central pool has surpassed 600 LMT, which puts the country in a comfortable position to meet its requirements for food grains.

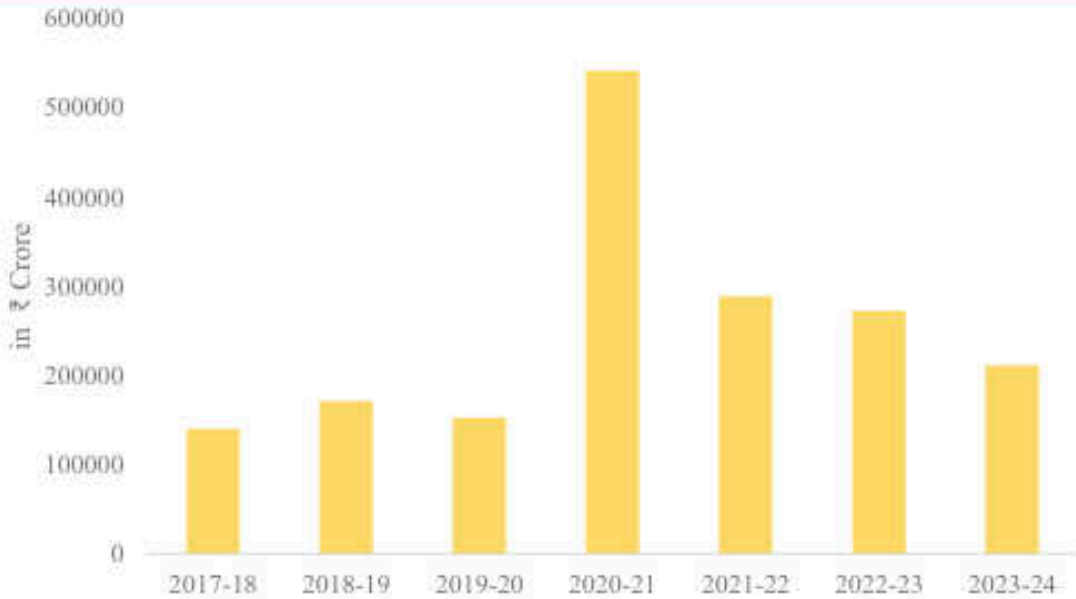
9.46 The distribution of food grains is undertaken primarily under the National Food Security Act, 2013 (NFSA) and other welfare schemes of the Government of India, including PMGKAY. The government has addressed the issue of food security at the household level for a long time through the public distribution system and the targeted public distribution system, as well as the enactment of the NFSA 2013. Further, the Government decided to continue to provide free food grains to about 81.35 crore beneficiaries (i.e., Antyodaya Anna Yojana (AAY) households and Priority Households (PHH) beneficiaries) under the PMGKAY for a further period of five years with effect from 01 January 2024, with an estimated total financial outlay of ₹11.80 lakh Crore to be borne by Central Government. The scheme provides a unified institutional mechanism with uniform prices and quantities across the country and removes difficulties for beneficiaries, especially migrants, under the One Nation One Ration Card (ONORC). Through this system, migrant beneficiaries can claim from any Fair Price Shop (FPS) of their choice, anywhere in the country, based on existing/same ration card in a seamless manner by using either their ration card or Aadhaar number after biometric/ Aadhaar authentication on ePoS device.

9.47 The procurement of foodgrains at MSP and distribution of foodgrains at less than economic cost have financial implications for the government. The economic cost[47] of both wheat and rice witnessed a significant increase during the last few years due to an increase in MSPs and a proportionate increase in the incidentals. The economic cost of rice and wheat for the year 2023-24(RE) is at ₹ 3931.34 per quintal and ₹ 2709.59 per quintal, respectively[48].

47 The economic cost of foodgrains consists of three components, namely, pooled cost of grains, procurement incidentals and the cost of distribution.

48 Based on FCI food bulletin data April, 2024

Chart: IX.10: Food subsidy released



Source: Department of Food and Public Distribution

Note:

- In addition to net subsidy released to FCI, Repayment of NSSF loan of ₹25,000 Crores in FY 2016-17, ₹ 40,000 Crore in FY 2017- 18, ₹70,000 Crore in FY 2018-19 & ₹44,164.02 Crore in FY 2019-20 by FCI. ₹3,39,236 Cr for FY 2020-21 released from food subsidy has been adjusted for repayment of NSSF loan. Excludes ₹11,436 crore repaid to FCI from DCP States Head.
- The RE, 2019-20 was ₹33508.35 crore. The subsidy released includes ₹11,436 crore (as part of the NSSF loan), released from FCI to DCP States and returned to FCI in 2020-21.
- During FY 2023-24, ₹336.64 Crores were re-appropriated in favour of NESA Division for Central Assistance.

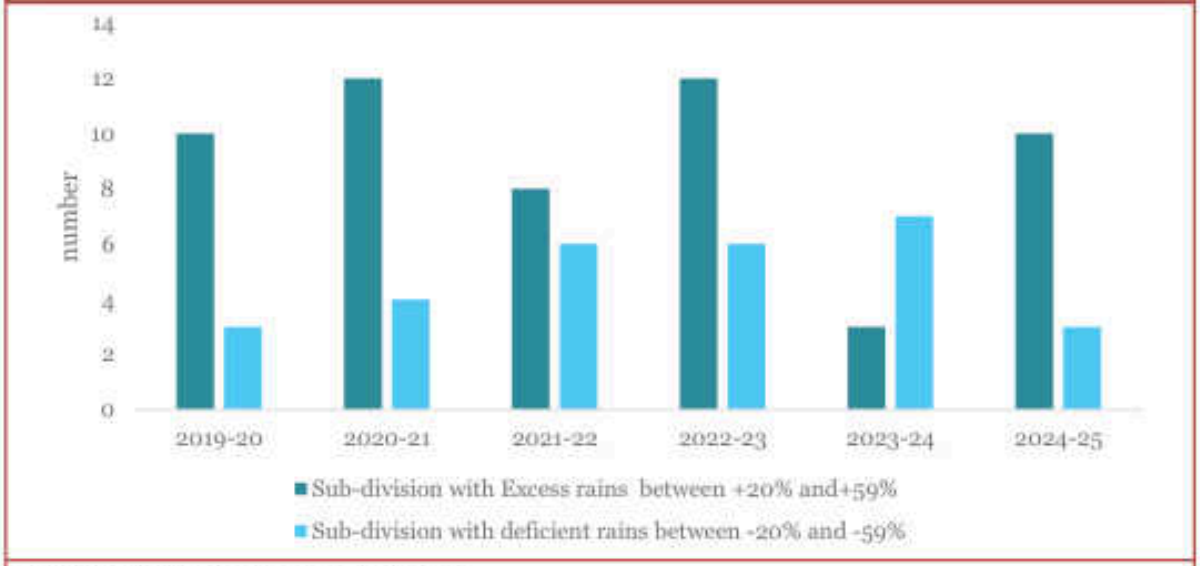
CONCLUSION

9.48 The performance of the agriculture sector remains critical for the economy's growth and has been growing at an average growth rate of 4.18 per cent over the last five years. The growing significance of allied sectors such as animal husbandry, dairying, and fisheries in enhancing farmers' income suggests that greater emphasis should be placed on tapping into the potential of these activities to boost farmers' incomes. Smallholder farmers' incomes cannot be increased by producing rice, wheat, or even millets, pulses and oilseeds. They need to move to high-value agriculture – fruits and vegetables, fisheries, poultry, dairy and buffalo meat. Once the incomes of smallholders increase, they will demand manufactured goods, spurring a manufacturing revolution. That is what happened in China between 1978 and 1984 when the real incomes of farmers doubled in just 6 years. India is well-placed to emulate this.

9.49 Promoting crop diversification towards oilseeds, pulses, and horticulture requires addressing critical issues such as investment in agri-infrastructure, credit accessibility and appropriate market institutions. MSP has incentivised crop diversification and there is evidence that MSP has a positive and statistically significant effect on retail prices of all crops, with a stronger effect for those crops where procurement is substantial, such as paddy and wheat.[49]

49 https://rbidocs.rbi.org.in/rdocs/AnnualReport/PDFs/0ANREPORT201718077745EC9A874DB-38C991F580ED14242.PDF

Chart IX.10: Subdivisions with excess/deficit rainfall in numbers



Source: Indian Meteorological Department

9.18 Several studies have also assessed the overall effect of climate change on agricultural production. Negi and Ramaswami (2024)[21] examined the relationship between crop yields and rainfall across India at the district level for nine key crops during the kharif season. They found a strong link between significant rainfall shortfalls and substantial crop yield losses. This statistical phenomenon is known as lower tail dependence, indicating that the correlation between yield losses and rainfall deficits is stronger for extreme rainfall deficiencies than for minor variations. Other studies have indicated that a potential 2°C rise in annual temperature and a 7 per cent increase in annual rainfall by 2099 could lead to an 8-12 per cent decline in Indian agricultural productivity[22]. Birthal et.al (2021)[23] found that heat stress harmed crop yield, which worsened over time. Most studies have suggested that droughts and heat waves negatively impact agricultural productivity compared to floods and cold waves in India. It is, therefore, pertinent to increase the area under irrigation and diversify towards heat and water-resistant crops.

9.19 Between FY16 and FY21, India experienced a significant increase in irrigation area coverage and intensity. The coverage of irrigation area has increased between FY16 and FY21 from 49.3 per cent to 55 per cent of the gross cropped area (GCA), while irrigation intensity has risen from 144.2 per cent to 154.5 per cent. States such as Punjab, Haryana, Uttar Pradesh, and Telangana exhibit high irrigation coverage of their gross cropped area, with figures around 98 per cent, 94 per cent, 84 per cent, and 86 per cent, respectively. In contrast, states like Jharkhand and Assam lag significantly, with irrigation coverage below 20 per cent, underscoring the necessity to improve irrigation

21 Negi, D. S., & Ramaswami, B. (2024). Basis risk and the demand for catastrophic rainfall insurance. QOpen, 4(1), qoae009.

22 NABARD Climate Change and Risk Management in Indian Agriculture, 2022.

23 Birthal, P S, J Hazrana, D S Negi and G Pandey (2021b): "Benefits of Irrigation against Heat Stress in Agriculture: Evidence from Wheat Crop in India", Agricultural Water Management, Vol. 255 , No. C.

9.36 To enhance efficiency in agricultural marketing and improve price discovery, the government has introduced the e-NAM Scheme. This initiative provides free software and financial assistance of ₹75 lakh per Agricultural Produce Market Committee (APMC) Mandi for essential hardware, which includes quality assaying equipment and the development of infrastructure for cleaning, grading, sorting, and packaging. Additionally, to support FPOs and empower farmers, the government launched a scheme in 2020 with a budget of ₹6,860 crore. As of October 31, 2024, over 1.78 crore farmers and 2.62 lakh traders have registered on the e-NAM portal. As of the same date, 9,204 FPOs have been registered, and 4,490 of these organisations have received equity grants amounting to ₹237 crore.

CLIMATE ACTION IN AGRICULTURE

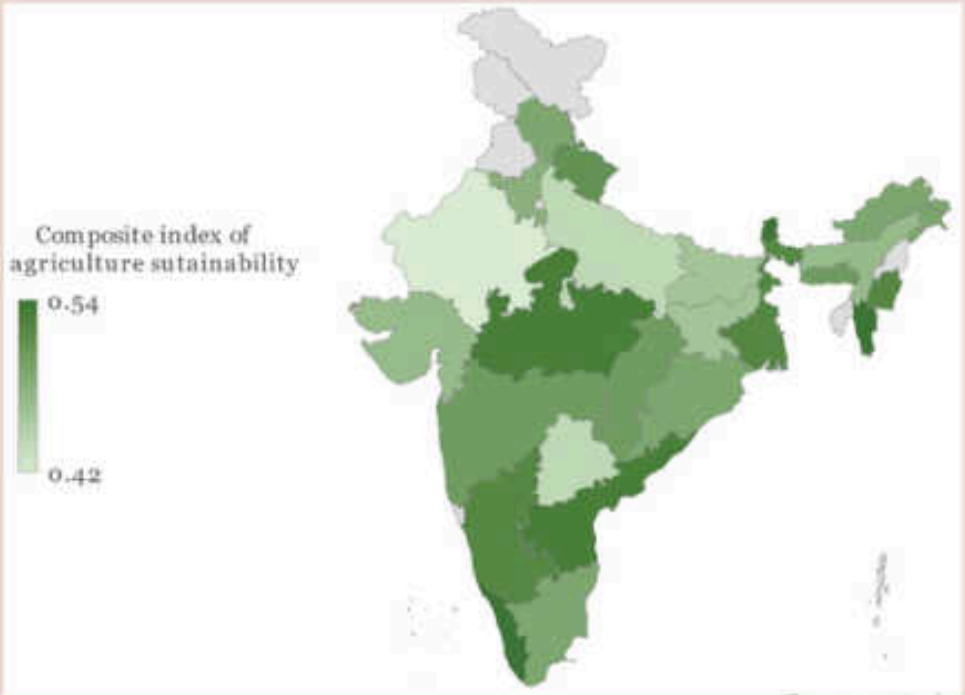
9.37 The NMSA is recognised as one of the nine missions within the National Action Plan on Climate Change (NAPCC). The NMSA has identified key climate adaptation strategies, including enhancing water efficiency, managing soil health and nutrients, providing crop insurance, offering credit support, strengthening value chains, delivering agro-advisories, promoting farm mechanisation, managing agricultural waste, developing integrated farming systems, and supporting organic and natural farming practices. It also emphasises climate-resilient varieties, livestock, and fish culture, promoting these initiatives through various government schemes.

9.38 To support organic farming, the government has implemented two dedicated schemes since 2015: the Paramparagat Krishi Vikas Yojana (PKVY) and the Mission Organic Value Chain Development for North Eastern Region (MOVCDNER). Under PKVY, 52,289 clusters covering 14.99 lakh hectares and 25.30 lakh farmers have been mobilised. Similarly, under MOVCDNER, 434 Farmer Producer Companies have been created, covering a total area of 1.73 lakh hectares and benefiting 2.19 lakh farmers.

BOX IX.7 : Spatial mapping of agricultural sustainability

A Composite Index of Agricultural Sustainability was calculated by ICAR using 51 indicators related to environmental health, soil and water quality, and socioeconomic development. The average estimated value of the Index is 0.49, suggesting that Indian agriculture is moderately sustainable. States like Mizoram, Kerala, Madhya Pradesh, Andhra Pradesh, Manipur, West Bengal, and Uttarakhand perform better than the national average.

Chart IX. 18: Composite index of agriculture sustainability-State wise



Source: A Spatial Assessment of Sustainability in Indian Agriculture, Policy paper 42, 2024, ICAR

In contrast, arid Rajasthan has the least sustainable agricultural practices. According to the report, the states with higher sustainability scores have experienced sizable crop diversification, improvement in agriculture infrastructure, farm credit and sustainable input usage. States in the Indo-Gangetic Plains, including Uttar Pradesh, Punjab, Bihar, and Haryana, as well as rice-dominant states like Jharkhand and Assam, are at a higher risk of climate change impacts.

ALLIED SECTORS: Potential to Build Resilience

9.39 The livestock sector has emerged as a significant engine of growth in agriculture, playing a vital role in the overall agricultural landscape. This importance is underscored by a remarkable increase in its contribution to the Gross Value Added (GVA) of agriculture and related sectors, which surged from 24.38 per cent in the fiscal year FY15 to an impressive 30.23 per cent by FY23. In the latter year, the livestock sector alone represented 5.5 per cent of the total GVA, reflecting its dynamic growth trajectory, with a robust CAGR of 12.99 per cent.

9.40 The economic significance of this sector is clearly illustrated by its escalating output value, which reached an astounding 17.25 lakh crore rupees (equivalent to US$205.81 billion) in FY23. Among the various branches of livestock production, the milk industry stands out, generating over ₹11.16 lakh crore (US$133.16 billion) in revenue. This figure highlights the sector's vitality and reveals its dominance, eclipsing the aggregate production value of staple crops like paddy and wheat. The livestock

receiving support from various federations. As part of the government's push for greater accessibility, 240 PACS have applied for retail petrol and diesel outlets, with 39 currently selected to operate, thus expanding their service offerings. Furthermore, a substantial number—35,293 PACS—are now functioning as Pradhan Mantri Kisan Samriddhi Kendras (PMKSK), which are dedicated to providing essential fertilisers and related services to farmers, directly impacting agricultural productivity. Further, 1,723 micro-ATMs have been distributed, facilitating doorstep financial services and enhancing accessibility for rural populations.

FOOD PROCESSING INDUSTRIES: Critical for the Economy

9.49 The food processing industry in India is one of the largest employers within organised manufacturing, accounting for 12.41 per cent of total employment in the organised sector. In the fiscal year FY24, the value of agri-food exports, which includes processed food exports, reached USD 46.44 billion, constituting roughly 11.7 per cent of India's total exports. Notably, the share of processed food exports within agri-food exports has risen from 14.9 per cent in FY18 to 23.4 per cent in FY24.

9.50 To foster growth in the food processing sector, the Indian government has initiated several key programs, including the Pradhan Mantri Kisan Sampada Yojana (PMKSY). This scheme focuses on developing modern infrastructure and optimising supply chains from farm to retail. By minimising post-harvest losses, increasing processing capabilities, and enhancing export levels, PMKSY aims to promote the overall advancement of the food processing industry. As of 31 October 2024, 1,079 PMKSY projects have been completed.

9.51 The Production Linked Incentive Scheme for Food Processing (PLISFPI), launched in 2021, seeks to cultivate globally competitive food processing leaders by facilitating branding and marketing initiatives in international markets. By 31 October 2024, 171 applications had been approved under this scheme, with beneficiaries investing ₹8,910 crore and receiving ₹1,084.01 crore in incentives.

9.52 Furthermore, to provide comprehensive support—including technical, financial, and business assistance for establishing or upgrading micro food processing enterprises—the Pradhan Mantri Formalisation of Micro Food Processing Enterprises (PMFME) scheme was launched in 2020. As of 31 October 2024, the scheme has received 407,819 applications, with loans totalling ₹8.63 thousand crore sanctioned to 108,580 applicants. Additionally, the programme has successfully trained 672 Master Trainers, 1,120 District Level Trainers, and 87,477 beneficiaries across 36 states and union territories.

FOOD MANAGEMENT: Enabling Food Security

9.53 The basic concept of food security is to ensure that there is access to basic food for their active and healthy lives. It is characterised by the availability, access, utilisation, and stability of food supply. While the government has long tackled household food

security through the Public Distribution System (PDS) and Targeted PDS (TPDS), the National Food Security Act (NFSA) 2013 and the Pradhan Mantri Garib Kalyan Anna Yojana (PMGKAY) marked a fundamental shift in the approach to food security. That is transitioning from a welfare-based to a rights-based approach. NFSA act legally entitles up to 75 per cent of the rural population and up to 50 per cent of the urban population to receive foodgrain, free of cost, under the Targeted Public Distribution System, which, as per Census 2011, comes to 81.35 crore persons. Therefore, about two-thirds of the population is covered under the Act to receive highly subsidised food grains. In addition, PMGKAY was introduced to alleviate the suffering of the poor and vulnerable caused by the COVID-19 pandemic. The allocation of free food grain under PMGKAY is in addition to the regular allocation for around 80 crore beneficiaries. The provision of free food grains under PMGKAY for another five years, with effect from 1st January 2024, reflects the long-term commitment and vision of the Government for addressing National food and nutrition security.

9.54 To enhance the efficiency of the Public Distribution System (PDS), the government is fully committed to achieving 100 per cent e-KYC compliance nationwide. This aligns with the One Nation, One Ration Card (ONORC) scheme, allowing beneficiaries to complete electronic know your customer e-KYC anywhere, regardless of their home state. Beneficiaries can authenticate their Aadhaar biometrics at any Fair Price Shop (FPS), making it especially convenient for migrant workers to access their entitlements. To facilitate post-harvest lending for farmers, especially small and marginal farmers, the government has approved the Credit Guarantee Scheme for electronic-negotiable warehouse receipt (e-NWR)-based Pledge Financing (CGS-NPF). Under this scheme, farmers can obtain loans against e-NWRs issued for agricultural and horticultural commodities stored in accredited warehouses. The scheme covers the loss incurred by the bank due to credit and warehouseman risk. The scheme will help in increasing post-harvest lending against e-NWRs and thereby play a role in improving farmers' income.

BOX IX.8 : Measures to support foodgrain storage infrastructure in the country

- To support and upgrade the storage infrastructure for foodgrains and to ramp up the storage capacity in India, steel silos are being created in PPP.
- The government is creating capacity under Hub and Spoke Model Silos, where "Hub" silos have a dedicated railway siding and container depot facility. While the transportation from "Spoke" Silos to "Hub" Silos is undertaken by road, transportation from Hub to Hub is via rail.
- To improve food grain storage, especially in hilly and remote areas, the government is exploring the use of Flospan, a type of Mobile Storage Unit (MSU), in collaboration with the World Food Programme (WFP). These units can be quickly erected and have a storage capacity of 400 metric tonnes. As a pilot project, WFP has installed Flospan in six states: Jammu & Kashmir, Himachal Pradesh, Rajasthan, Mizoram, Uttarakhand, and Chhattisgarh.
- To modernise government grain warehouses, the government partnered with WFP and IGMRI to pilot a 'Smart Warehouse'. This warehouse uses sensors to monitor temperature, humidity, airflow, and rodent activity, providing real-time data to improve storage and reduce losses.

Table I.3: Performance of high-frequency indicators indicates resilient domestic demand (YoY growth, per cent)

Sector	Indicators	Q1 FY26	Q2 FY26	Q3 FY26	Monthly Avg YoY Growth (FY16-FY20)
Urban Demand	UPI Transaction	33.3	33.0	28.7	-
	FMCG Vol. Sales (Urban)	4.6	5.6	NA	NA
	Passenger Vehicle Sales	0.0	-2.9	20.5	1.4
	Domestic Air Passenger*	5.3	-1.9	5.3	14.9
Rural Demand	FMCG Vol. Sales (Rural)	8.4	7.7	NA	NA
	Two-wheeler Vehicle Sales	-6.2	7.4	16.9	2.5
	Three-wheeler Vehicle Sales	0.1	9.8	14.0	9.2
	Tractor Sales	9.2	30.7	23.2	7.3

Source: SIAM, NPCI, Tractor and Mechanisation Association, NielsenIQ, Airport Authority of India
Note: *Data up to November 2025. NA: Not available

1.12. Along with consumption, investment has continued to anchor growth in FY26, with the share of gross fixed capital formation (GFCF) estimated at 30.0 per cent. Investment activity strengthened in the first half of the year, with, GFCF expanding by 7.6 per cent, exceeding the pace recorded in the corresponding period last year and remaining above the pre-pandemic average of 7.1 per cent (Chart I.9). This momentum was buoyed by sustained public capital expenditure[12] and a revival in private investment activity as evident from corporate announcements.[13] Reflecting this strength, the share of GFCF in GDP remained steady at 30.5 per cent in H1 of FY26, well above the pre-pandemic average of 28.6 per cent (Chart I.10). Together, these developments indicate a strengthening of the investment cycle, supporting growth.

Chart I.9: Growth in GFCF remains robust

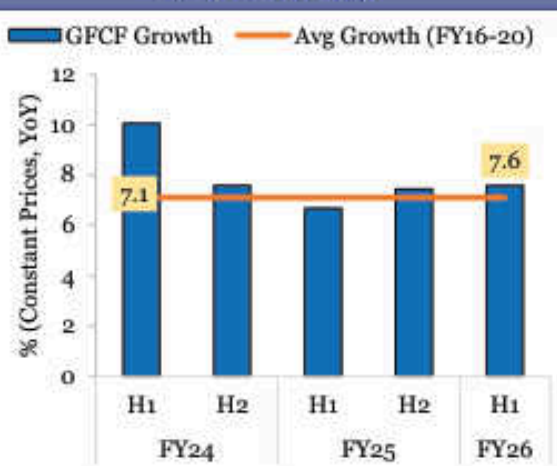
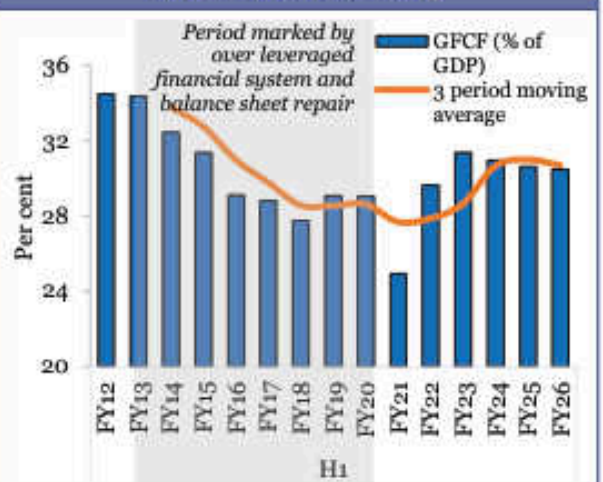


Chart I.10: GFCF contribution to GDP remains steady



Source: MoSPI

12 During the current fiscal year, capex recorded a strong rebound, rising by 28 per cent YoY during April-November 2025.

13 As per CMIE database, private corporate investment announcements totalled ₹ 14.6 lakh crore in H1 of FY26 Vs ₹ 7.9 lakh crore in H1 of FY25 and the previous decadal peak of ₹ 11.4 lakh crore in H1 of FY24.

1.13. The implicit H2 estimate of GFCF for FY26[14] indicates a firming of the investment cycle, likely driven by continued growth in non-food bank credit, coupled with capacity utilisation remaining above its long-term average. High-frequency indicators for the third quarter also suggest healthy investment activity. In particular, the Index of Industrial Production (IIP) shows sustained strength in capital goods, while imports of capital goods continue to record strong growth. (Table I.4).

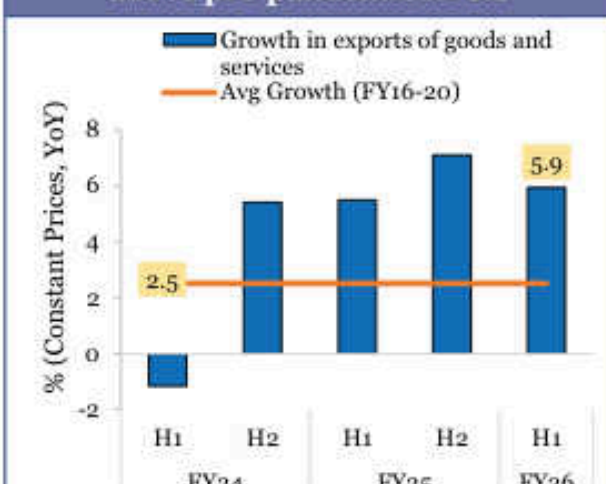
Table I.4: High-frequency indicators signalling firming investment momentum (YoY growth, per cent)

Indicators	Q1 FY26	Q2 FY26	Q3 FY26	Monthly Avg YoY Growth (FY16-FY20)
Non-food bank credit	10.2	10.4	14.4	9.7
IIP Capital goods*	9.8	5.6	6.2	0.2
Capital Goods Imports*	6.6	9.2	13.4	7.1
Capacity Utilisation (in %)	74.1	74.8	NA	72.9

Source: MoSPI, Ministry of Commerce and Industry, RBI
Note: *Data up to November 2025.
Capital goods imports are defined as the aggregate of imports of electric machinery and equipment; base metals excluding iron and steel; industrial machinery, including machinery for dairy and allied uses; machine tools; other construction machinery; project goods; and transport equipment.

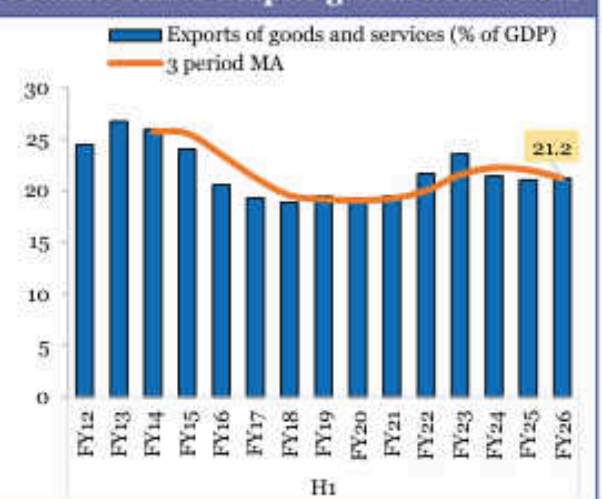
1.14. While domestic drivers remained the primary source of growth in FY26, external demand[15], with a share of 21.6 per cent of GDP, also supported growth. In the first half of FY26, exports of goods and services grew by 5.9 per cent, exceeding the growth seen in the same period last year, and remaining above the pre-pandemic average, supported by trade diversification (Chart I.11). Services exports have continued to provide a stable anchor for growth, partially offsetting the greater volatility in goods exports, amid tariff related uncertainties. The broader external sector dynamics are examined in detail in Chapter 4.

Chart I.11: Export growth sustains above pre-pandemic levels



Source: MoSPI

Chart I.12: Exports share in GDP remain stable despite global headwinds



Source: MoSPI

14 Calculated using FAE and H1 of FY26

15 i.e., the share of exports of goods and services in GDP

which experienced a 13.6 per cent (YoY) increase. At the same time, the country's total imports also increased by 7.4 per cent (YoY) and stood at USD 919.9 billion. The surplus in services trade cushioned the merchandise trade deficit, and consequently, in FY25, the total trade deficit amounted to USD 94.7 billion. This trade pattern reflects the impact of stronger import demand amid an expanding economy and resilient consumption.

Chart IV.5: India's total (merchandise and services) trade



Source: Data for FY20-FY25 is retrieved from FTPA, Department of Commerce (https://tinyurl.com/mw864w62) and DBIE, RBI (https://tinyurl.com/58jae338).
Data for April-Dec 24 and April-Dec 2025 is retrieved from the PIB press release of the Ministry of Commerce and Industry, dated 15 January 2026 (https://tinyurl.com/3fp93r6d).

4.21 This strong momentum has been carried forward into FY26 as both Q1 (Apr-June 2025), Q2 (July-Sept 2025) and Q3 (Oct-Dec 2025) recorded their highest-ever export levels for their respective quarters, despite ongoing global uncertainties. Furthermore, during April-December 2025, exports totalled USD 634.3 billion, a 4.3 per cent (YoY) increase, whereas imports increased by 4.9 per cent (YoY) to USD 730.8 billion. During the same period, the trade deficit increased to USD 96.6 billion compared to USD 88.4 billion in the corresponding period of the previous year.

Merchandise trade

4.22 The merchandise trade performance exhibits the economy's ability to hold ground amid a volatile global environment. It reaffirms its position as a reliable trading partner and an emerging force in global trade. Chart IV.6 depicts the trends in the country's merchandise trade in recent years. In FY25, India's merchandise exports totalled USD 437.7 billion, maintaining similar levels to those observed in FY24. While aggregate merchandise export growth may appear subdued, it is imperative to acknowledge

that non-petroleum, non-gems and jewellery exports, accounting for 78.7 per cent of aggregate merchandise exports, exhibited a robust growth rate of 7.5 per cent (YoY) during the same period. Further, the non-petroleum exports reached a historic high of USD 374.3 billion. This underscores the inherent strength of the export sector, which volatility in the exports of gems and jewellery and petroleum may obscure. Consequently, the perceived stagnation in total merchandise exports can largely be attributed to the disproportionate influence of petroleum and gems and jewellery, which often masks the more favourable trends present within the broader export sector.

4.23 In FY25, the export composition was predominantly concentrated in petroleum products, telecom instruments, drug formulations and biologicals. Collectively, these sectors account for approximately a quarter of the total exports. The exports of petroleum products declined by 24.7 per cent (YoY), amid softer crude oil prices, which decreased by 15.4 per cent (YoY).[20] Exports of telecom instruments witnessed a significant growth of 51.2 per cent (YoY) while those of drug formulations and biologicals experienced a growth of 11.2 per cent (YoY). These trends are a testament to the country's increasing strength in electronic manufacturing and reaffirm its position as a global pharmaceutical hub. Among the other principal commodities, steady gains were observed in exports of electric machinery and equipment, ready-made garments made from cotton, and motor vehicles/cars.

4.24 On the import side, merchandise imports increased by 6.3 per cent (YoY) in FY25, amounting to USD 721.2 billion. This increase was primarily driven by a rise in non-petroleum, non-gems and jewellery imports, which increased to USD 446.5 billion from USD 421 billion in FY24. This trend is attributed to a higher demand for critical intermediate inputs and capital goods, indicating a resilient domestic demand.

4.25 In FY25, India's import composition continues to be dominated by petroleum crude, gold and petroleum products, with these sectors accounting for over one-third of total imports. Imports of petroleum crude increased marginally by 2.7 per cent (YoY) amid softer crude oil prices, reflecting stable energy demand, while gold imports increased by 27.4 per cent (YoY). The increase in gold imports may be attributed to a rise in gold prices, increasing by 38.2 per cent (YoY) and driven by strong domestic consumption.[21] Additionally, imports of electronic components, telecom instruments and computer hardware also experienced significant growth.

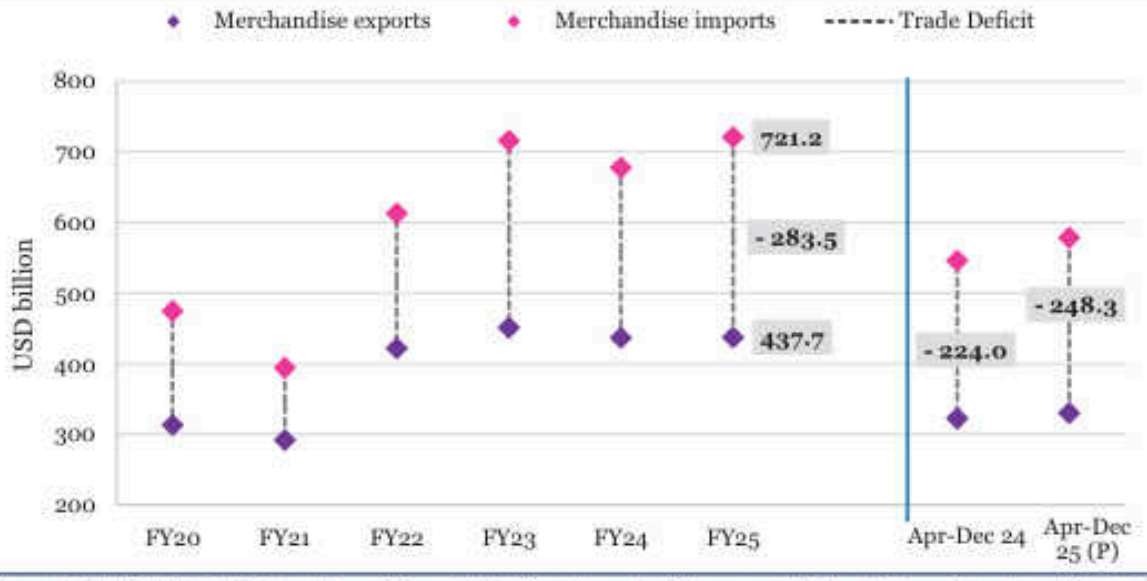
4.26 This trend suggests that as India's exports of finished products rise, there is a concomitant increase in the import of their intermediate products, implying a continuous

20 World Bank. (n.d.). Commodity price data (The Pink Sheet), accessed on 4 November 2025, https://tinyurl.com/ywzwfpc2.

21 Ibid note 20.

improvement in the country's manufacturing abilities. A rise in merchandise imports in conjunction with stable merchandise exports resulted in a widening of the merchandise trade deficit, which reached USD 283.5 billion, marking a 17.6 per cent (YoY) increase in FY25.

Chart IV.6: India's merchandise trade performance



Source: Data for FY20-FY25 is retrieved from FTPA, Department of Commerce, (https://tinyurl.com/mw864w62). Data for April-Dec 24 and April-Dec 2025 is retrieved from the PIB press release of the Ministry of Commerce and Industry, dated 15 January 2026 (https://tinyurl.com/3fp93r6d).
Note: P- Provisional.

4.27 The merchandise trade performance in FY26 so far reflects a continuation of the trends prevailing in FY25. During April-December 2025, merchandise exports have registered a growth of 2.4 per cent (YoY), while non-petroleum, non-gems and jewellery exports increased by 6.0 per cent (YoY), signalling sustained strength in India's core export basket. This increase in exports is driven by electronics goods, which grew by 35.1 per cent (YoY). The exports of mica, coal, and other ores and minerals, including processed minerals, grew by 14.7 per cent (YoY), whereas marine product exports grew by 15.5 per cent (YoY).

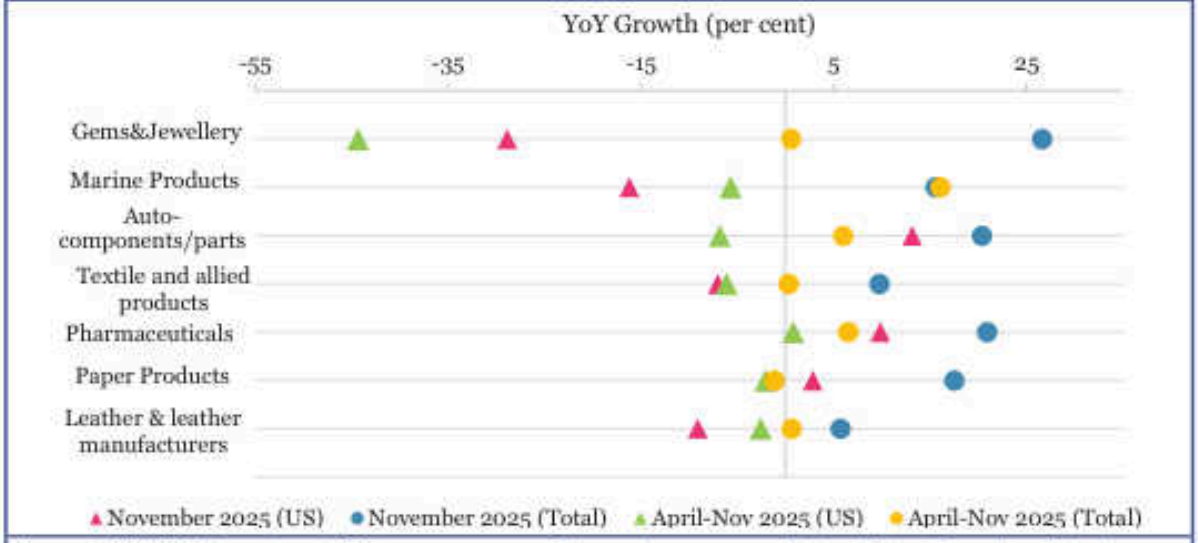
4.28 On the imports side, merchandise imports amounted to USD 578.6 billion, registering a 5.9 per cent (YoY) increase. Furthermore, non-petroleum and non-gems and jewellery imports witnessed a higher growth of 9.5 per cent (YoY) in the same period, reflecting resilient domestic demand for both consumption and production. This increase is attributed to an increase in imports of electronic goods (17.1 per cent YoY) and machinery, electrical and non-electrical (14.2 per cent YoY), which support the country's manufacturing and technology sectors. Furthermore, a substantial increase in imports of chemical material and products (44.4 per cent) suggests steady industrial demand in the economy. As a result, the merchandise trade deficit widened to USD

4.39 In particular, the telecom sector has achieved notable success, with exports rising by 1.5 per cent (AAGR) while imports have concurrently declined by 18.5 per cent (AAGR). This trend indicates an early accomplishment in import substitution, aligning with the PLI framework's overarching objective of enhancing market penetration abroad. Collectively, these developments illustrate the success of the PLI scheme in boosting the country's export growth.

Diversification of India's export destinations and import sources

4.40 India is currently subject to an effective export tariff rate of 50 per cent on goods exported to the US, and this rate is among the highest imposed on any country.[26] There has been progress in the negotiations of the trade deal between the two countries.[27] While there has been some specific impact of the tariffs, Chart IV.9 illustrates the resilience of India's overall exports in some labour-intensive and small-scale sectors, primarily attributed to diversification towards alternative destinations. This is evident in the chart, which shows a YoY decline in exports to the US during the period April-November 2025 (indicated by the green triangle). In contrast, India's exports to the world have shown positive growth (YoY) (indicated by the yellow dots). These trends of diversification are presented in Table IV.1.

Chart IV.9: Exports from labour-intensive sectors to the US and India's total exports



Source: FTSPCC, Department of Commerce, accessed on 15 January 2025, https://tinyurl.com/ywybytmj.

26 US President's Executive Order: Further modifying the reciprocal tariff rates dated July 31, 2025 (https://tinyurl.com/29aphjt9) and US President's Executive Order: Addressing threats to the United States by the government of the Russian Federation dated August 06, 2025 (https://tinyurl.com/ykzcr8tw).

27 Currently, six rounds of negotiations have already been held between India and the US. Both nations have expressed a commitment to continue their dialogue with the aim of reaching an early conclusion of a mutually beneficial trade deal. Further, the recent US decision to exempt over 200 agricultural and food products from elevated tariffs marks a constructive shift in India–US trade dynamics. The move restores price competitiveness for key Indian exports, including spices, tea, coffee, nuts, and processed foods, helping offset earlier tariff pressures that had constrained market access. Source: As per information from Department of Commerce and PIB Press release of the Ministry of Commerce and Industry dated 26 September, 2025: https://tinyurl.com/5dhy99ez.

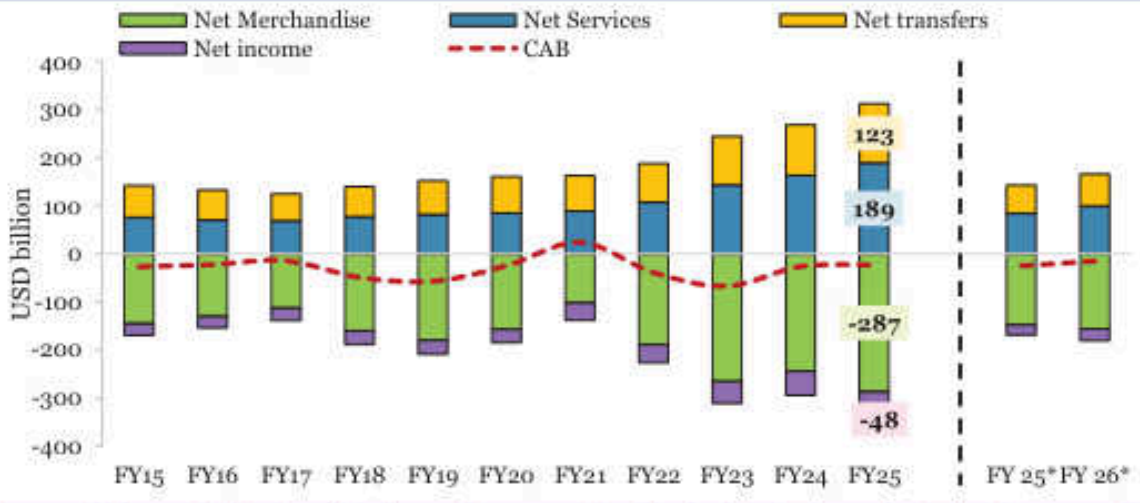
INDIA'S BALANCE OF PAYMENTS

4.60 The Balance of Payments (BoP) statement provides a consolidated record of India's transactions with the rest of the world, capturing developments in the current account and capital account and offering a comprehensive view of the economy's external sector.

Developments in the current account

4.61 India's current account structure reflects a merchandise trade deficit offset by strong net inflows of invisibles, led by rising surpluses in services and private transfers (Chart IV.12). The services surplus has expanded steadily, driven by India's growing role in global IT, business, and professional services. Remittance inflows have also increased consistently, providing a stable, counter-cyclical buffer. These components have collectively kept the current account deficit (CAD) within manageable levels. In H1 FY26, the CAD moderated to USD 15 billion (0.8 per cent of GDP) from USD 25.3 billion (1.3 per cent of GDP) in H1 FY25.

Chart IV.12: India's current account balance



Source: Table 127 India's Overall Balance of Payments, Handbook of Statistics on the Indian Economy, RBI, https://tinyurl.com/52wb2c7f and RBI's Press release dated 1 December 2025, https://tinyurl.com/4fjr67sc.
Note: * - (April-September of FY)

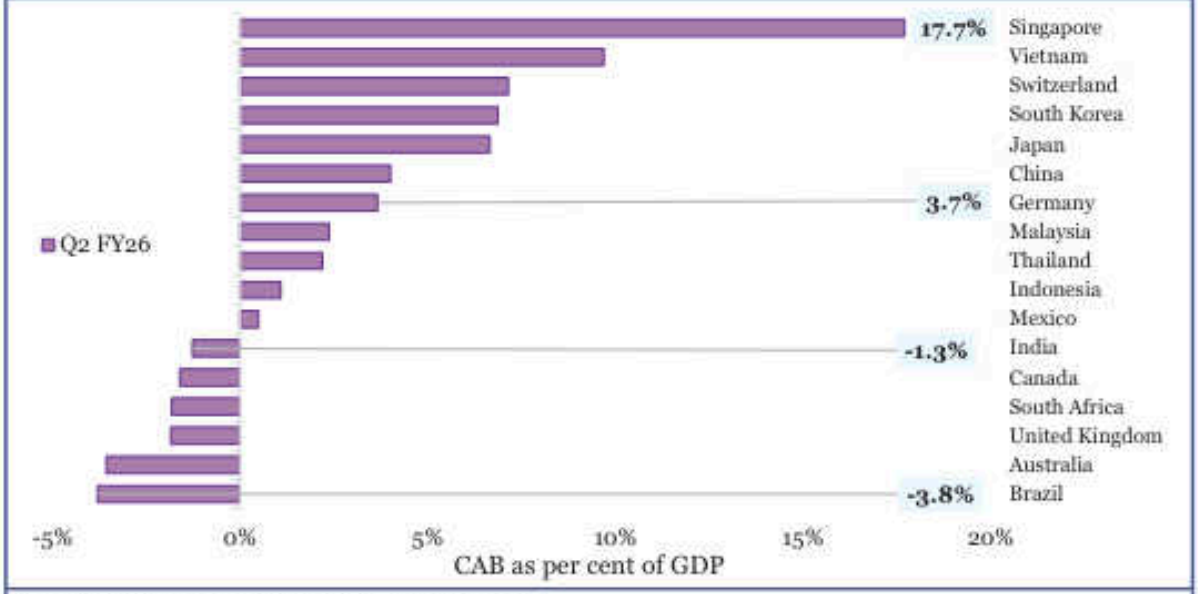
4.62 Countries with sizeable current account surpluses tend to be export-driven, while deficits are common in import-heavy economies. Major surplus economies such as Singapore and Germany run substantial trade surpluses in high-value manufacturing and related sectors.[68,69] Comparing India's CAD with that of other countries in Q2 FY26, India's modest deficit of about 1.3 per cent of GDP places it among economies with modest and sustainable deficits. It is better positioned than its high-deficit peers, such

68 Singapore Department of Statistics, https://tinyurl.com/3cbc4rn2.

69 Deutsche Bundesbank Euro system, https://tinyurl.com/ypp47v2a.

as New Zealand (-7.7 per cent), Brazil (-3.8 per cent), Australia (-3.6 per cent), the UK (-1.8 per cent), and Canada (-1.6 per cent).

Chart IV.13: Current account balance (as per cent of GDP) (India vs select countries)



Source: CEIC Data, https://tinyurl.com/52kavkdh.

Remittances

4.63 Private transfer receipts, mainly representing remittances by Indians employed overseas, remained a key source of external-sector strength in FY26. Remittances increased to USD 73 billion in H1 FY26, up from USD 64.7 billion in the same period the previous year.[70]

4.64 This near-term performance is consistent with India's longer-term position as the world's largest recipient of remittances. The latest available data indicate that remittance inflows increased steadily from USD 55.6 billion in FY11 to USD 135.4 billion (provisional) in FY25, accounting for approximately 3.5 per cent of GDP in FY25 (provisional). Correspondingly, net transfers[71] have financed a substantial portion of the merchandise trade deficit during this period. Furthermore, in most years, remittances (private transfer receipts) have surpassed gross FDI inflows, highlighting their importance as a dependable source of external funding (Chart IV.14).

70 Remittances are proxied by private transfer inflows, which denote total private transfer credits in the BoP statistics.

71 Defined as the difference between total (private and official) transfer receipts and transfer payments

data centres in developed countries,[74] underscoring the increasing concentration of global capital in technology-intensive sectors.[75]

4.70 International project finance, which is central to infrastructure investment in developing economies, also experienced a contraction, with its total value declining by 8 per cent in H1 2025, continuing the downward trajectory observed in CY 2023 and CY 2024. This weakness was more pronounced in renewable energy and power, reflecting the impact of higher global interest rates that have raised the cost of capital for long-gestation projects.

4.71 FDI inflows to developed countries (excluding European conduit economies) declined by 7 per cent in the H1 of 2025. The number of greenfield project announcements in these nations fell by 20 per cent over the same period, primarily driven by decreases in manufacturing sectors reliant on supply chains. The US remained a notable exception, supported by substantial commitments in the AI and semiconductor sectors. Taken together, these trends indicate a shift in global investment flows away from capital-intensive, long-gestation infrastructure projects towards shorter-cycle, technology-driven investments.

Trends in investment flows into India

4.72 Cross-border financial flows, which include flows in securities (e.g., stocks and bonds) and in FDI, are an important mechanism of India's integration with global capital markets. India has consistently attracted sizeable gross investment inflows, amounting to 18.5 per cent of GDP in FY25 and 16.9 per cent in H1 FY26, even amid tightening global financial conditions.[76]

4.73 According to UNCTAD data, while India recorded a marginal decline of 2 per cent in FDI inflows in CY 2024, it remained the largest recipient of gross inflows in South Asia and surpassed major Asian peers such as Indonesia and Vietnam. The decline was relatively modest when viewed against the sharp contractions experienced by several large emerging markets.

4.74 This resilience is also evident in greenfield investment activity, which provides a forward-looking signal of global investor sentiment. India ranked fourth globally in CY 2024, with 1,080 announced greenfield projects, and has emerged as the largest destination for greenfield digital investments between CY 2020 and CY 2024,

74 The number of greenfield announcements fell by 12 per cent, while the total project value dropped by 37 per cent in developing countries.

75 Ibid note 72.

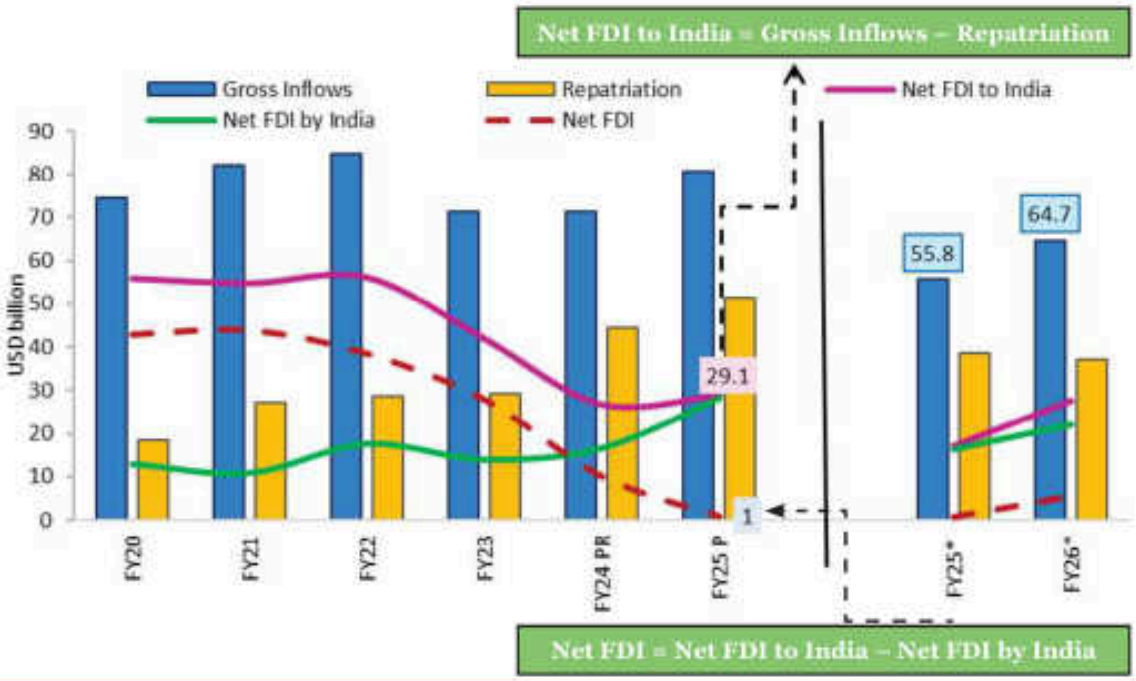
76 Gross investment inflows refer to credit-side foreign investment flows, comprising gross FDI and FPI inflows, as reported in RBI's BoP statistics.

attracting USD 114 billion. This is well ahead of Malaysia (USD 74 billion), Singapore (USD 39 billion), Vietnam (USD 32 billion), Mexico (USD 29 billion), and China (USD 24 billion).[77] Notably, India attracted investments across various categories, including digital services, data centres, and IT infrastructure.

4.75 According to RBI data, in FY25, gross FDI inflows into India stood at USD 81.0 billion, representing a 13 per cent increase from USD 71.3 billion in FY24[78]. Of these gross FDI inflows, equity inflows accounted for USD 51 billion, and approximately 60 per cent of these flows were directed towards services, computer software and hardware, trading, non-conventional energy, construction, and the automobile sector.[79]

4.76 In April-November 2025, gross FDI inflows strengthened further to USD 64.7 billion, compared with USD 55.8 billion in April-November 2024. The magnitude of inflows during the first eight months of the year highlights sustained investor confidence despite a subdued global environment and reflects the underlying strength of India's digital economy, as well as the continued policy emphasis on manufacturing and infrastructure.

Chart IV.16: Trends in India's FDI



Source: Table Foreign Investment Inflows, DBIE, RBI. (https://tinyurl.com/4rk4n8r8).
Note: *: (April-November); P = Provisional; PR = Provisionally Revised.

77 UNCTAD's World Investment Report, 2025: https://tinyurl.com/28zfatdw

78 RBI Bulletin (December 2025): https://tinyurl.com/4b7y7zc7

79 Factsheet on FDI Inflow, DPIIT: https://tinyurl.com/38vfrd49

4.77 FDI outflows from India increased to USD 22.1 billion during April-November 2025. Despite this, improvements in gross inflows along with a moderation in profit repatriation by foreign investors led to a strengthening of net FDI. Net FDI increased nearly sevenfold to USD 5.6 billion during April-November 2025, up from USD 0.8 billion in the same period a year earlier.

4.78 It is important to examine the return on inward FDI, while there is a recent slowdown in net FDI, which would indicate the profitability of the investments. Box IV.2 presents evidence on India's inward FDI returns, providing a useful complement to the analysis of gross and net inflows.

Box IV.2: Returns on inward Foreign Direct Investment

Returns on inward FDI are estimated as FDI equity income receipts expressed as a percentage of the total inward FDI stock.[80] This indicator provides information on the profitability of direct investment enterprises. For example, when the rate of return on inward FDI increases, it implies that resident direct investment enterprises are more profitable and more competitive for investors.

To account for the lag between investment and income generation, the inward FDI stock is measured with a two-year lag, recognising that returns typically materialise after an initial gestation period. This is particularly relevant given the multi-year gestation periods typical of investments in manufacturing, infrastructure and digital services. Accordingly, annual returns on inward FDI are computed as:

$$\text{FDI Return}_t = \frac{\text{Direct Investment Income}_t}{\text{Inward FDI Stock}_{t-2}}$$

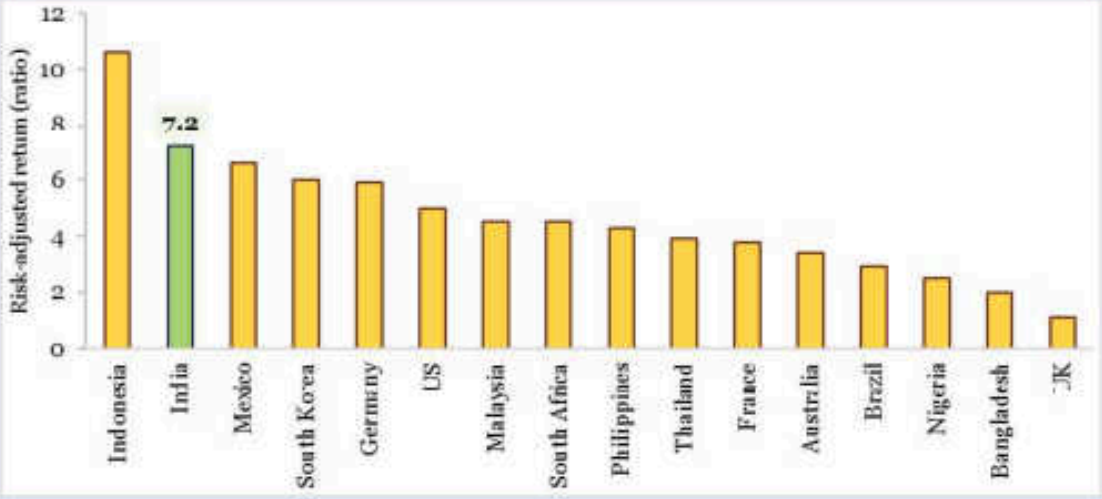
Using this approach, CareEdge analysis[81] for 2014-2023 shows that India's average return on inward FDI is approximately 7.3 per cent, higher than that of some major emerging economies, such as Thailand (7 per cent), Brazil (6.1 per cent), Mexico (4.3 per cent), and South Africa (4.3 per cent). It is also higher than that of some AEs, such as France (3.6 per cent), Germany (3.0 per cent), the US (2.9 per cent), and the UK (2.2 per cent).

This analysis is further extended to compute the risk-adjusted return, defined as the ratio of the 10-year average return to its standard deviation. India ranks second only to Indonesia among the major economies examined, with a ratio above 7, compared with around 5 or lower for several large AEs. This combination of higher returns and comparatively low volatility underscores India's positioning as a destination offering reasonable yield without exposing investors to extreme earnings swings. The average FDI returns in India remain attractive.

80 OECD Methodology: https://tinyurl.com/4652bhy5 and Eurostat's Methodology: https://tinyurl.com/y4mxr6ey

81 CaredEdge Ratings Analysis (December, 2025): https://tinyurl.com/bddpjnta

Chart IV.17: Risk-adjusted return on inward FDI
(Average Return [2014-23] ÷ Standard Deviation)



Source: CareEdge Estimates.

4.79 Net FDI, which has traditionally been driven by fluctuations in gross inflows, has also reflected the impact of repatriation and Overseas Direct Investment (ODI) by Indian firms in recent times. While these outflows mechanically reduce net FDI, they align with the growing internationalisation of Indian enterprises as they expand their business presence abroad, enabling them to compete in the global market and strengthening the Indian economy in the long term. Furthermore, the rising trend of repatriation suggests that India is not only attracting foreign capital but also delivering strong returns, which enhances its reputation as a reliable investment destination.

4.80 Going forward, the challenge is to sustain FDI inflows in an environment of heightened global volatility, which underscores the need for a multi-pronged strategy that strengthens the investment climate by addressing both structural and cyclical factors that determine capital flows. Box IV.3 discusses this challenge and suggests a way forward.

Box IV.3: India's FDI Challenge: Turning success into sustainable growth

EMDEs, including Indonesia, Mexico, Poland, Morocco, and Vietnam, have effectively leveraged the trade and investment reorientation strategies employed by China and the US. By positioning themselves as connector countries,[82] these nations have reaped the benefits of FDI, driven by tariff avoidance and the intent to serve as export platforms. Connector countries serve as a base for multinational enterprises that establish production in the host

82 "Connector countries" are geopolitically aligned countries or those that satisfy the criteria for political stability, regulatory quality, and other structural factors conducive to investment. They serve as conduits in trade and investment flows between geopolitical blocs.

country primarily to export goods or services to other markets, rather than mainly catering to the domestic market.[83]

Countries have also been attracting FDI through a mix of policies and incentives, with the latter being used to stimulate investments in strategic sectors. For example, Vietnam has been able to attract greater FDI by leveraging corporate tax rate reductions for investors, well-developed industrial zones, and a deep network of FTAs. Its geographic proximity to East Asia's top emerging economies and business-friendly policies make it a preferable host for investments.[84] Vietnam has introduced a policy to accelerate investment project approvals, "Decree 19", which establishes a special investment procedure that drastically streamlines the investment licensing process, aiming to attract high-quality capital, particularly in high-tech industries.[85] To attract investment into strategic sectors like artificial intelligence and semiconductors, Vietnam, Brazil, and Chile have been incentivising investment in research and development centres.[86]

Malaysia undertook the facilitation route to attract investments by launching a 'Golden Pass' scheme to attract unicorn start-ups and venture capitalists, streamlining visa, employment, and licensing procedures.[87] Similarly, the Philippines has enacted the 'Create More Act', which empowers the Investment Promotion Agency to issue special visas to foreign nationals possessing highly specialised skills or occupying executive roles. These efforts collectively contribute to the favourable investment climate in these regions.[88]

Another model through which many countries have successfully attracted foreign investment involves the establishment of structures headed by higher levels of government. Some examples are detailed in the table below.[89]:

Country	Institutional structure to attract FDI	Tools/Measures
Vietnam	Foreign Investment Agency and Special Working Group headed by the Minister for investment promotion and attracting multinational manufacturing projects Prime Minister (PM) level meetings and monitoring of investments	Tax holidays; customs exemptions; industrial park preparedness; land access; proactively courting investments; roadshows to headquarters

83 Bloomberg. (2023, November 1). These five countries are key economic 'connectors' in a fragmenting world, https://tinyurl.com/42d5myev.

84 Dezan Shira and Associates, Doing Business in Vietnam, accessed on 09 December 2025, https://tinyurl.com/46emvss2.

85 Government of Vietnam, Decree No. 19/2025/ND-CP, dated 10 February 2025, https://tinyurl.com/54rzw6k3.

86 Ibid note 77.

87 Government of Malaysia, keynote address by the Hon'ble Rafizi Ramli, Minister of Economy at KL20 Summit 2024, https://tinyurl.com/yr6uw3bb.

88 2025 Investment Climate Statements: Philippines, https://tinyurl.com/48akrxaj.

89 Ministry of Planning and Investment of the Socialist Republic of Vietnam: https://tinyurl.com/yaexy2hm, Malaysia Investment Development Authority: https://tinyurl.com/4btnp925, Thailand Board of Investment – Investment Promotion Guide 2025: https://tinyurl.com/epbt9n6m, InvesTaiwan: https://tinyurl.com/y3t2rjtt, Australian Government – Department of Foreign Affairs and Trade: https://tinyurl.com/32ukcazu.

Malaysia	Malaysian Investment Development Authority, chaired by Minister/senior industrialist National Committee on Investment PM chairs National Investment Council	Investment missions; tailor-made tax incentives; more extended tax holidays up to 10 years; short-term special incentives; entry facilitation for businesses
Thailand	Board of Investment; Eastern Economic Corridor initiative (chaired by the PM) Special task force in high-tech industries Quasi-regulatory authority for the grant of incentives	Tax cuts- personal, corporate; import exemptions; largest project approvals by the Board chaired by the PM
Taiwan	InvesTaiwan initiative; focus on high-value manufacturing; Single-window service centre for investors	Low-interest loans, tax breaks and subsidies, simplified tax procedures
Australia	Global business and talent attraction taskforce Cross-agency task force PM level monitoring	Visa concessions, R&D tax incentives, and fast-tracked project approvals; permanent residency to highly skilled individuals in target sectors

A common thread running across these senior empowered structures is their authority to negotiate investment deals using incentives. India's task to attract FDI is even more complex, as it is not just competing with the world's current largest exporter for shifting GVCs to India, but also with emerging FDI destinations (as per the Table above), which have already been wooing investors with interventions at the PM level.

India has what it takes to attract FDI, given that institutions like Invest India are already in place. However, the integration of efforts, culminating in a structure led by the highest levels of the government with robust Centre-State coordination, is what would make large investors believe in our intention to host them. Furthermore, a strategic approach to identifying priority sectors, along with a mix of incentives and reforms, will help prevent the dilution of policy efforts and improve overall effectiveness.

Challenges and way forward

Global surveys show that political stability and strong macroeconomic fundamentals are key drivers of FDI. India excels in this area but could better leverage its strengths. Despite a clear government intent and proven economic management, FDI inflows remain below their potential, especially for infrastructure needs. Proactive reforms are essential to attract more foreign investment.

This approach involves developing a targeted strategy that identifies a specific set of GVC anchors and establishes a state apparatus that collaborates directly with them as partners. The direct engagement will help resolve cross-agency issues and provide customised and time-bound solutions. Additionally, it is crucial for India not only to offer compelling incentives but also to ensure these incentives are reliably implemented. This strategy will

help mitigate execution risks for involved firms, fostering a more stable and attractive investment environment. Establishing a single, empowered centre of accountability will position India as a credible alternative production hub capable of handling large volumes, integrating with global suppliers, meeting regulatory and compliance standards, and, importantly, providing predictability over a multi-year timeframe.

Creating a task force to engage top global companies and promote India's advantages - stability, macroeconomic strength, sustained growth and market size - could boost FDI, especially in targeted sectors. Proactive diplomacy, highlighting these strengths, can help offset tariff challenges.

Efforts to improve the investment environment by simplifying processes and procedures to attract FDI will need to be kept up.[90] As foreign investors prioritise predictability and sustainability in policies, every policy change in the country must pass the necessity test to meet both these parameters.

Further enhancements are required on the regulatory environment front, as well as improvements in logistics and workforce development. The World Bank's Logistics Performance Index, for example, placed India at 38th position in 2023.[91] While on the one hand, these gaps indicate how far India is from achieving the world's best in terms of logistics, they also open doors to vast improvement opportunities that could yield substantial returns in terms of FDI.

The momentum created by improvements in the investment climate, streamlined regulations, and reduced bureaucratic procedures, thereby enhancing the ease of doing business, must be sustained and accelerated. India's credit rating upgrades can be leveraged for additional momentum, as foreign investment and sovereign ratings demonstrate a strong historical correlation globally. India's continued economic resilience, steadfast commitment to fiscal consolidation, and reform trajectory position it well for unlocking investor interest.

On bilateral initiatives, the signing of the Bilateral Investment Treaty between India and Israel, along with the binding commitment of USD 100 billion in investment under the India-European Free Trade Association (EFTA) Trade and Economic Partnership Agreement (TEPA), bodes well for India's FDI inflows, provided these agreements are effectively implemented. This along with the signing of the India-UK Comprehensive Economic and Trade Agreement, have the potential to increase FDI inflows.[92] While it is essential to recognise the role of trade and investment agreements as enablers, there is a pressing need for action across departments and agencies to streamline India's trade policy and secure a greater role in GVCs.

Companies harvest profits from established operations in India while hesitating to commit fresh capital amid global uncertainty. The solution lies in mobilising new investment through policy stability, aggressive investor engagement, and scaling proven state-level models nationwide.

90 WTO. (2023, December 21). Statement By India on Agenda Item 18, General Council Meeting 13-15 December 2023. WT/GC/262, https://tinyurl.com/2fbwbjh3.

91 World Bank's Logistics Performance Index, https://tinyurl.com/bdfr8a5z.

92 PIB press release India-EFTA Trade Pact: Boosting USD 100 billion investment and 1 million jobs, dated 11 October 2025, https://tinyurl.com/23m4ezv7.

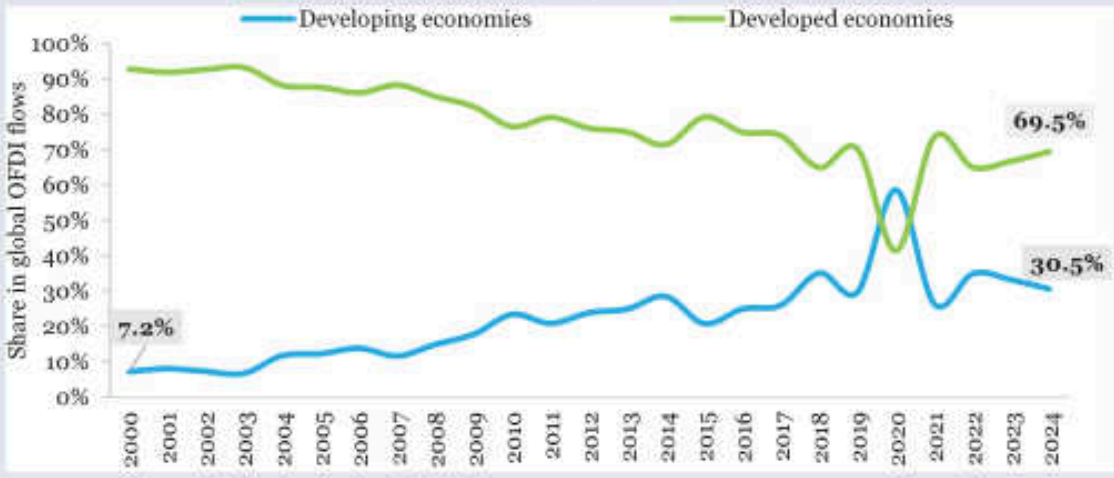
India's fundamentals, such as strong growth, expanding markets, and a demographic dividend, remain compelling magnets for investment, yet FDI strategies must become bolder and offer the best to global investors. The window for action is still open, but it will not remain so indefinitely. There is a need to move decisively to transform the FDI challenge into the next chapter of our economic growth story.

4.81 As India deepens its integration with GVCs, outward investment has emerged as an important complement to inward FDI in shaping India's external sector. The steady rise in ODI reflects the growing competitiveness of Indian firms, their efforts to secure technology, markets, and resources abroad and the enabling effect of the liberalised ODI framework introduced in 2022. These trends underscore a structural shift to an economy whose enterprises are increasingly engaging in global expansion.[93] Box IV.4 examines the evolving contours of India's ODI and its implications for long-term growth.

BOX IV.4: Impact of Outward FDI on the home country

International investment flows reflect growing economic integration, with inward FDI signalling investor confidence and outward FDI reflecting the internationalisation and competitiveness of domestic firms. In recent decades, developing economies have become increasingly prominent sources of outward FDI (OFDI), with their share in global FDI outflows rising from nearly 7 per cent in 2000 to close to 30 per cent by 2024, signifying the internationalisation of enterprises from traditionally capital-scarce regions.

Chart IV.18: Share of developed & developing economies in global OFDI flows, 2000-2024 (%)



Source: UNCTAD Stat database, https://tinyurl.com/2tatc2u4.

For India, where capital formation needs remain substantial, a key policy question is whether OFDI diverts investible resources from the domestic economy or strengthens long-term

93 PIB Press Release of the Ministry of Finance, dated 22 August 2022, https://tinyurl.com/mvdjajkr.

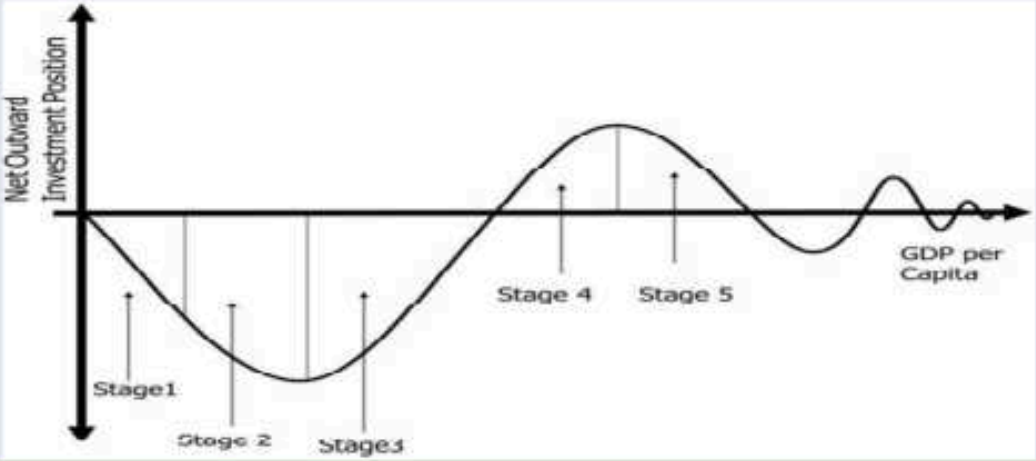
competitiveness through knowledge internalisation, capability enhancement, and greater participation in GVCs. Assessing home-country effects of OFDI, therefore, requires moving beyond concerns of capital flight towards evaluating longer-term productivity and innovation impacts.

Channels of impact

Knoerich (2017)[94] outlines three principal channels through which OFDI can generate home-country gains: (i) financial returns through repatriated profits, dividends payments and intra-firm transfers that strengthen foreign exchange earnings and potentially relax financing constraints for parent firms; (ii) intangible capability gains as firm's overseas operations expose firms to new technologies, managerial know-how, regulatory & quality standards, and global customer preferences. (iii) tangible capacity gains from acquiring resources, brands, and distribution networks, which deepen domestic production networks, foster product upgrading, and facilitate entry into higher value-added segments of GVCs.

The overall effect of OFDI on the home economy depends on domestic absorptive capacity (skills, R&D, institutions, and financial depth) as well as policies that support the domestic assimilation of overseas knowledge and investment.

Chart IV. 19: Investment Development Path



Net Outward Investment becomes persistently positive only at later stages as domestic firms accumulate capabilities.
Note: IDP stages adapted from Dunning & Narula (1996)

The Investment Development Path (IDP) framework (Dunning, 1981[95]; Dunning and Narula, 1996[96]) posits that countries evolve from being predominantly inward FDI recipients in early development stages to net outward investors as their productive capabilities, technological depth, and institutional maturity improve. India is generally assessed to be in Stage III of

94 Knoerich, J. (2017). How does outward foreign direct investment contribute to economic development in less advanced home countries. Oxford Development Studies, 45(4), 443-459.

95 Dunning, J.H. (1981). "Explaining the International Direct Investment Position of Countries: Towards a Dynamic or Developmental Approach." Weltwirtschaftliches Archive, 117(1), 30-64.

96 Dunning, J. H., & Narula, R. (1996). The investment development path revisited. Foreign direct investment and governments: Catalysts for economic restructuring, 1-41.

the IDP, with outward investments increasingly concentrated in technology-intensive sectors, indicating a gradual shift from primarily market-seeking investments to asset and capability-enhancing overseas ventures [Mohanty et al. (2024)].[97]

This pattern aligns with cross-country evidence from BRICS and Next-11 economies, where GDP growth, trade openness, and financial development are major drivers of OFDI, while human capital and inward FDI further reinforce the process in more advanced emerging markets such as BRICS. In turn, higher OFDI is associated with improved economic dynamism, suggesting a two-way relationship between external investment and domestic performance [Mohanty, Sethi and Dash, 2024].[98]

Several studies find that OFDI may substitute for domestic investment in the short run but complements it in the long run, as firms reinvest earnings, upgrade domestic facilities, and strengthen supply chains (Herzer, 2010[99]). Overall, international evidence suggests that outcomes are heterogeneous, shaped by sectoral structure, development stage, and the nature of outward investment (market-seeking vs. efficiency or asset-seeking).

India's experience: Empirical evidence from India indicates that OFDI has complemented, rather than displaced, domestic investment. Sahoo and Bishnoi (2021)[100] use macro-panel and time-series approaches to find that outward investment by Indian firms is positively associated with domestic investment. Outward-oriented firms have tended to expand exports and enhance R&D intensity (Das, 2015[101]), benefiting from learning and access to global markets that mitigate concerns regarding capital outflows and support productivity gains at home.

These patterns are consistent with OFDI being used as a mechanism for learning and upgrading, rather than as a pure relocation of existing activities. In India, post-liberalisation policy reforms since the 1990s, such as the relaxation of OFDI norms, simplification of approval processes, and a more open foreign exchange regime, have enabled Indian firms to undertake strategic asset-seeking investments abroad.

Trends in India's ODI[102]

India's recent outward investment trajectory reflects its progression along the Investment Development Path, with firms increasingly using overseas ventures to acquire capabilities, strategic assets and access to global production systems. Outward investment increased from USD 14.4 billion in FY24 to USD 23.6 billion in FY25, driven by greenfield projects,

97 Mohanty, S., Sethi, N., & Dash, D. P. (2024). What determines outward FDI in developing blocs? A new empirical comparative macroeconomic perspective of post-1990s. Heliyon, 10(23).

98 Ibid note 97.

99 Herzer, D. (2010). "Outward FDI and Economic Growth." Journal of Economic Studies, 37(5), 476-494.

100 Sahoo, P., & Bishnoi, A. (2021). Impact of outward foreign direct investment: Evidence from Asia. Journal of Policy Modelling, 43(5), 1131-1148.

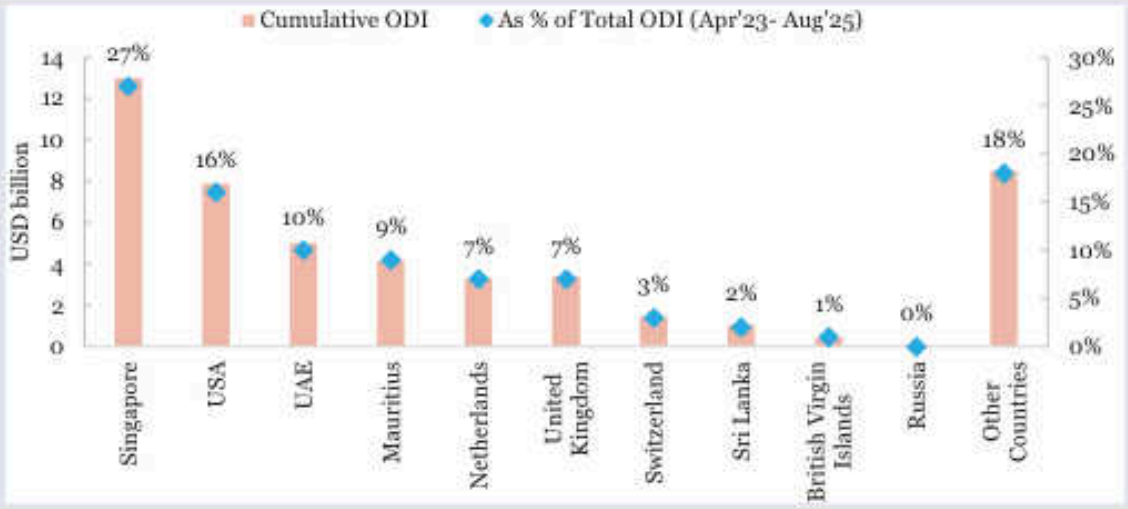
101 Das, K. C. (2015). Outward FDI by Indian manufacturing MNEs: impacts and implications (No. 148). ARTNeT working paper series.

102 OFDI refers to outward FDI as defined by international statistical standards such as UNCTAD. ODI denotes overseas direct investment by Indian residents as reported by the DEA, Ministry of Finance, and captures actual outward flows in the form of equity, loans, and guarantees.

joint ventures, and acquisitions. It increased further to USD 10.2 billion during April-August FY26, compared with USD 7.9 billion in the corresponding period last year.[103] The persistence of outflows despite global financial uncertainties underscores ongoing internationalisation by Indian enterprises.

Between April 2023- August 2025, nearly 82 per cent of cumulative ODI of USD 48.3 billion was directed to the top ten destination countries of Singapore (27 per cent), the United States (16 per cent), UAE (10 per cent), Mauritius (9 per cent) and the Netherlands and the United Kingdom (7 per cent each), reflecting both strategic-asset seeking in advanced markets and the use of international financial hubs.

Chart IV.20: Geographical composition of India's ODI (April 2023-August 2025)



Source: Overseas Direct Investment Data Fact Sheet, DEA (https://tinyurl.com/57wur3kk)

Conclusion

In sum, the home-country effects of OFDI are multi-dimensional and context-dependent. A common pattern across international and Indian evidence is that outward investment tends to reinforce domestic economic strength when underlying capabilities and institutions are robust. In contrast, weak absorptive capacity may result in resource diversion without adequate spillovers.

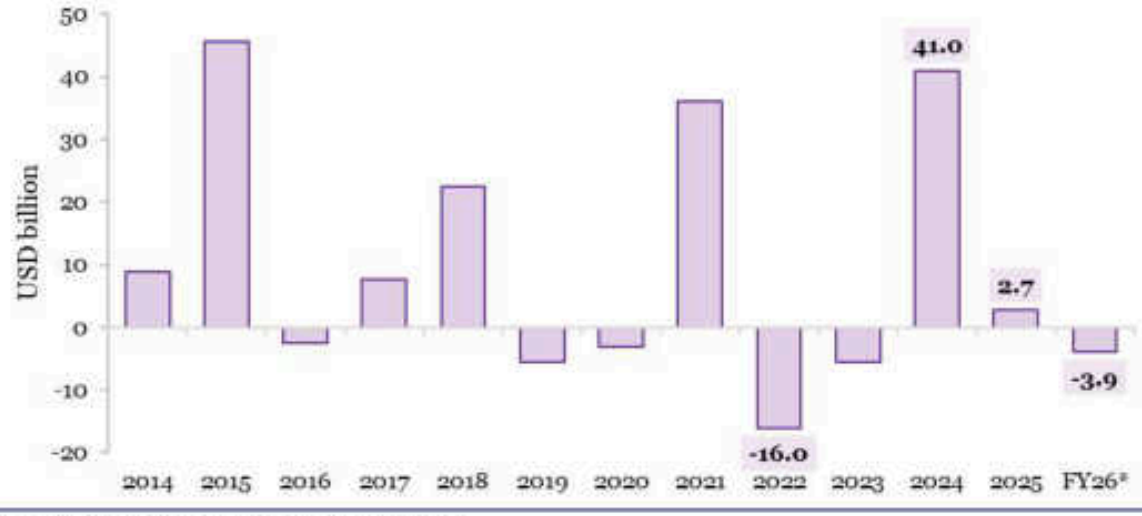
As Indian firms expand into higher-value segments, OFDI can serve as a strategic instrument for innovation, risk diversification, and global market positioning. The policy focus may therefore lie in enhancing repatriation and diffusion of knowledge, easing access to finance for globalising firms, and strengthening institutional support for international R&D collaboration.

103 Overseas Direct Investment Data Fact Sheet August 2025, DEA: https://tinyurl.com/57wur3kk

Foreign portfolio investments

4.82 India's FPI pattern shows recurring cycles of inflows and outflows, with significant shifts often linked to global financial changes. While India attracts substantial foreign investment, episodes of sharp outflows, such as after the global financial crisis, during the 2013 'taper tantrum', the 2018 tightening cycle, and pandemic volatility, reflect global financial-cycle sensitivities. These fluctuations underscore how FPI flows, a key external capital source, also serve as channels through which global shocks can impact domestic markets.

Chart IV.21: Net FPI inflows into India



Source: NSDL (https://tinyurl.com/tetk7n7e).
Note: FY26*: FY (April-December 2025)

4.83 A breakdown of FPI components shows how foreign participation has evolved. Equity flows tend to dominate in positive years, especially during periods of strong earnings and favourable valuations, whereas debt flows are more sensitive to interest rates and global risk. The Voluntary Retention Route (VRR)[104] and Fully Accessible Route (FAR)[105] routes have stabilised debt flows, particularly when traditional flows were weak.

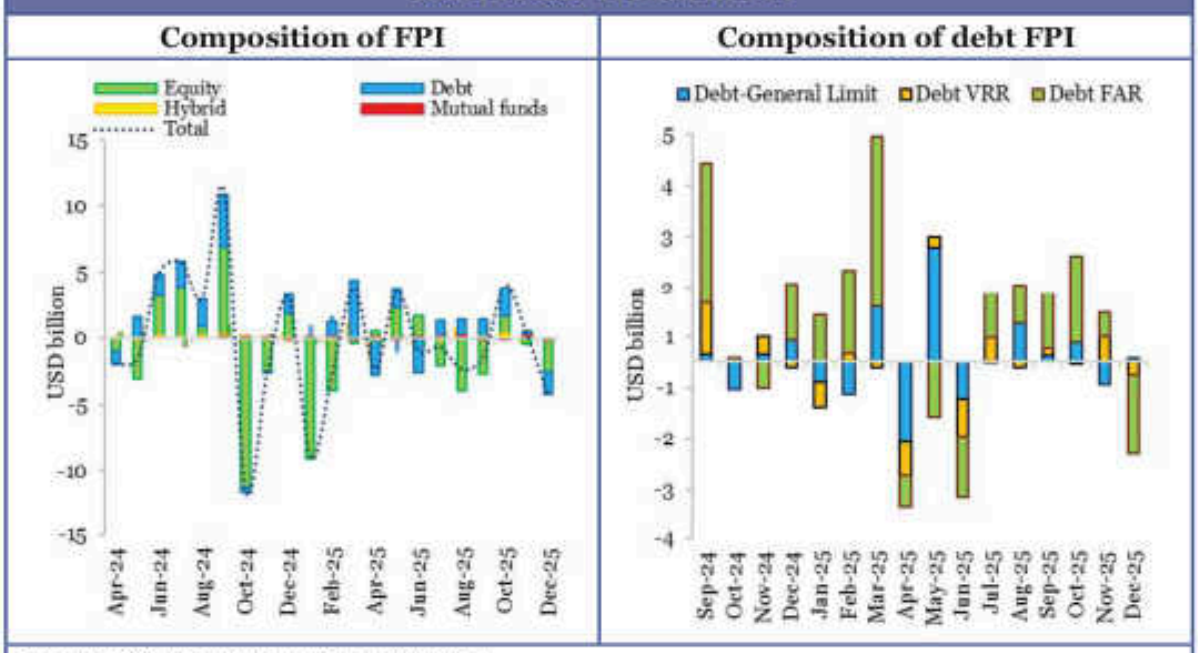
4.84 FPIs during FY26 (April-December) have fluctuated, mainly influenced by global financial conditions rather than domestic macroeconomic factors. The data indicate volatility, with six months of net outflows and three months of sizable net inflows, resulting in a modest net balance for the year-to-date. The outflow periods in FY26 coincided with periods of tightening global financial conditions, such as rising

104 Under VRR, FPIs invest in Indian debt instruments with the condition that they will retain at least 75% of their investment for a minimum period (currently three years).

105 FAR permits FPIs, NRIs, OCIs, and other eligible non-residents to invest without quantitative caps, repatriation limits, or sectoral restrictions in designated G-secs (known as FAR bonds).

US Treasury yields, which often trigger 'flight-to-safety' flows into advanced-economy assets. During these periods, foreign investors reduced their holdings in both Indian equities and debt; even categories that typically showed greater stability, such as VRR, experienced occasional selling.

Chart IV.22: Trends in FPI



Source: NSDL (https://tinyurl.com/tetk7n7e).

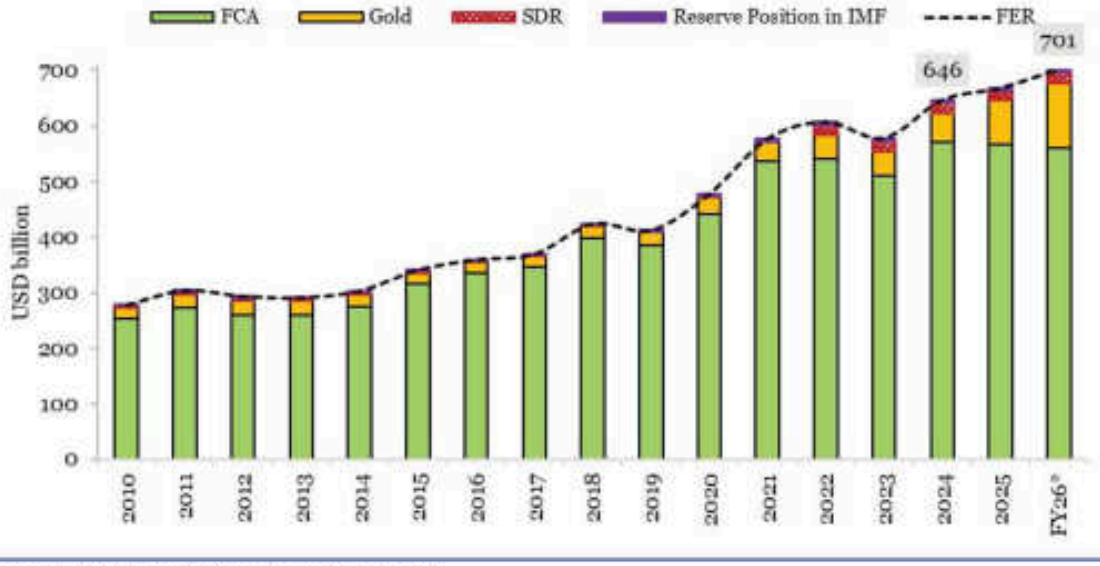
4.85 The three months of inflows in FY26 occurred during periods when global markets reassessed the interest-rate outlook and became more receptive to emerging market risks (May, October and November). These rebounds were mainly driven by the equity segment, supported by renewed interest in India's corporate earnings trajectory and flows into the debt market as global long-term yields declined. The swift return of inflows during these periods highlights that foreign investors' medium-term view of India remains positive, even though their short-term allocations are influenced by still-high valuations of Indian stocks and uncertainty about the trajectory of the India-US relationship, amid the high tariffs that Indian goods face in the United States.

Foreign exchange reserves

4.86 India's foreign exchange reserves increased to USD 701.4 billion as of 16 January 2026, up from USD 668.3.4 billion as of the end of March 2025. The increase has come at a time when global interest-rate differentials, capital flows and commodity prices remain volatile. This expansion in reserves has helped steady India's external position despite shifts in global risk sentiment and episodes of portfolio outflows. In terms of adequacy, the reserves as of 16 January 2026 are sufficient to cover around 11 months

of goods imports and about 94 per cent of the external debt outstanding at the end of September 2025, providing a comfortable liquidity buffer.

Chart IV.23: Composition of India's foreign exchange reserves



Source: RBI (https://tinyurl.com/235dzyy9).
Note: FY26*- As of 16 January 2026.

4.87 Foreign Currency Assets (FCA), which form the liquid core of reserves, softened slightly from USD 567.6 billion in end March 2025 to USD 560.5 billion as of 16 January 2026. In contrast, the gold component rose sharply to USD 117.5 billion as of 16 January 2026, compared with USD 78.2 billion at the end of March 2025. This increase reflects both valuation gains during a period of elevated global gold prices and a continued preference among central banks for diversifying into non-dollar reserve assets.[106]

4.88 The growing share of gold in reserves aligns with a broader international pattern where many emerging markets have increased gold holdings amid geopolitical uncertainty and shifts in the global interest-rate cycle. SDRs remained stable at USD 18.7 billion, while India's reserve position in the IMF increased to USD 4.7 billion.

Exchange rate

4.89 Between 1 April 2025 and 15 January 2026, the Indian rupee (INR) depreciated by approximately 5.4 per cent against the US dollar. This movement was somewhat larger than in the immediately preceding period, making INR one of the most depreciated currencies alongside the Japanese Yen (-5.5). Other Asian peers experienced relatively lower depreciation, such as the Philippine peso (-3.8 per cent) and the Indonesian rupiah (-1.9 per cent). In contrast, currencies of economies such as Australia, Brazil and

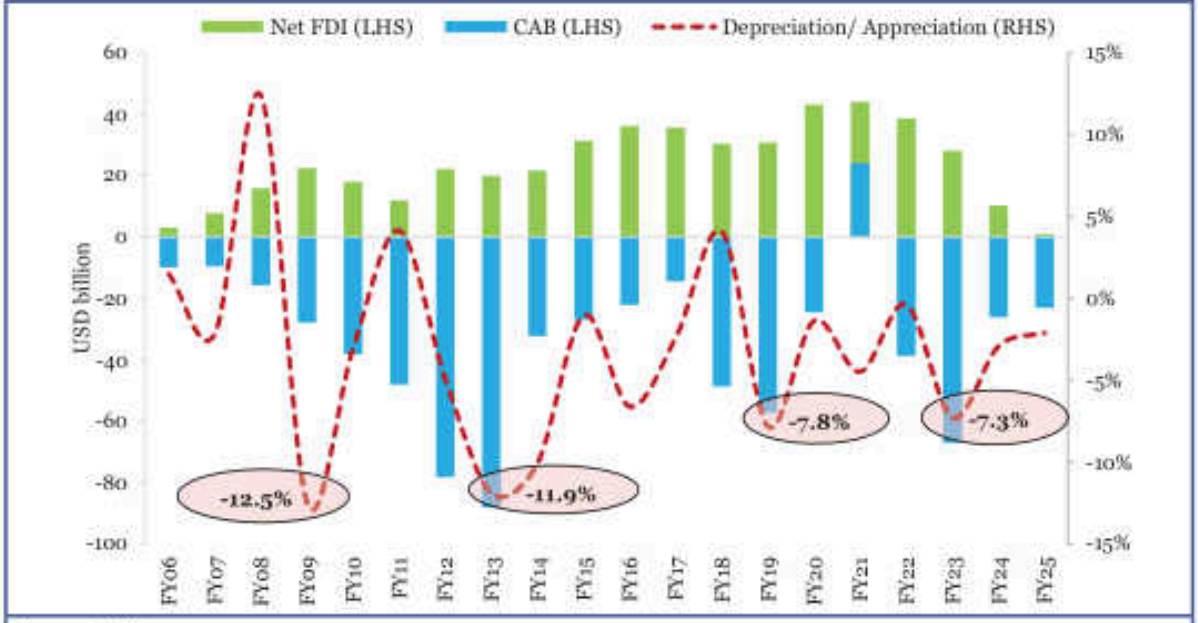
106 J.P. Morgan Global Research, https://tinyurl.com/36kb33s6

South Africa appreciated during this period, supported by favourable terms-of-trade movements.

4.90 In the short run (1 April 2025 – 15 January 2026), exchange rate dynamics in India were shaped by persistent outflows of foreign investment and evolving bilateral trade developments, including tariff-related measures, which together contributed to cautious investor sentiment. These pressures were accompanied by episodes of elevated foreign-currency demand in both spot and forward markets, reflecting hedging behaviour and import-related adjustments.[107] Such factors amplified short-term currency movements even when underlying macroeconomic fundamentals remained stable.

4.91 Traditional BoP inflows, such as exports and foreign capital, that have historically arrested currency depreciation in EMEs, are increasingly influenced by global shifts in geopolitical sentiment.[108] At the same time, import growth, which exerts downward pressure on currencies, tends to rise as these economies advance along their development path. With the certainty of import growth and narrower prospects for the kind of export and capital inflow growth seen during the hyper-globalisation phase, emerging-market currencies can be expected to face persistent depreciation pressures.

Chart IV.24: Exchange rate movements in relation to the current account balance & net FDI



Source: RBI
Note: A negative CAB indicates CAD.

4.92 Against this backdrop, India's historical experience also indicates that exchange-rate outcomes are shaped by the interaction between the current account position and

107 SBI Research: https://tinyurl.com/nr75py78.

108 Global Financial Stability Report, IMF: https://tinyurl.com/52neuvxk; CEPR: https://tinyurl.com/bdepraz4.

capital account dynamics. As shown in Chart IV.24, periods of elevated CAD have generally coincided with sharper rupee depreciation, as observed in FY09, FY13, FY19 and FY23, reflecting pressures from widening merchandise trade deficits. However, there have been notable episodes in which rising imports and a widening trade deficit did not translate into currency depreciation. In FY08, FY18 and FY22, despite a rapid expansion in imports, the rupee appreciated, supported by robust net foreign investment inflows. These episodes underscore that exchange-rate movements are influenced not merely by the size of the trade deficit, but by the composition and stability of capital inflows.

4.93 This historical association between BoP position and exchange rate movements raises an important question about the relative strength of the trade and financial channels through which these effects are transmitted. Box IV.5 extends this analysis by empirically examining how exchange rate variations are transmitted across different components of India's external sector.

BOX IV.5: Trade and financial channels of the exchange rate

This is a continuation of the box item (Box IV.7: Trade and Financial channels of the Exchange rate) seen in the Economic Survey of 2023-24.

The previous analysis (published in July 2024) explored the contrasting impact that a weaker currency has on the two key components of the BoP – trade and finance.[109] The trade channel of the BoP gained from a weaker currency, due to a rise in the cost of imports and reducing export prices. On the other hand, a weaker currency does not benefit the financial channel of the BoP. Prolonged currency weakness can raise the cost of foreign lending and lead to a net capital outflow. The box concluded that for India, the benefits stemming from a weaker currency on the trade channel far outweighed the costs arising from the financial channel. On balance, therefore, a weaker currency was beneficial for India's BoP.

The discussion on whether a weaker currency supports the BoP as opposed to a stronger one continues to be a relevant one. Building on the previous box, this analysis offers a more disaggregated picture of the trade channel of the exchange rate by analysing the impact of exchange rate movements on both merchandise and services trade. The analysis utilises external sector data at a quarterly frequency between Q1 of FY12 to Q2 of FY26, employing an autoregressive distributed lag (ARDL) model, which is well-suited for small sample regressions and controls for endogeneity.

The study employs four separate ARDL regressions using dependent variables such as net total trade, net merchandise trade, net services trade and net capital balance. The basic form of the regression is as shown below:

$$\Delta \ln Y_t = \gamma_i \sum Y_{t-i} + \beta_i \sum \Delta Exch\ Rate_t + \theta X_t + \varepsilon_t$$

109 For a detailed exploration of the trade and financial channel of the exchange rate, see for instance, Kearns, J., & Patel, N. (2016). Does the financial channel of exchange rates offset the trade channel?. BIS Quarterly Review December, https://tinyurl.com/m74y6t4u.

7.2. The rapid expansion of digitally and remotely deliverable services has allowed the services sector to scale swiftly by transcending traditional geographical and operational constraints. This has enabled new forms of value creation, particularly in knowledge-intensive and technology-enabled activities, helping even smaller economies leave a global imprint. In contrast to the typical manufacturing-led growth paths followed by other countries at a comparable stage of development, India has experienced service-led growth at a significantly lower level of per capita income.

7.3. Currently, India's services sector contributes more than half of the Gross Value Added and serves as a major driver of exports and employment in the country. Not only does it continue to underpin domestic growth, but services have also emerged as the most stable and resilient component of GDP, acting as a high-growth, low-volatility anchor, as is the case across the globe. The sector has recorded average annual growth of around 7-8 per cent year after year, in sharp contrast to the more pronounced cyclical fluctuations observed in agriculture and industry.[2] India is the world's seventh-largest exporter of services, with its share in global services trade more than doubling from 2 per cent in 2005 to 4.3 per cent in 2024[3], and the sector continues to be the largest recipient of foreign direct investment inflows.

7.4. Even as services continue to anchor growth, the rapid technological progress which helped services ascend scales is now outpacing the growth in firm-level and worker-level adaptation, creating skill gaps and disruptions. Coupled with tighter immigration, data protection/localisation norms and remittance rules, the promise of being the 'stabilising force' is now under challenge.

7.5. Building on this context, the chapter first reviews global trends in the services sector, placing India's performance in an international context. It then assesses recent developments in India's services sector during the current year, followed by an analysis of performance across major sub-sectors, highlighting key strengths and emerging opportunities. The chapter concludes with a discussion of the main challenges and policy priorities for sustaining services-led growth.

GLOBAL SERVICES TRENDS AND INDIA'S EXPERIENCE

Structural Rebalancing and Evolving Growth Dynamics

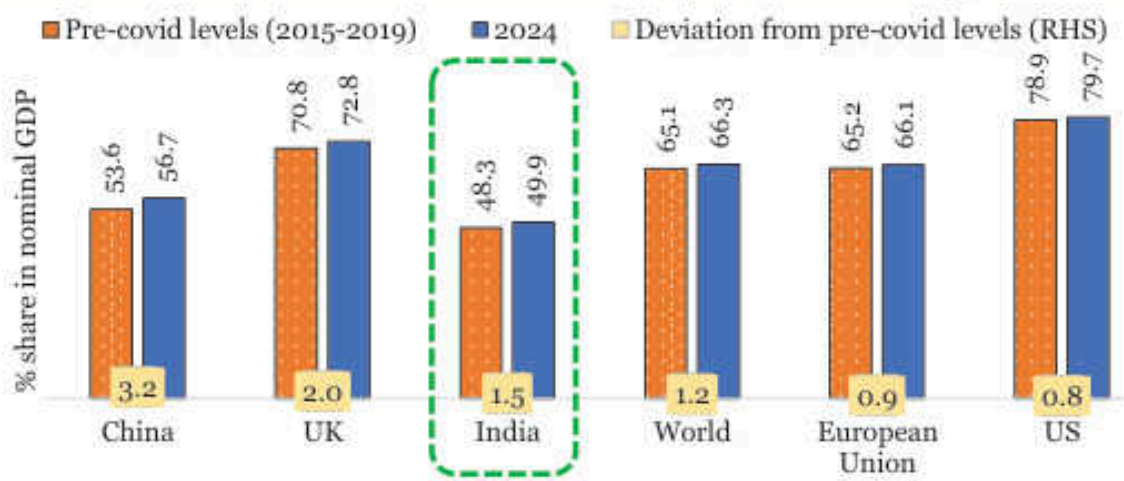
7.6. The COVID-19 pandemic severely disrupted contact-intensive services, such as tourism, hospitality, and transport, while accelerating the expansion of digitally

2 The standard deviation of growth rate for services during FY 05-25 is 1.22 compared to 4.20 for manufacturing, 3.07 for industry and 1.5 for overall GDP.

3 As per latest available data from WTO.

delivered services, including IT, finance, and professional services. This has thereby reshaped the structure and growth dynamics of the services sector. Since the latter is of high value, this compositional shift post-pandemic has raised the share of services in global gross value added, even as overall services growth has moderated relative to the pre-pandemic period (2015-19).[4] India has followed this trajectory with greater momentum relative to most peers (Chart VII.1). For instance, as per World Bank data, India's share of services in GDP rose by about 1.5 percentage points above its pre-pandemic average to reach 49.9 per cent in 2024, an increase larger than that observed globally and across major advanced economies.

Chart VII.1: India's Services Share in GDP strengthens in line with global trends



Source: World Development Indicators, World Bank.
Note: For India, data for 2024 pertains to 2024-25.

7.7. As mentioned above, notwithstanding the rising share of services in GDP, services growth in the post-pandemic period has faced headwinds from slower global growth, weaker expansion of trade and investment-linked services, and subdued productivity gains, particularly in labour-intensive segments. China has experienced the sharpest deceleration among major economies. India, by contrast, has remained relatively resilient, with services growth in 2024 staying close to its pre-pandemic levels (Chart VII.2).

4 Throughout the chapter, FY16-FY20 is taken as pre-pandemic period (for global comparison the same period corresponds to calendar year 2015-19) and the post-pandemic period is assessed using FY23-FY25, with FY21 and FY22 excluded due to base effects arising from the COVID shock.

KEY INITIATIVES TO PROMOTE MANUFACTURING

Performance Linked Incentive (PLI) scheme

8.39. In alignment with India's vision of 'Aatmanirbhar Bharat', the Production Linked Incentive (PLI) Scheme was launched in 2020 and now spanning 14 key sectors, with an outlay of ₹1.97 lakh crore. The core purpose of these schemes is to attract investments in key sectors and introduce cutting-edge technology, ensure efficiency, and achieve economies of size and scale, thereby making Indian companies and manufacturers globally competitive. Till September 2025, under the PLI scheme, an actual investment of over ₹2.0 lakh crore has been realised, leading to incremental production/sales of over ₹18.70 lakh crore and employment generation of over 12.60 lakhs (direct and indirect). Cumulative incentives of ₹23,946 crore have been disbursed across 12 sectors, with 806 applications approved across all 14 sectors. The PLI Schemes have witnessed exports surpassing ₹8.20 lakh crore, driven significantly by sectors like electronics, pharmaceuticals, and telecom & networking products.[28]

8.40. Sectoral gains are tangible and worth noting. The electronics sector has emerged as a flagship success story under the PLI strategy. The PLI Scheme has encouraged major smartphone companies to re-locate their production to India. As a result, India has become a major mobile phone manufacturing hub.[29] In the first three years, pharmaceutical sales under PLI Scheme for Pharmaceuticals crossed ₹2.63 lakh crore, including exports worth ₹1.69 lakh crore and overall, Domestic Value Addition in the sector has been 83.74 per cent as on March 2025.[30] Further, the PLI scheme for automobile and auto-components has attracted cumulative investments worth ₹35,657 crore, resulting in creation of 48,974 jobs, until September 2025.[31]

National Manufacturing Mission

8.41. Complementing PLI, the National Manufacturing Mission (NMM), was announced in Union Budget 2025–26. The NMM aims to boost innovation, raise competitiveness, and expand manufacturing capacity across priority sectors, complementing Make in India and the Atmanirbhar Bharat push. It emphasises faster technology adoption and deeper MSME integration into value chains.

28 Department for Promotion of Industry and Internal Trade (DPIIT).

29 Source: PIB Release available at https://tinyurl.com/3ffsxy57.

30 Department of Pharmaceuticals

31 Ministry of Heavy Industries.

based industrial development. Similarly, with lower operating costs and improved road, rail and air connectivity with wider supply chains - a diversified manufacturing base anchored in steel processing, metal fabrication and consumer durables has been developed at Hubballi–Dharwad.[58] India's industrial future thus is likely to be facilitated by its smaller cities across various states.

SCALING UP MSMEs: DEEPENING COMPETITIVENESS, EXPANDING MARKET ACCESS, AND BRIDGING THE CREDIT GAP

8.60. Micro, Small, and Medium Enterprises (MSMEs) form the backbone of India's industrial economy, accounting for approximately 35.4 per cent of manufacturing, around 48.58 per cent of exports, and 31.1 per cent of GDP in the country.[59] With over 7.47 crore enterprises employing over 32.82 crore persons,[60] the sector holds its position as the second-largest employer after agriculture. Globally, MSMEs make up about 90 per cent of businesses and are responsible for over 50 per cent of the total global employment.[61] With India's manufacturing sector positioned for greater global integration, MSME sector's role is critical in enabling effective supply-chain participation, fostering local value addition, and supporting inclusive regional growth.

8.61. Despite expanding credit footprints and rising digital integration, access to formal credit remains a binding constraint for many micro-enterprises due to limited collateral and documentation readiness, as is generally the case worldwide. This challenge was also highlighted in the World Bank's Financial Sector Assessment Report for India (2025),[62] which noted that 27 per cent of MSMEs identify finance as their biggest obstacle. Women-owned MSMEs, in particular, account for a small fraction of commercial credit, though formalisation under Udyam and targeted credit guidelines are gradually addressing this gap. Furthermore, MSME credit has maintained a positive trajectory in recent times, bolstered by several government interventions aimed at enhancing credit flow to the sector.

8.62. MSME credit remained the primary driver of industrial credit growth during H1FY26. In recent periods, the overall MSME credit growth year-on-year (Y-o-Y)

58 https://tinyurl.com/5bmuz77n.

59 Source: National Statistical Office, Ministry of Statistics & Programme Implementation for the year 2023-24. Data for exports is sourced from the portal of Directorate General of Commercial Intelligence and Statistics for the year 2024-25

60 Total employment reported by the MSMEs on the Udyam Registration Portal as on 09.01.2026, https://tinyurl.com/t557udxr.

61 NITI Aayog Report on Enhancing MSMEs Competitiveness in India (2025) available at https://tinyurl.com/2x443e58.

62 World Bank- India Financial Sector Assessment Report (June 2025) https://tinyurl.com/dv2bsduc.

has significantly outpaced the Y-o-Y growth observed in large industry credit (Table VIII.3).[63] This acceleration can partly be attributed to policy measures, including the revised MSME classification thresholds implemented in April 2025, which expanded the eligibility for priority sector lending, guidelines enabling jewellery-backed small business loans and targeted Budget schemes. Further, NBFCs continued to play a critical role as last-mile credit providers, maintaining strong double-digit lending growth to MSMEs.[64]

Table VIII.3: Growth in deployment of gross bank credit to MSMEs (Y-o-Y per cent), select periods

Segment	March 2024	August 2024	March 2025	August 2025
Micro & Small Enterprises	14.7	13.4	8.8	20.9
Medium Enterprises	13.3	19.2	18.6	13.1
Total MSME Credit	**14.3**	**15.1**	**11.7**	**18.5**
Large Industry (comparison)	6.4	7.7	6.2	1.8
Source: Monetary Policy Report, RBI Bulletin (October 2025)				

8.63. The SME public markets have also witnessed a dramatic expansion over the past two years, driven by buoyant market conditions and digital retail participation. Between 2022–23 and 2024–25, SME IPO listings surged by 87.2 per cent, with the total issue amount growing by 52.7 per cent over the previous year. This boom was characterised by a sharp rise in retail investor participation, facilitated by UPI-based ASBA (Application Supported by Blocked Amount) system, and significant listing gains.[65] While capital-intensive manufacturing dominated issue sizes, the services sector also saw a rise in listings, indicating the SME segment's enhanced role in mobilising capital for growth.

Box VIII.10: Maharashtra's Pioneering Defence & Aerospace Venture Fund— A State-Led Innovation Finance Model

Maharashtra launched a pioneering Defence & Aerospace Venture Fund (MDAVF), a registered Category II-Alternative Investment Fund (AIF) worth ₹330 crore[66] in 2018, aimed at catalysing manufacturing capabilities among MSMEs in strategic, high-growth sectors. The Fund, managed by IDBI Capital Markets & Securities, focuses on supporting enterprises engaged in precision engineering, aerospace components, UAV systems, and defence sub-systems. This model blends State capital

63 RBI Bulletin, October 2025: https://tinyurl.com/2d59t9vx.

64 ibid.

65 Chattopadhyay, B., & Ganguly, S. (2025, October 20). Fundraising by Indian Small and Medium Enterprises through IPO: Recent Trends and Developments. RBI Bulletin, (October 2025) available at https://tinyurl.com/bdumbhz8.

66 https://tinyurl.com/5n97nfcj.

with professional fund management to provide crucial early-stage and growth-stage support to MSMEs. The Fund invests in companies through various instruments, including equity or equity-linked investments, convertible instruments, and debentures, to achieve risk-adjusted returns.[67]

The Fund has attracted significant private capital into high-technology MSMEs through co-investment and risk-sharing mechanisms and supported MSMEs in acquiring vital defence certifications, facilitating vendor development processes, and achieving their export readiness. The initiative is explicitly aligned with the national indigenisation priorities outlined under the Defence Production and Export Promotion Policy (DPEPP). It leverages strong linkages to Maharashtra's established industrial corridors and emerging defence clusters.

The Fund has extended capital support to a broad set of MSMEs, with investments spanning missile systems, aerospace technologies, naval platforms, and electronic warfare. It has built a diversified portfolio, including investments across multiple enterprises and selective full and partial exits, reflecting a maturing investment cycle. This Fund demonstrates how State-level innovation finance can accelerate MSME technological capabilities in sunrise sectors. Such funds can indeed accelerate MSME participation in strategic value chains by catalysing private investment, facilitating certification and R&D and anchoring enterprises within emerging industrial corridors.

8.64. To strengthen credit linkages, the Credit Guarantee Scheme (CGS) for Micro and Small Enterprises (MSEs) was revamped, effective from 01 April 2023, following a corpus infusion of ₹9,000 crore in the Credit Guarantee Fund Trust for Micro and Small Enterprises (CGTMSE), enhancing the ceiling for guarantee coverage from ₹2 crore to ₹5 crore and reducing the annual guarantee fees to as low as 0.37 per cent. Key enhancements include increasing the extent of guarantee coverage for women-owned MSEs from 85 per cent to 90 per cent. Additionally, effective from 01 April 2025, the government has doubled the ceiling for guarantee coverage from ₹5 crore to ₹10 crore and rationalised the annual guarantee fee for coverage exceeding ₹1 crore. [68]

8.65. The Self-Reliant India (SRI) Fund, launched to infuse ₹50,000 crore as equity funding in MSMEs, has assisted 682 MSMEs by way of investment worth ₹15,442 crore, as of 30 November 2025.[69] Additionally, the Prime Minister's Employment Generation Programme (PMEGP) assists micro-entrepreneurs by providing margin money subsidies on bank loans. The programme has been expanded to include higher project costs and an enhanced scope of activities. Further, to bolster competitiveness, the MSME Champions Scheme promotes "Zero Defect, Zero Effect" practices through ZED Certification and productivity improvements via the MSME Competitive (Lean)

67 https://tinyurl.com/2b87ffef.

68 Ministry of Micro, Small and Medium Enterprises.

69 ibid.

CHAPTER 09

INVESTMENT AND INFRASTRUCTURE: STRENGTHENING CONNECTIVITY, CAPACITY AND COMPETITIVENESS

Infrastructure continues to be central to India's growth strategy, with public capital expenditure following a sustained upward trajectory since FY15 and gaining further momentum in recent years. Large-scale investments across roads, railways, ports, power, aviation and digital infrastructure have strengthened connectivity, expanded capacity and improved logistics efficiency, generating strong multiplier effects for growth and productivity. This phase has been characterised not only by rapid asset creation but also by a shift towards integrated, system-level development.

A defining feature of this transition has been the institutionalisation of multimodal planning through PM GatiShakti, complemented by the National Logistics Policy and digital platforms that are reducing transaction costs and execution risks. Reforms in infrastructure financing—spanning project finance regulation, Public-Private Partnership (PPP) frameworks, asset monetisation and capital market instruments—are crowding-in private investment. At the same time, India's infrastructure base is broadening to include digital public infrastructure (DPI), clean energy, resilient water systems and emerging sectors. Sustaining investment momentum while aligning infrastructure with efficiency, sustainability and competitiveness will continue to remain key towards supporting India's medium-term growth and its long-term vision of Viksit Bharat @2047.

INTRODUCTION: INFRASTRUCTURE AS THE ENGINE OF GROWTH

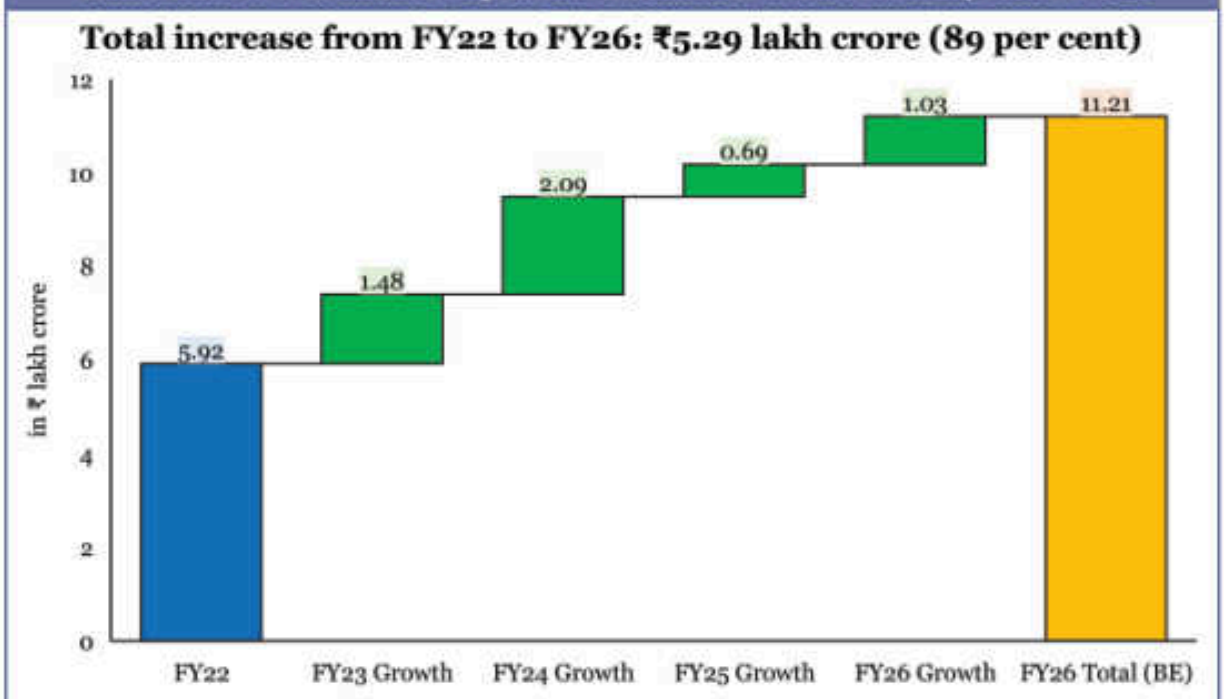
9.1 The decisive role of infrastructure investment in India's growth strategy has been a defining feature in recent years. Union Budget priorities since FY20 reflect a sustained shift toward large-scale public capital expenditure, supported by integrated planning, and the systemic modernisation of logistics and connectivity networks.

9.2 This period has been marked not just by the accelerated expansion of traditional infrastructure, such as highways, railways, ports, and energy systems, but also by the expansion of digital public infrastructure (DPI), data systems, and assets aligned with renewable energy. Together, these developments signal a shift in how India plans and finances infrastructure projects, moving beyond mere capacity addition to focusing on enhancing efficiency, competitiveness and network integration.

9.3 A major element of this shift has been the substantial increase in public capital expenditure. Between FY19 and FY22, the Government of India's capital expenditure increased by 92 per cent, from ₹3.07 lakh crore to ₹5.92 lakh crore. This momentum has been sustained over the subsequent fiscal years, reinforcing the objective of expanding access to quality infrastructure across the country.

9.4 The Government of India's capital outlay has increased by nearly 89 per cent, from ₹5.92 lakh crore in FY22 to a budgeted allocation of ₹11.21 lakh crore for FY26, recognising the strong multiplier effects that infrastructure generates on the economy. Public expenditure on infrastructure has a high multiplier effect, estimated by studies to be around 2.5 to 3.5 times the GDP over the medium term.[1] This means, for every rupee spent by the government in creating infrastructure, GDP gains worth ₹2.5-₹3.5 accrue.[2] The scale and consistency of this investment momentum have positioned infrastructure as a cornerstone of India's growth engine.

Chart IX.1: Government Capex has continued to rise steadily since FY22



Source: Union Budget Documents.

9.5 This investment push has been paired with major institutional and technological reforms designed to enhance planning quality and reduce execution risks also through private participation. The PM GatiShakti National Master Plan, launched in 2021, has successfully institutionalised multimodal, GIS-enabled planning. The recent launch of PM GatiShakti Public has further democratised the platform, providing regulated

1 Fiscal Multipliers for India (by Sukanya Bose & N. R. Bhanumurthy, NIPFP, 2013) https://tinyurl.com/4972rumk.

2 Speeding up with Gati Shakti (October 2021), NITI Aayog, available at https://www.niti.gov.in/speeding-gati-shakti.

access to geospatial data for private developers and researchers to leverage advanced analytics for infrastructure planning and investment decisions.[3]

9.6 Complementary initiatives under the National Logistics Policy (NLP) have supported the emergence of a more predictable and digitised logistics environment. The use of the Unified Logistics Interface Platform (ULIP) and the insights drawn from the Logistics Ease Across Different States (LEADS) framework are actively improving logistics efficiency, which directly affects the competitiveness of India's manufacturing sector. A detailed analysis of the impact of the PM GatiShakti framework, the National Logistics Policy, and other initiatives aimed at reducing logistics costs and improving competitiveness is presented in the Infrastructure and Logistics section in Chapter 8 on Industry.

9.7 Beyond physical assets, DPI acts as an essential force multiplier across the economy. The expansion of BharatNet, the deployment of 5G networks, and the continued growth of platforms like Unified Payments Interface (UPI) and Aadhaar, ULIP, DigiYatra, FASTag, form the essential digital backbone upon which smart, efficient, and inclusive infrastructure services are delivered at scale.

9.8 Overall, infrastructure remains a key lever for sustaining medium-term growth through (i) continued public capex and improved project planning and execution, (ii) crowding-in of private investment through a stronger Public-Private Partnership (PPP) pipeline, (iii) deeper and more resilient long-term financing via capital markets and prudentially sound bank/NBFC credit.

9.9 While public capital expenditure on infrastructure has expanded significantly in recent years, its effectiveness increasingly depends on the quality of project planning, prioritisation and execution. Evidence from various studies suggests that time and cost overruns in large infrastructure projects are often linked to gaps in upstream project preparation, including weak feasibility assessments, incomplete Detailed Project Reports (DPRs), rigid procurement practices, and delays in dispute resolution. Strengthening institutional capacity across the project lifecycle, including improved project preparation, lifecycle-based costing, standardised contracts, faster dispute resolution, wider adoption of technology, and skill development are critical to delivering infrastructure faster, at scale, and within budget.

9.10 The subsequent sections of this chapter discuss measures to enhance infrastructure financing and provide a detailed review of progress, outlook, and challenges across key infrastructure sub-sectors, including core physical infrastructure, energy sector, digital infrastructure, rural infrastructure and emerging

3 PIB Release available at https://tinyurl.com/nm83nmw7.

9.13 A major regulatory milestone in infrastructure credit is the RBI (Project Finance) Directions 2025, effective from 01 October 2025. These guidelines introduce a unified framework for project lending across all financial institutions, ensuring a consistent approach to large-scale financing. A key feature of these directions is the revised treatment of the Date of Commencement of Commercial Operations (DCCO). This provides a more realistic way to handle project delays, which helps in better identifying actual stress and preventing the artificial 'evergreening' of loans. Furthermore, by aligning the definition of the 'Infrastructure Sector' with the government's Harmonised Master List (HML) of infrastructure sub-sectors, the RBI has ensured regulatory clarity and policy synchronisation across the entire financial ecosystem.

9.14 Capital-market channels for long-term funding and asset recycling have also been strengthened through incremental reforms. SEBI's Small and Medium Real Estate Investment Trusts (SM REIT) framework reduced the minimum asset size relative to existing REITs, from ₹500 crore to ₹50 crore. It also introduced a scheme-based structure, allowing multiple schemes to be managed under one SM REIT. Each scheme's proposed assets are generally required to be at least ₹50 crore and less than ₹500 crore. This broadens the universe of monetizable real estate assets and can support urban regeneration/commercial infrastructure by bringing smaller, stabilised assets into a regulated pooled vehicle.

9.15 Additionally, SEBI has decided that from 01 January 2026, investments by Mutual Funds and Specialised Investment Funds (SIFs) in REITs will be treated as equity-related instruments, which is expected to ease participation constraints and potentially improve secondary market liquidity. From April to November 2025, ₹13,893 crore was raised by listed Real Estate Investment Trusts (REITs) and Infrastructure Investment Trusts (InvITs).[5]

Public-Private Partnerships

9.16 PPPs are vital mechanisms through which governments can leverage private sector expertise and resources to address critical infrastructure needs. They play a significant role in bridging infrastructure gaps and enhancing the efficiency of service delivery. The success of PPPs is largely dependent on the robustness of institutional frameworks, financial support, and the adoption of standardised documents, such as the Model Request for Qualification (RfQ), Model Request for Proposal (RfP), and Model Concession Agreements (MCAs).

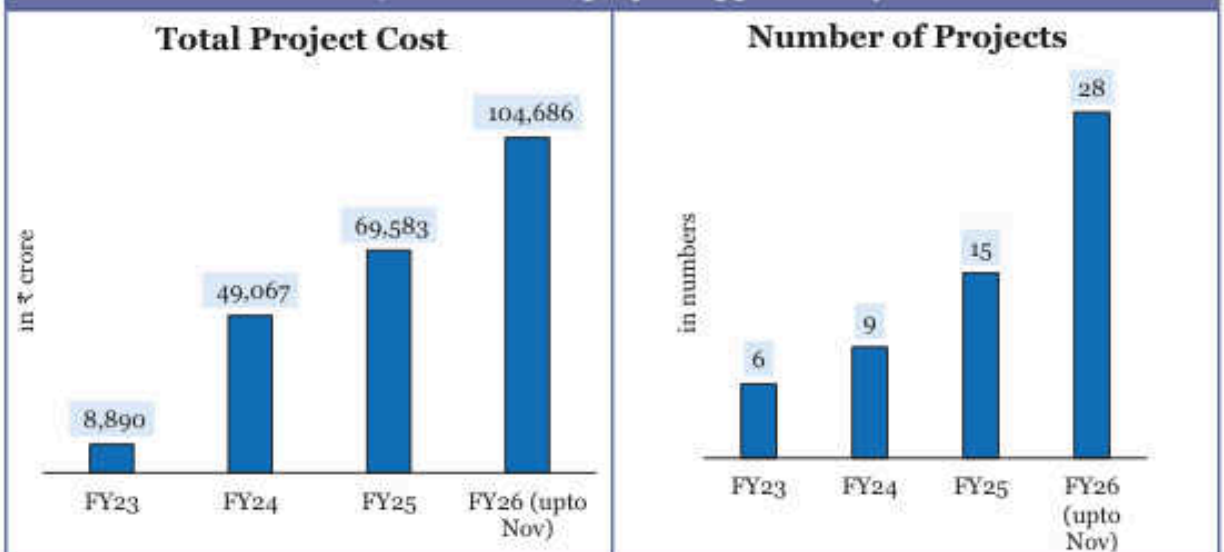
9.17 India's infrastructure programs support a variety of PPP models, including

5 SEBI statistics on fund raising by REITs and InvITs available at https://tinyurl.com/57u2rjtb. Note: Total funds mobilised by REITs & InvITs includes funds raised through public issue, private placement, preferential issue, institutional placement, rights issue.

management contracts like Build-Operate-Transfer (BOT), Design-Build-Finance-Operate-Transfer (DBFOT), Hybrid Annuity Model (HAM), and Toll-Operate-Transfer (TOT). Within the BOT framework, two variants exist—BOT (Toll) and BOT (Annuity)—distinguished by the allocation of traffic risk.

9.18 According to the World Bank's Private Participation in Infrastructure (PPI) Report 2024,[6] India has consistently ranked among the top five countries globally in terms of private investment in infrastructure among low- and middle-income economies. India also emerged as the largest recipient of PPI investment in South Asia, accounting for over 90 per cent of the region's total private infrastructure investment. This strong global standing is reflected domestically in the marked increase in project approvals by the Public-Private Partnership Appraisal Committee (PPPAC), as illustrated in Chart IX.4. The key institutional and policy mechanisms underpinning India's PPP framework are discussed in Box IX.1, while the priorities shaping its next phase are outlined in Box IX.2

Chart IX.4: Increase in project approvals by PPPAC



Source: PPP Database, Department of Economic Affairs, pppinindia.gov.in.

Box IX.1: Key Institutional Mechanisms to support PPPs in India

Public Private Partnership Appraisal Committee (PPPAC):

- Apex body for appraising and recommending central sector PPP projects. Chaired by the Secretary, Department of Economic Affairs (DEA), its members include Secretaries from the Department of Expenditure, Department of Legal Affairs, the sponsoring ministry/department, and the CEO of NITI Aayog.

6 Private Participation in Infrastructure (PPI) Annual Report (2024), World Bank available at https://tinyurl.com/muywbmcb.

- 129 projects with a Total Project Cost (TPC) of ₹5,60,965.71 crore have been recommended by the PPPAC [From 2014-15 to 2025-26 (up to 04 December 2025)].

Viability Gap Funding (VGF) Scheme:

- Provides financial assistance to projects that are economically desirable but commercially unviable.
- Economic sector projects may receive up to 40 per cent of capital expenditure (Capex) as a VGF grant, while social sector projects are eligible for grant up to 80 per cent of Capex and 50 per cent of operational expenditure for five years post-commercial operations.
- 72 projects have been approved, with a GoI VGF of ₹7,941.838 crore sanctioned (up to 04 December 2025). Out of this, ₹6,314.86 crore has already been disbursed.

India Infrastructure Project Development Fund (IIPDF):

- Aims to create a pipeline of viable, bankable projects by funding transaction advisers for central and state authorities.
- Notified in November 2022 with a total outlay of ₹150 crore for three years from FY23 to FY25.

Development of a Three-Year PPP Project Pipeline:

- Pursuant to the announcement in the Union Budget 2025-26, a 3-year PPP project pipeline has been created.
- The PPP project pipeline comprises 852 projects across central infrastructure ministries and States/Union Territories with a combined total project cost of over ₹17 lakh crore.

Standardisation and Institutional Capacity Building:

- Development of reference guides on risk allocation, PPP appraisal, and project implementation mode (Waterfall Framework) including sector-specific PPP structuring toolkits for Roads, Ports, Solid Waste Management, and Water & Sanitation have been done.
- Capacity Building Initiatives for States/UTs through workshops, seminars, and one-to-one guidance are ongoing.
- National Infrastructure Enablement Index (NIEI) has been developed to assess the institutional preparedness of States/UTs and Central Ministries/Departments.
- Model RfP (Request for Proposal) for single-stage PPP projects has been developed to embed best practices, ensure consistency, and enhance flexibility in procurement and contract management.

Box IX.2: Public–Private Partnerships (PPP)- Strengthening the Partnership Paradigm

India's PPP programme has traversed a full learning curve since the first wave of projects in the early 2000s. In the initial phase, multiple PPP models were tested across sectors, with some projects encountering stress or failure. These experiences informed successive improvements in model concession agreements, policy guidelines, and institutional practices by addressing contractual weaknesses, risk-allocation errors, and regulatory gaps. Over time, this iterative process has converted early experimentation into codified frameworks and institutional memory.

As a result, the environment for private participation in infrastructure is materially stronger in select core sectors. Roads, ports, power, and renewable energy now attract sustained private capital, supported by more stable policy regimes, standardised contracts, and clearer regulatory architectures. These sectors have transitioned from infancy to relative maturity through deliberate sequencing of reforms. For instance, tariff regulation in ports was initially institutionalised through the Tariff Authority for Major Ports (TAMP) and subsequently withdrawn as market depth and competition improved, underscoring the need for regulation to evolve with sector maturity.

The next challenge is to extend this maturity to a new generation of PPPs in emerging and socially critical sectors such as health, education, warehousing, sanitation, urban infrastructure, green hydrogen, and the broader energy transition. In these areas, conventional concession-style risk transfer is often insufficient. PPPs must increasingly reflect the third "P" — Partnership — where public and private actors co-design projects, share early-stage risks, and align incentives around long-term service outcomes rather than narrow financial closure.

Accordingly, India's PPP framework needs to move from transaction-centric execution toward system-level market building, with a sharper focus on reducing structural uncertainty. This requires clearer sectoral pipelines with multi-year visibility, a tighter linkage between national programmes (PM GatiShakti, National Infrastructure Pipeline, sector missions), and bankable project preparation, as well as disciplined pre-construction risk closure by the public authority. PPP outcomes have been weakest where land acquisition, statutory clearances, demand assessment, or utility shifting have remained unresolved. In the coming decade, a credible PPP regime will be defined less by risk transfer on paper and more by the State's capacity to absorb early-stage risks that private capital cannot efficiently price.

While PPP frameworks have matured at the central level and in select sectors, challenges persist at the sub-national level. The distinction between PPPs and Engineering Procurement and Construction (EPC) contracts—namely that PPPs are partnerships rather than vendor arrangements—is not always fully understood. Trust deficits and a limited understanding of risk–reward principles continue to constrain the uptake of PPPs in several states and urban local bodies. These challenges are compounded by uneven institutional capacity, even as states and cities account for a rising share of infrastructure demand. The next reform frontier lies in professionalising PPP cells, adopting programmatic approaches instead of project-by-project experimentation, and leveraging data platforms to track performance, renegotiation, and lifecycle outcomes. In this context, the Department of Economic Affairs'

schemes such as the India Infrastructure Project Development Fund (IIPDF) and Viability Gap Funding (VGF) are important.

An often-overlooked aspect relates to public perception. PPPs are sometimes perceived as 'selling of assets' rather than long-term service partnerships, particularly at the state level. Improving communication and transparency is therefore important for sustaining public acceptance and investor confidence, especially for projects with long concession periods. The perception principle has improved by a long margin among central projects as compared with States. The lesson is often lost that wrong signalling in this regard creates two problems – adversely affecting the public mood and driving away investors willing to stomach the entirety of risks across a long concession period (30-60 years).

India has also consciously deepened the financial architecture supporting PPPs since 2014. Long-term domestic capital, including pension funds, insurance funds, InvITs and Alternative Investment Funds (AIFs), has increasingly participated in brownfield assets. However, greenfield PPPs continue to rely heavily on bank balance sheets. Expanding construction-period risk mitigation tools, partial credit guarantees in social sectors, take-out financing, and standardised refinancing frameworks can materially lower the cost of capital. Equally important is creating predictable exit and restructuring pathways for stressed but viable projects, so that failure does not permanently impair investor confidence. A renegotiation framework for complex cases to prevent project termination is worth exploring. PPPs should be treated as evolving financial assets rather than one-time contracts frozen at financial close.

Finally, private sector participants that have benefitted from de-risked brownfield assets should be encouraged to increase their risk appetite for greenfield projects, particularly through pure-play PPP models such as BOT, especially in mature sectors like roads.

9.19 As also highlighted in the Economic Survey 2024-25, public sector efforts alone cannot meet India's growing infrastructure requirements. While public investment remains crucial, a multi-pronged financing approach is essential to attract the requisite investments from the private sector and long-term institutional investors. This strategy requires strengthening resource mobilisation across all levels through innovative measures, including viable user charges and empowering municipal bodies to float bonds for localised resource generation. It also involves accelerating private participation including through public-private partnerships (PPPs). A wider array of PPP models with nuanced risk-sharing, right-price infrastructure services and a pipeline of bankable projects can facilitate infrastructure development significantly. Improved project preparation, standardised contracts and dispute resolution mechanisms are also critical to sustaining investor confidence and scaling PPPs.

CORE PHYSICAL INFRASTRUCTURE

Roadways & Highways

9.20 The roads and highways sector continues to be a primary driver of India's infrastructure, transitioning from rapid capacity expansion to a focus on logistic

programme has been strengthened through focused initiatives, including a dedicated road connectivity vertical under PM-JANMAN (Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan) to serve Particularly Vulnerable Tribal Groups (PTVGs), targeting 8,000 km of roads over the period 2023-28. As of 31 December 2025, 2,495 roads (7,323.96 km of road length) and 163 bridges have been sanctioned under PM-JANMAN for connecting 2,909 habitations, out of which 248 roads (1242.41 km of road length) has been completed.[10]

9.22 Further momentum has been reinforced through PMGSY-IV, launched in September 2024, which aims to connect 25,000 unconnected habitations by constructing or upgrading 62,500 km of roads and bridges at an estimated cost of ₹70,125 crore during FY25–FY29. The phase adopts international standards, GIS-based surveys, and the PM GatiShakti framework, while converging with the Dharti Abha–Janjatiya Gram Utkarsh Abhiyan (DA-JGUA) scheme to prioritise habitations with high Scheduled Tribes population. So far, 2,490 roads (8,655.378 km of road length) and 01 bridge have been sanctioned under PMGSY-IV, which will provide connectivity to 2,734 new habitations.

9.23 **Outlook:** India's roads and highways sector is moving from rapid network expansion towards greater logistics efficiency and quality. Sustained capital investment, expansion of high-speed corridors, multimodal integration under PM GatiShakti and reforms in project delivery are strengthening capacity and reliability. This infrastructure-led push is central for reducing logistics costs, easing congestion and improving connectivity.

Railways

9.24 Indian Railways continue to play a pivotal role in India's infrastructure landscape by expanding network capacity, modernising assets, and strengthening multimodal connectivity. As of March 2025, the rail network has expanded to 69,439 Rkm. During FY26, it is targeted to extend the network further by 3,500 km. Electrification has reached 99.1 per cent of the network by October 2025.[11] Sustained investments have enabled faster network expansion, with commissioning rates more than doubling in the post-2014 period compared to the previous decade (Chart IX.9).

10 ibid.

11 Ministry of Railways.

Chart IX.9: Average Commissioning of railway network per year

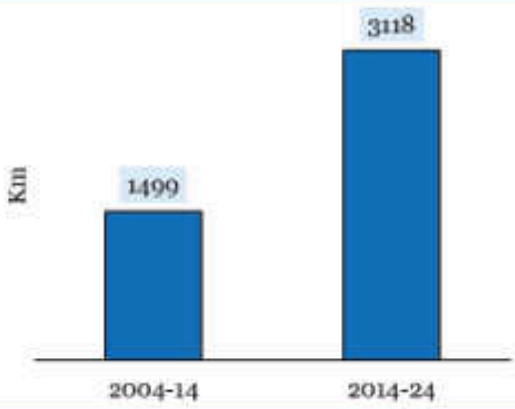
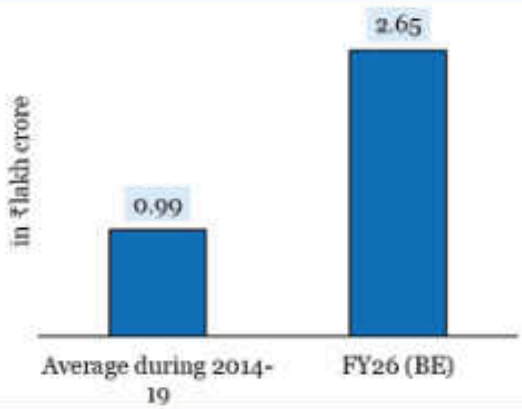


Chart IX.10: Capital Expenditure on Railways (in ₹ Lakh crore)



Source: Ministry of Railways.

9.25 A defining feature of recent years has been the record capital expenditure on railway infrastructure, with a focus on new lines, doubling and multi-tracking, rolling stock augmentation, signalling, and safety-related works. In FY26 (BE), capital outlay has been maintained at historically high levels, to accelerate capacity creation in a time bound manner (Chart IX.10).

9.26 Railways have also emerged as a backbone of India's freight and energy logistics, supporting coal movement, industrial supply chains, and containerised traffic through dedicated corridors, modern terminals, and first-mile connectivity projects. Parallel investments in stations, signalling, telecom and digital systems are improving network reliability, safety, and user experience (Box IX.4).

Box IX.4: Key Infrastructure Initiatives in the Railways Sector

- **Economic Railway Corridors (PM GatiShakti):** Three corridor programmes—Energy, Mineral & Cement; Port Connectivity; and High Traffic Density routes—are being implemented to strengthen multimodal connectivity and logistics efficiency. 434 projects with an outlay of ₹11.17 lakh crore have been identified under three corridors and mapped on the PM GatiShakti portal. Of these, 122 projects covering 12,150 km have been sanctioned, and 198 projects spanning 19,779 km are under different stages of appraisal.
- **Major ongoing projects:**
  - **Mumbai–Ahmedabad High Speed Rail (MAHSR):** Over 55 per cent physical progress has been achieved as on October 2025, with land acquisition completed and most civil packages awarded, marking a major step towards introducing high-speed rail infrastructure in India.
  - **Dedicated Freight Corridors (DFCs):** About 2,741 km (96.4 per cent) of the 2,843 km DFC network has been commissioned as of October 2025, with the Eastern

DFC (1,337 km) fully completed and 1,404 km of the 1,506 km Western DFC completed. These corridors are easing congestion on the passenger network, significantly reducing freight transit times and contributing to lowering logistics costs.

- **Station Redevelopment:** Under the Amrit Bharat Station Scheme, 1,337 stations have been identified for phased redevelopment, with works completed at select stations and progress underway across the country to improve capacity, accessibility and multimodal integration. Further, 15 stations have been identified to develop/redevelop on PPP mode.
- **Safety & Technology Upgradation:** Large-scale deployment of Kavach [Advance Train Protection (ATP) System], electronic interlocking, automatic block signalling, and track renewal are enhancing network safety and throughput.
- **Track upgradation:** More than 78 per cent of railway tracks have been upgraded for sectional speed of 110 kmph and above.
- **Public–Private Partnerships (PPPs):** 18 projects worth ₹16,636 crore have been completed and seven projects worth ₹16,334 crore are under implementation as on September 2025.[12]

9.27 **Outlook:** Indian Railways is undergoing a transformation, driven by sustained capital investment, rapid network expansion, near-universal electrification and a corridor-based approach to capacity creation. Continued focus on dedicated freight corridors, economic rail corridors under PM GatiShakti, and modern signalling and station infrastructure is improving throughput, reliability and multimodal integration. These infrastructure-led initiatives will be central for reducing logistics costs, and strengthening connectivity.

Civil Aviation

9.28 India has emerged as the world's third-largest domestic aviation market. The number of airports increased from 74 in 2014 to 164 in 2025.[13] In FY25, Indian airports handled 412 million passengers, and the same is projected to increase to 665 million by FY31. Further, air cargo volume grew from 2.53 million metric tonnes (MMT) in FY15 to 3.72 MMT in FY25, and 2.95 MMT handled in FY26 (until December), driven by several key policy initiatives and reforms (Box IX.5).[14]

12 Ministry of Railways.

13 PIB Release: https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2181865&reg=3&lang=2.

14 Ministry of Civil Aviation.

Chart IX.11: Air cargo handling (International and Domestic)

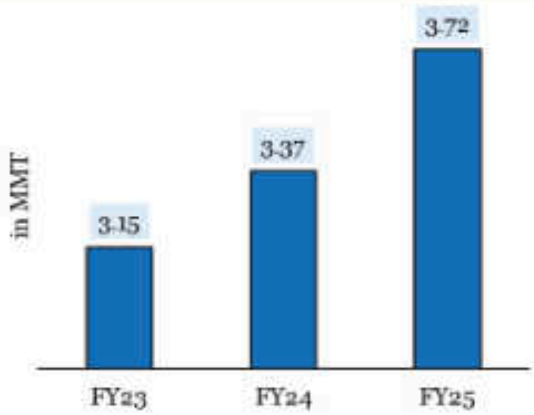
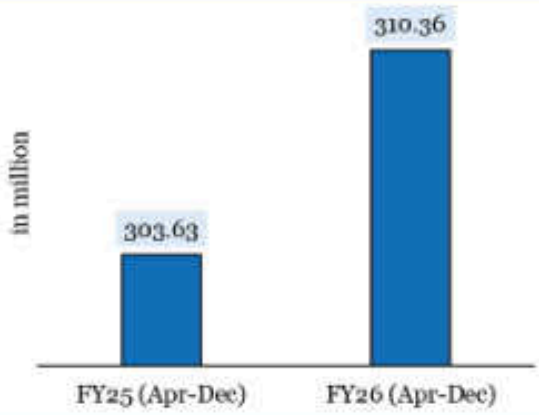


Chart IX.12: Passenger traffic at all airports



Source: Ministry of Civil Aviation.

Box IX.5: Key policy initiatives driving civil aviation growth

- **Regional Connectivity Scheme- Ude Desh ka Aam Nagrik (RCS-UDAN):** Operationalisation of 657 routes connecting 93 airports, including heliports and water aerodromes, has made air travel affordable while improving access to remote and aspirational regions. Further, the modified UDAN Scheme has been announced to increase regional connectivity to 120 new destinations and cater to 4 crore passengers over the next 10 years.
- **Greenfield Airports Policy:** In-principle approval for 24 Greenfield airports, with 13 already operational airports, including the Navi Mumbai International Airport, is expanding urban and regional capacity.
- **Airport Modernisation & Capacity Expansion:** Since FY20, modernisation projects at several airports across the country have successfully raised combined passenger-handling capacity of operational airports to approximately 575 million passengers per annum.
- **Digital & Technological Initiatives:** Expansion of Digi Yatra, liberalised drone regulations, PLI support for drone manufacturing, and focus on Advanced Air Mobility (AAM) have been ongoing.
- **Legislative Reforms:**
  - The **Bharatiya Vayuyan Vidheyak, 2024**, replacing the Aircraft Act, 1934, aims to modernize India's aviation sector by enhancing safety, innovation, growth, and global compliance.
  - The **Protection of Interests in Aircraft Objects Act, 2025**, aims to align India's aviation leasing laws with global standards to reduce leasing costs.

9.29 **Outlook:** India's civil aviation sector is on a sustained growth trajectory, supported by a conducive policy environment, rising demand and steady infrastructure expansion. While the sector remains sensitive to global economic cycles and the need for continuous capacity upgradation, the current passenger volumes represent only a fraction of India's potential. Further, India currently operates approximately 0.11 airports per million people, significantly lower than the US (47.35) and China (0.39)[15], signalling substantial headroom for further growth. Expansion in India's airport and air navigation infrastructure and a growing ancillary ecosystem, including Maintenance, Repair, and Overhaul (MRO) and leasing, are strengthening the sector. These developments, along with technology integration, positions civil aviation as a key driver of nationwide economic connectivity and integration.

Ports and Shipping

9.30 India's maritime sector has undergone significant modernisation and is emerging as an essential pillar of logistics competitiveness, industrial growth and global supply-chain integration. Under Maritime India Vision 2030 and Maritime Amrit Kaal Vision 2047, substantial progress has been made in upgrading port infrastructure, enhancing regulatory frameworks, improving operational efficiency, and increasing private sector participation.

9.31 India's ports have demonstrated strong growth momentum, driven by accelerated capacity creation and a substantial increase in cargo throughput (Chart IX.13). Furthermore, mechanisation, smart port initiatives, and digital trade facilitation have significantly improved operational efficiency across major ports, with the average container vessel turnaround time achieving near-global best standards (Chart IX.14). India's growing competitiveness is reflected in global rankings, with two Indian ports now featuring among the top 30 and seven among the top 100 in the World Bank's Container Port Performance Index 2024. Recent legislative reforms in the ports and shipping sector may be seen at Box IX.6.

15 Airport density figures are calculated based on the total number of operational airports (commercial and general aviation) as reported in the CIA World Factbook 2025 and International Civil Aviation Organization (ICAO) Data Plus (2025 estimates), divided by the mid-year population projections from the World Bank Open Data 2025. High values for the USA (~47 per million) reflect a mature market with a high concentration of general aviation and municipal airfields (over 15,000). For India, the focus is primarily on commercial hubs and regional airports under active development schemes, indicating significant structural headroom for future expansion.

Chart IX.13: Port Infrastructure Expansion and Record Cargo Performance

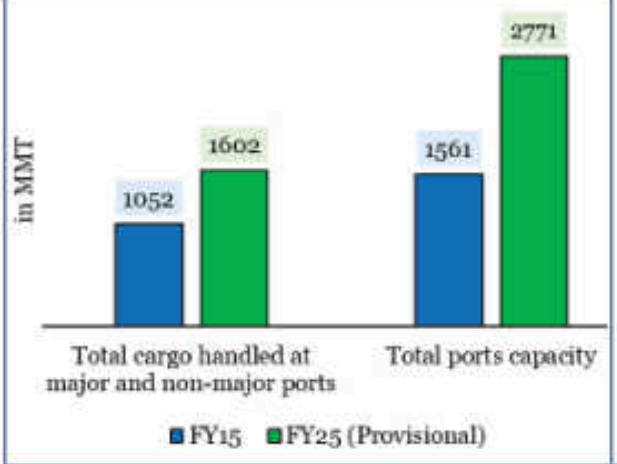
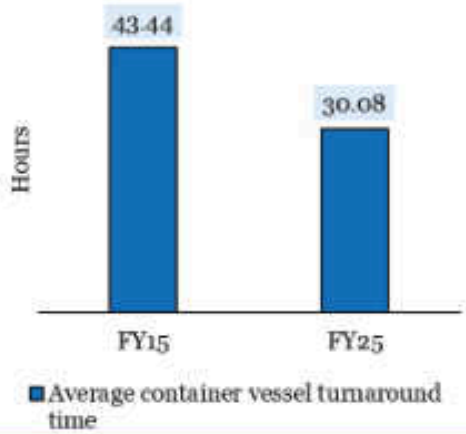


Chart IX.14: Reduction in average container vessel turnaround time



Source: Ministry of Ports, Shipping and Waterways.

9.32 India has further strengthened the landlord port model[16] to catalyse private investments in port development and operations. This strategic shift is evident in the substantial growth of PPP projects. The number of PPP projects awarded rose from 37 in FY15 to 87 in FY25, with the total value of PPP projects increasing from ₹16,180 crore to ₹61,029 crore, reflecting a 377 per cent rise. Currently, 57 operational PPP projects valued at ₹42,235 crore have increased port capacity by approximately 660 MTPA.

9.33 **Outlook:** PPP and captive operators are projected to handle 80 per cent of all cargo at major ports by 2030. A pipeline of 48 PPP projects worth approximately ₹23,000 crore (excluding the Development of Vadhvan Port Project of worth ₹76,220 crore) has been identified for the next five years (FY26 to FY31). These projects will further enhance the capacity and efficiency of India's Major Ports.[17]

Box IX.6: Recent Legislative Reforms in Ports and Shipping Sector

- **Merchant Shipping Act, 2025:** It aligns Indian maritime laws with global standards and International Maritime Organisation (IMO) conventions, promotes ease of doing business, ensures seafarer welfare and training, and enhances safety, environmental protection and emergency preparedness.
- **Coastal Shipping Act, 2025:** Promotes coastal shipping as an economical and eco-friendly mode, removes licensing requirements for Indian vessels in coasting trade and

16 In line with the Major Port Authorities Act, 2021, the landlord port model provides for port authorities to retain ownership of port land and core infrastructure while awarding long-term concessions to private operators for terminal development and operations, with an aim to enhance operational efficiency and modernize infrastructure through private investment.

17 Ministry of Ports, Shipping and Waterways.

Box IX.9: Shipbuilding: Reforms, Achievements, Key Projects and Initiatives

- **Shipbuilding Financial Assistance Scheme (SBFAS):** The Scheme, with a corpus of ₹24,736 crore and validity up to 31 March 2036, provides structured incentives for shipbuilding within India, including a shipbreaking credit note of ₹4,001 crore to promote environmentally responsible recycling and green shipbuilding practices.
- **Maritime Financing and Investment Framework:** A dedicated Maritime Development Fund (MDF) of ₹25,000 crore has been approved to strengthen long-term sectoral financing, attract investment and promote PPPs. This includes:
  i. Maritime Investment Fund of ₹20,000 crore, with 49 per cent participation from the Government of India, and
  ii. Interest Incentivization Fund of ₹5,000 crore to lower financing costs and enhance project viability.
- **Shipbuilding Development Scheme (SbDS):** With an outlay of ₹19,989 crore, the SbDS seeks to enhance shipbuilding capacity to 4.5 million Gross Tonnage (GT) annually and support development of greenfield mega shipbuilding clusters, modernization of existing shipyards, risk and insurance support for shipbuilding projects, and establishment of an India Ship Technology Centre to help develop India's capabilities related to shipbuilding design, training of manpower, R&D in shipbuilding and co-ordination for testing facilities.
- **Inclusion of Ships in Infrastructure Category:** Large ships (above a specified size) have been included in the Infrastructure Harmonised Master List (HML) in September 2025.

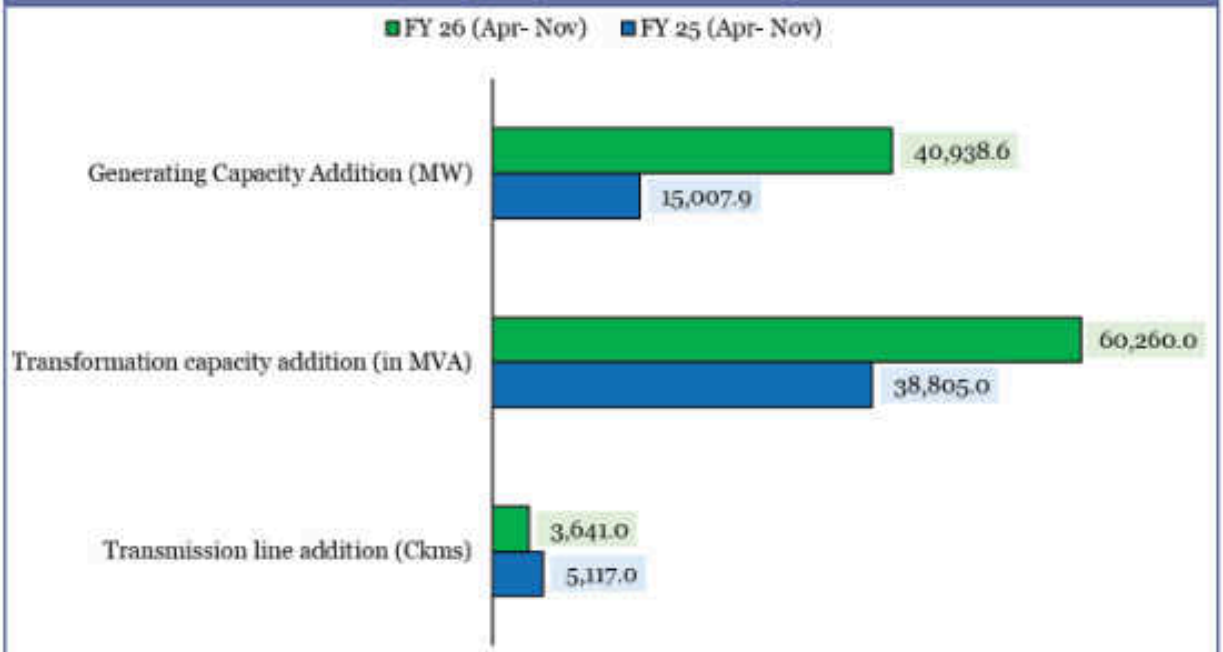
ENERGY SECTOR

Power

9.37 The power sector continues to expand, with the installed capacity rising by 11.6 per cent year-on-year to 509.74 GW as of November 2025.[23] The addition of transformation capacity also gained momentum this year (Chart IX.15).

23 Installed Capacity Report (November 2025), Central Electricity Authority available at https://cea.nic.in/installed-capacity-report/?lang=en.

Chart IX.15: Capacity addition in the power sector



Source: Executive Summary Report (November 2025), Central Electricity Authority.

9.38 The Government of India implemented multiple initiatives aimed at supporting States/distribution utilities for providing uninterrupted power supply to every household. Under the Deen Dayal Upadhyaya Gram Jyoti Yojana (DDUGJY), the Integrated Power Development Scheme (IPDS), introduced in 2014, and the Pradhan Mantri Sahaj Bijli Har Ghar Yojana (SAUBHAGYA), introduced in 2017, about ₹1.85 lakh crore has also been invested to boost distribution infrastructure across various states. Around, 18,374 villages were electrified under DDUGJY, and 2.86 crore households have gained access to electricity during SAUBHAGYA period. To further support States to improve financial sustainability and operational efficiency of distribution utilities, the Revamped Distribution Sector Scheme was launched in 2021 with an outlay of ₹3.03 lakh crore, wherein projects worth ₹2.8 lakh crore have been approved to improve distribution infrastructure and implement smart metering solutions.[24]

9.39 These interventions have contributed significantly in improving power availability and reliability, particularly in rural areas (Chart IX.16). The gap between energy demand and supply has also declined from 4.2 per cent in FY14 to nil by November 2025.[25] Besides, several initiatives have been undertaken to strengthen the viability and operational efficiency of Distribution Companies (DISCOMs) (Box IX.10).

24 Ministry of Power.

25 ibid.

9.40 To further strengthen distribution sector, the Government has proposed the Electricity (Amendment) Bill, 2026, with the objective of enhancing efficiency, competition, and financial discipline in the power sector. The Bill seeks to enable regulated competition in electricity distribution, promote cost-reflective tariffs, strengthen payment security for generators, and improve regulatory accountability, while continuing to safeguard subsidised supply for vulnerable consumers.[26]

Chart IX.16: Improvement in Daily Average Power Supply



Source: Ministry of Power.

Box IX.10: Reforming the last mile - Key initiatives to strengthen DISCOM finances

The distribution sector remains the most critical yet financially vulnerable segment of India's power value chain. Operating as regulatory monopolies, many DISCOMs face operational inefficiency and financial distress. Between 2020-21 and 2024-25, accumulated losses rose from ₹5.5 lakh crore to ₹6.47 lakh crore, with outstanding debt increasing to ₹7.26 lakh crore. This is rooted in non-cost-reflective tariffs, delayed state subsidies payment, and high Aggregate Technical and Commercial (AT&C) losses. To safeguard the long-term sustainability of the sector, the Government of India has introduced several reforms to strengthen the DISCOM finances, reduce losses, and ensure long-term sector sustainability including the following:-

1. **Late Payment Surcharge (LPS) Rules:** Streamlined payment discipline has sharply reduced outstanding dues from ₹1.4 lakh crore (June 2022) to ₹4,927 crore (January 2026).
2. **Automatic Monthly Fuel & Power Purchase Cost Adjustment:** Rules to provide for formula-based monthly tariff adjustments were notified to prevent cash-flow gaps for DISCOMs.
3. **Recognition of Prudent Costs:** Rules notified to allow pass through of prudent power procurement and distribution network costs, subject to regulatory approval.

26 PIB Release available at https://tinyurl.com/f5cst2k6.

11.14. India has made notable gains in school enrolment by strengthening infrastructure and teacher capacity, with schemes like Poshan Shakti Nirman and Samagra Shiksha Abhiyan promoting access and equity. Further action is required, especially as the focus shifts from enrolment to learning outcomes. Policy interventions to expand composite and integrated schools, upgrading schools up to Class XII, and strengthening open schooling are vital for improving retention and optimising resources. Improving infrastructure, teacher skills through strengthened DIETs (District Institute of Education & Training), and SCERTs (State Council of Educational Research and Training), and involving parents and communities in governance can create an inclusive, learner-focused environment. Combining these strategies with curriculum and assessment reforms aligned with NEP and the use of digital platforms such as PM e-Vidya can provide high-quality, equitable education, even in remote areas. **Box XI.1** highlights how community participation improves educational outcomes.

Box XI.1: Empowering change in Indian education through community participation

Education occurs not only in schools, but also within families, communities, and society as a whole, and evidence shows that community-school partnerships boost accountability[14],[15], enrolment, and children's FLN.[16] Parental involvement in education encourages children to work harder and achieve greater academic success.[17]

Community mobilisation for enhancing enrolment and learning outcomes is not new in India and has been part of the education landscape for decades. Campaigns such as 'Chaduvula Panduga' in Andhra Pradesh (launched during 1999–2002) and 'Alokar Jatra' in Assam (launched in 2002) illustrate how large-scale community mobilisation can improve access and enrolment and create local databases on children's educational status.[18] The 'Aao School Chalein' initiative in Haryana addresses student enrolment challenges through a micro-improvement approach and collaboration. Empowering school leaders, leveraging technology, and engaging stakeholders promote community participation to increase enrolment.[19]

14 Hevia, Felipe & Vergara-Lope, Samana. (2019). Educational Accountability or Social Accountability in Education? Similarities, Tensions, and Differences: Accountability Working Paper: Educational Accountability or Social Accountability in Education? Similarities, Tensions, and Differences. 10.13140/RG.2.2.18090.16326.

15 Lant Pritchett, Amanda Beatty, Slow down, you're going too fast: Matching curricula to student skill levels, International Journal of Educational Development, Volume 40, 2015, Pages 276-288, ISSN 0738-0593, https://doi.org/10.1016/j.ijedudev.2014.11.013

16 Kumar D, Sunder N, Ricardo, Wadhwa W, Improving children's foundational learning through community-school participation: Experimental evidence from rural India, Labour Economics, Volume 91, 2024, 102615, ISSN 0927-5371, https://doi.org/10.1016/j.labeco.2024.102615.

17 Epstein, J. L. (1995). School-Family-Community Partnerships: Caring for the children we share. Phi Delta Kappan, 76, 701-712. https://tinyurl.com/nhjsmw3e

18 Ministry of Education. Concept note on community participation: https://tinyurl.com/mt2amy93

19 OECD. Boosting School Enrolment Through Micro-Improvements: https://tinyurl.com/3e8x4j3z

Another example is the Zilla Parishad School Jalindarnagar, a public primary school in Taluka Khed, Maharashtra, which has pioneered a peer-learning 'Subject Friend' model in response to teacher shortages. The method requires grouping mixed-age students under older students for support in lessons and homework, thereby strengthening comprehension, confidence, and participation. This student-led approach has fostered inclusive, collaborative classrooms, expanded learning to advanced subjects such as coding, robotics, electronics, and multilingual education, increased enrolment to 120 students, and catalysed strong community engagement, with parents and local experts volunteering to teach vocational and academic skills and to co-maintain school infrastructure, leading as a model for wider replication.[20] The school's innovative intervention has received global recognition, with the institution being named the World's Best School in 2025 for the Community Choice Award by the T4 Education platform.[21]

Community engagement serves purposes beyond accountability. Teacher communities provide crucial peer support, boost motivation, and facilitate the implementation of policy initiatives at the ground level. Teachers in isolated Zilla Parishad schools often feel disconnected and demotivated due to the limited opportunities for recognition. This highlights the importance of building motivated teacher communities by leveraging technology and behavioural insights, creating platforms for peer learning and practice-sharing, and institutionalising recognition of teachers through appreciation at block- and district-level meetings.[22]

Community-based learning offers students real-world, practical experiences outside traditional schools. This method helps students link their academic knowledge with community needs, promoting a deeper understanding and greater engagement. Additionally, it fosters empathy, social responsibility, and leadership skills by exposing students to a variety of community issues.

Improvement in learning outcomes

11.15. In the evolving landscape of education, learning assessments are vital for evaluating the education systems by identifying gaps and for setting priorities for improvement and innovation. Since 2001, National Achievement Surveys (NAS) conducted by the National Council of Educational Research and Training (NCERT) have provided valuable insights into the school education system. Building on this and with a focus on competency-based learning, PARAKH (Performance Assessment, Review, and Analysis of Knowledge for Holistic Development) Rashtriya Sarvekshan 2024 was

20 https://t4.education/5-prizes-finalists-winners/zp-school-jalindarnagar/

21 T4 Education is a global platform bringing together a community of over 200,000 teachers from more than 100 countries to transform education.

22 NGOs, like the Open Links Foundation, have been working with the district administration on building such teacher communities (https://www.openlinksfoundation.org/)

launched.[23] Its main objective is to evaluate students' learning outcomes, competency, and holistic progress across key grades and provide a guide for policy, curriculum, and resource planning.

11.16. The PARAKH 2024 findings inform that Grade III results show promising recovery post-COVID. Girls slightly outperformed boys in Language (65 per cent vs. 63 per cent), while both scored equally in Mathematics (60 per cent). Rural students outperformed their urban peers, and state government schools led the way in foundational outcomes, validating the impact of NIPUN Bharat. Compared to NAS 2021 and 2017, Grade III proficiency levels have rebounded significantly, with 65 per cent of students proficient in Mathematics (up from 42 per cent in 2021) and 57 per cent in Language (up from 39 per cent).

11.17. To translate these assessment insights into sustained learning improvements, the rollout of Vidya Samiksha Kendras across all states/UTs will play a key role in standardising administrative practices and regulations, enabling real-time, data-driven monitoring of every child's learning outcomes, thereby supporting timely interventions. While the PARAKH Rashtriya Sarvekshan has strengthened learning assessment in India, going forward, it is essential to move beyond aggregate scores to identify underlying learning gaps and areas of conceptual confusion among students. This calls for further strengthening of the assessment system and strategy, so that assessment results effectively support improvements in learning outcomes through enhanced teacher capacity and pedagogy, as discussed in Box XI.2.

Box XI.2: Reimagining school education: Strengthening meaningful assessments and enhancing accountability

The present education policy landscape is being shaped by two complementary imperatives: deepening educational quality and reducing regulatory frictions that constrain innovation and choice. The next phase of educational transformation necessitates a shift toward enhanced accountability, learning measurement, and diversified, participatory delivery models. The demographic dividend can only be fully realised if learning outcomes improve across socioeconomic groups. There is an opportunity to institutionalise systemic assessments, evaluation, benchmarking and widened stakeholder participation to improve the quality of education.

23 The PARAKH Rashtriya Sarvekshan 2024 (formerly National Achievement Survey) was undertaken by the National Assessment Centre, PARAKH, NCERT under the aegis of the Department of School Education & Literacy, Ministry of Education to understand the baseline performance under the NEP in development of competencies among students at the end of the Foundational, Preparatory, and Middle stages (Grades 3, 6, and 9 respectively) of schooling. Nationwide, over 21.15 lakh students, 2.70 lakh teachers, and school leaders from more than 74,000 schools across 781 districts from all States/UTs participated in the assessment. https://dashboard.parakh.ncert.gov.in/en

of the workforce. It discusses policy suggestions for boosting female labour force participation and increasing involvement in the manufacturing sector. The second section examines how the recently notified Labour Codes can catalyse job creation. It also discusses the evolving nature of the gig workforce. The third section addresses the challenge of re-strategising the vocational education framework to prepare the youth for the evolving demands of industry. The last section concludes with an outlook for employment and skilling landscape.

EMPLOYMENT OVERVIEW

12.6 Employment is not a standalone metric but a downstream result of a thriving economy. As the 'State of Economy' chapter 1 of the Survey describes, India's economic growth remains resilient and stable, leading to improved employment opportunities and better labour market parameters. Over the past year, India has introduced several significant reforms, including a major overhaul of its tax regime and various deregulation measures (discussed in Chapter 16, Part II of this survey) aimed at boosting business growth and job creation.

12.7 As a result of these reforms, the labour market continues to show steady activity as revealed by official monthly and quarterly Periodic Labour Force Survey (PLFS) data alongside the establishment-level data from the Annual Survey of Industries (ASI) for the manufacturing sector and the Quarterly Bulletin of Unincorporated Sector Enterprises (QBUSE) for the non-agricultural sector. Together, these sources highlight improving labour force participation, falling unemployment, and strong job creation in both the organised and unorganised sectors.

12.8 The quarterly and monthly PLFS data show a steady labour market with seasonal variations.[7] It indicates that the period from April to September 2025 (H1 FY26) saw a declining unemployment rate (UR)[8] in the current weekly status (CWS)[9] with a stabilising labour force participation rate (LFPR)[10], and substantial employment levels, signalling an improvement in employment conditions. A total of 56.2 crore people (aged 15 years and above) were employed in Q2 (July to September 2025) FY26, reflecting a creation of 8.7 lakh new jobs in Q2 compared to Q1 (April to June 2025) of FY26.[11]

7 The Ministry of Statistics and Programme Implementation (MoSPI) published the first monthly bulletin of the revised PLFS in April 2025. The bulletin presents key labour market indicators, for both rural and urban areas, based on the Current Weekly Status (CWS) of individuals. Along with the monthly PLFS, MoSPI has revamped the quarterly survey to include all-India and rural estimates, as well as urban estimates. https://tinyurl.com/ap6twwkw

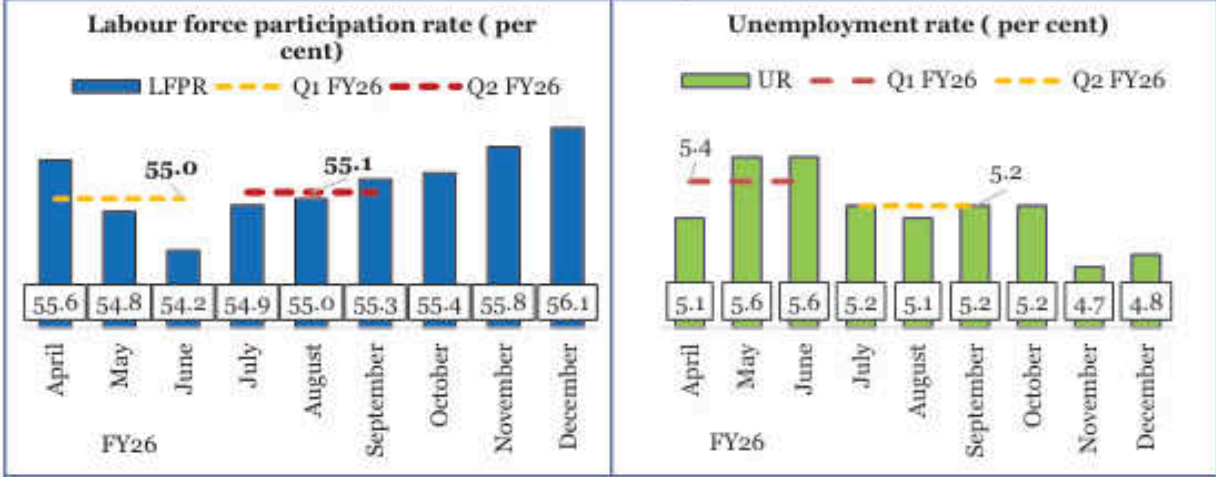
8 UR is defined as the percentage of persons unemployed among the persons in the labour force.

9 The activity status determined on the basis of a reference period of the last 7 days preceding the date of the survey is known as the CWS of the person.

10 LFPR is defined as the percentage of persons in the labour force (i.e. working or seeking or available for work) in the population.

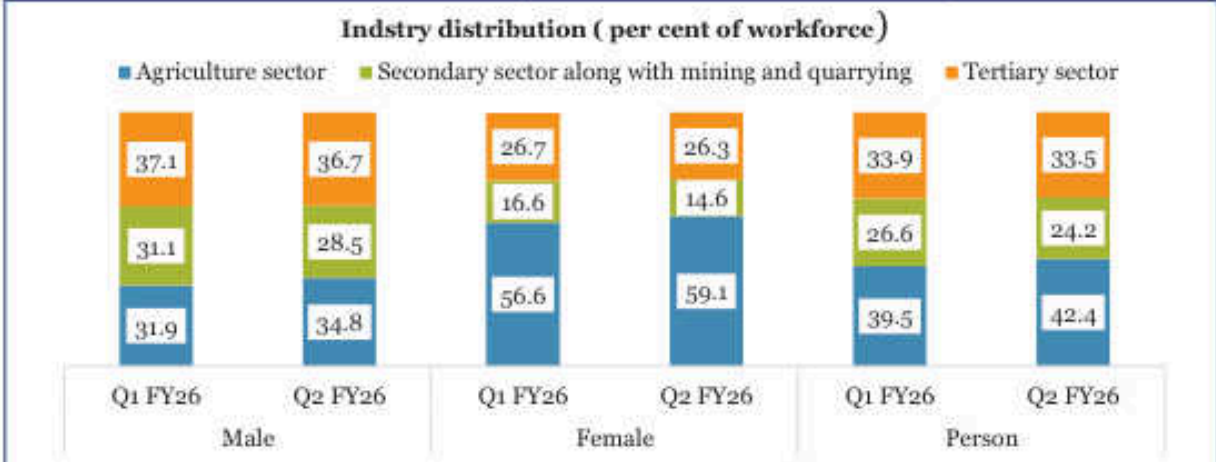
11 PLFS Quarterly bulletin July- September 2025: https://tinyurl.com/3bn85esj

Chart XII. 1: Labour market dynamics for H1 FY26



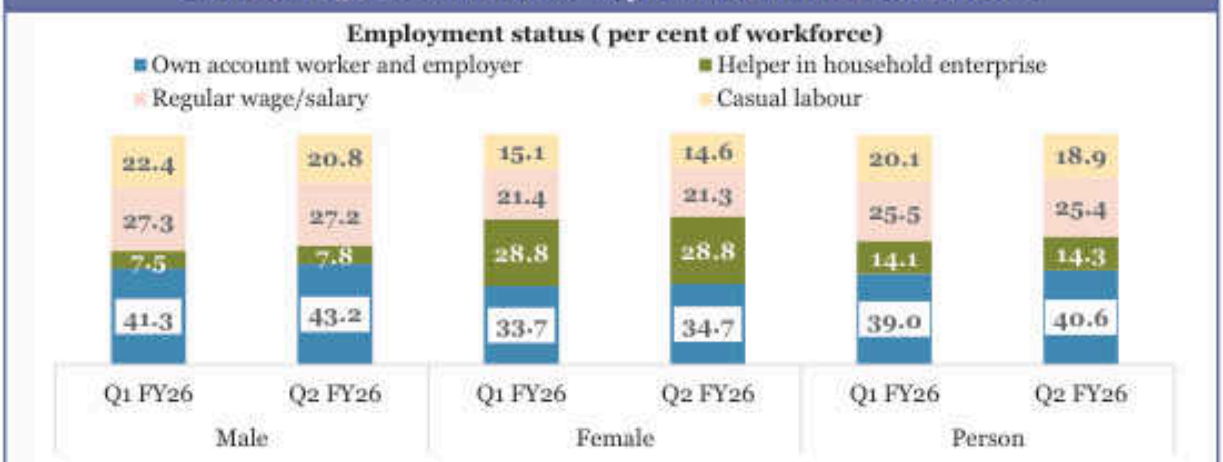
Source: PLFS Monthly and Quarterly Bulletin; Note: Age 15 years and above in CWS

Chart XII. 2: Labour market dynamics in H1 FY26: Industry distribution



Source: PLFS Quarterly Bulletin; Note: Age 15 years and above in CWS

Chart XII. 3: Labour market dynamics in H1 FY26: Status



Source: PLFS Quarterly Bulletin; Note: Age 15 years and above in CWS

and an increase in worker productivity, positioning financial well-being as a factor yielding clear returns for both employees and employers.[58]

Physical well-being through sexual and reproductive health (SRH) awareness: The physical well-being of women workers is often impacted by low awareness and social norms surrounding SRH, which limit their access to information and care. This can be addressed through a capacity-building programme at the workplace. Evidence suggests that SRH training can improve general health behaviours and result in a higher rate of contraceptive use. Notably, there was also a significant increase in the reporting of sexual harassment experienced in public spaces, reflecting greater awareness and confidence.[59] These models demonstrate the effectiveness of workplace-based programs in overcoming information barriers, offering a scalable pathway to improve women's health and well-being.[60]

These examples demonstrate that a focus on holistic well-being (mental, financial, and physical) creates a more resilient, secure, and productive workforce. Such a foundation of a supported and healthy workforce is a critical prerequisite for tackling broader national priorities, such as increasing women's participation in manufacturing sectors, and ensuring India's industrial growth is both sustainable and inclusive.

CATALYSING JOB GROWTH

12.29 The four Labour Codes- Code on Wages 2019, Industrial Relations Code 2020, CSS and Occupational Safety, Health and Working Conditions Code 2020 have consolidated 29 central laws to streamline regulations and extend protections to workers. The implementation of these Codes was notified on 21 November 2025. The Codes have attempted to strike a balance between regulation and flexibility, while protecting worker rights and ensuring social security for workers. They are a result of the deliberations held in the tripartite meeting of the government, employers, industry representatives, and various trade unions from 2015 to 2019. [61]

12.30 The Codes have introduced formal recognition for gig workers (CSS) (Sections 2(35) & 2(60)), mandatory appointment letters for all (Occupational Safety, Health and Working Condition Code,2020), portable social security for migrant workers (CSS), equal benefits (including gratuity and leave after one year) for fixed-term and contract

58 The study was conducted by GBL. GBL developed a simple, Android-based tablet application for the workplace, which enabled workers to withdraw their earned wages via direct transfer to their bank account with just a few taps. It also conducted an RCT with 408 women in the treatment group and 426 women in the control group, along with baseline, midline, and endline surveys to assess the state of household finances. Furthermore, GBL analysed worker-level administrative data, including tenure, daily attendance, wages, and output, as well as an analysis of EWA tool utilisation, including per-worker transaction frequency and transaction size.

59 GBL (2025). Sexual and reproductive health program. https://tinyurl.com/ycy3s9kh

60 The Programmes | Sexual Reproductive Health Training. (2021). UNFPA. https://tinyurl.com/fdcz4bpz

61 PIB release dated 21 November 2025: https://tinyurl.com/2s7er4sa

workers, a minimum wage, and timely payments, as well as a national floor wage (Code on wages, 2019). The simplified compliance with a single licence for contract staffing is poised to accelerate formalisation and inclusive growth.

12.31 Complementing the roll-out of the Centre's Labour Codes, 32 states/UTs have published draft rules under the Codes. Notably, the Uttar Pradesh (UP) government has recently notified new rules allowing women to work night shifts, provided they are accompanied by mandatory safeguards, including employer-provided transportation, health facilities, CCTV surveillance, security personnel, and double wages for overtime work. The quarterly overtime cap has been raised to 144 hours.

12.32 In UP, the amendment to the Factories Act, 1948, enables women to work in all 29 hazardous sectors, promoting gender equality, industrial modernisation, and increased female workforce participation. The reform is supported by enhanced safety infrastructure, including helplines, police beats, and dedicated anti-harassment squads.[62]

12.33 Implementing the Codes marks the first step towards the labour market transformation. The transition will require coordination and investment from the private sector. Companies must enhance their systems, update policies, reevaluate workforce models, and improve their digital readiness to remain competitive. While the Codes offer a unified framework, it is up to the private sector to integrate this framework into daily operations. The impact of the Labour Code on the labour market is discussed in **Box XII.4**.

Box XII.4: Economics of the Labour Codes

Labour is a concurrent subject in the Constitution of India, and until recently was regulated by more than 140 laws (Central and State) with overlapping jurisdictions and inconsistent definitions.[63] The multiplicity of laws led to difficulties in compliance and multiplicity of authorities in different labour laws, leading to complexity and difficulty in enforcement. This resulted in a complex regulatory framework, underscoring the need for a flexible, streamlined and efficient reform to encourage entrepreneurship, support firm growth, and promote productive employment.

Government regulatory measures are as influential in resource allocation as the invisible hand of the market. Empirical studies highlight the policy trade-offs between labour market flexibility and regulation, showing that while regulations protect workers from exploitation,

62 The Factories (Uttar Pradesh Amendment) Act, 2024. Gazette notification No. 185(2)/LXXIX-V-1–2025-1-ka-17-2024 Dated 3 October 2025

63 Confederation of Indian Industry. (July 2025). Policies for a competitive India: CII recommendations. https://tinyurl.com/2fxp9f5p

rather than self-employed contractors, and establishing rules for the use of algorithms in the workplace.[83] The European Union Platform Workers' Directive (2024) was implemented to correct the employment status of gig workers who are misclassified as self-employed, improve transparency, and regulate the use of algorithms and data in decision-making processes involving platform workers.[84] In 2025, the ILO initiated formal discussions on establishing international standards for decent work in the platform economy, with a focus on fair wages, working conditions, and social protections.[85] Over the years, cities such as Seattle[86] and New York[87] have implemented worker protections, including anti-retaliation measures, deactivation appeals, and minimum wage laws, for app-based workers.

Historically, in India, gig and platform workers have been classified as informal labour, often not covered by existing labour laws. The CSS provides legal acknowledgement to these workers and lays the foundation for establishing their rights and protections.

While the law recognises gig and platform workers as a distinct category, it treats them as a largely homogeneous group, whereas in practice the workforce is highly segmented by skill. According to a NITI Aayog report, the share of high-skilled gig workers is expected to be 27.5 per cent by 2030, while for low-skilled workers, it is projected to be 33.8 per cent by 2030. [88]

McKinsey categorises gig workers in the US and EU into four categories based on income reliance and preference.[89] These included i) Free agents (primary income and prefer gig work); ii) Casual earners (gig work for supplementary income or skills); iii) Reluctants (rely on gig work but prefer regular jobs), and iv) the financially strapped (gigs to make ends meet). While preferences shape the labour supply of gig work, demand is driven by task type and skill level, ranging from low-skill-high turnover (jobs like retail, food service, etc) to middle-skilled generalists (e.g., taxi drivers, beauticians) and highly specialised professionals (e.g., programmers, physicians, lawyers) serving multiple employers. Policy should aim to move gig workers from categories 'iii' and 'iv' to 'i' and 'ii', making gig work a choice rather than a necessity. The CSS formally recognises gig workers; policy should now prioritise upward mobility of low-skilled workers through upskilling, move into better-paying jobs, and make gig work a stable, reliable income source.

Another major challenge for gig workers is limited access to credit and productive assets needed to move into better gigs. Many cannot upgrade from low- to medium-skilled gigs

83 Spain's rider law. https://tinyurl.com/mrxw3jyu

84 Directive (EU) 2024/2831 of the European Parliament: https://tinyurl.com/bdzvj7mh
European Parliament. Gig economy: How the EU improves platform workers' rights. https://tinyurl.com/4x5f4bv8

85 Decent work in the platform economy (ILO, 2025). https://tinyurl.com/ykecu9zw

86 Seattle, Office of Labour Standards (2025): https://tinyurl.com/mrxbn88w

87 New York City council (2025): https://tinyurl.com/fh9p5xmw

88 NITI Aayog. (2022). India's booming gig and platform economy: Perspectives and recommendations on the future of work: https://tinyurl.com/chct6udx

89 Manyika et al. (2016). Independent work: Choice, necessity, and the gig economy. McKinsey Global Institute. https://tinyurl.com/3tnvswyv

because they lack tools (for example, a bike, car, or specialised equipment). Encouraging platforms and employers to co-invest in assets and training could help workers progress into more secure, higher-quality jobs.[90] It is important to help gig workers manage unstable incomes. They require basic financial planning support, including access to low-cost emergency savings schemes, portable social security benefits, and budgeting or financial-literacy programmes. The market should be encouraged to offer more flexible and agile financial products that consider the income patterns of gig workers.

Platforms have become essential gig-market infrastructure for finding workers and work. This concentration of power raises concerns over fees, algorithms, and worker protections. Policy should address this through competition rules, data access, and algorithmic transparency, while reorganising the social contract so that gig work benefits workers more fairly. Policy can reduce the cost gap between regular and gig work by limiting incentives to avoid mandatory benefits and by setting minimum per-hour or per-task earnings (including waiting time), encouraging formal employment and raising incomes for low- and medium-skilled gig workers. Taken together, the goal of gig-economy policy should be to reshape the terms so that workers exercise real choice rather than being pushed into gigs due to weak demand, skill mismatch, or the absence of a safety net.

SKILL ECOSYSTEM OVERVIEW

12.40 Expanding access to skills and improving their quality requires a well-integrated skilling ecosystem. Skills policy sits at the crossroads of education, labour markets, and industry, making close coordination and collaboration essential among a broad range of stakeholders, including multiple institutions, ministries, government at various levels, students, educators, workers, employers, trade unions, and other relevant actors. As India seeks to harness its demographic dividend and respond to evolving labour market needs, a well-integrated and forward-looking skilling system is critical to enabling the workforce to capitalise on emerging economic opportunities.

12.41 The PLFS 2023-24 findings show that the share of individuals (in the 15–59 age group) having acquired some form of vocational or technical training has increased from 8.1 per cent in 2017-18 to 34.7 per cent in 2023-24, reflecting the positive impact of skilling initiatives in India.[91] However, there is a need to expand access to formal skilling as only 4.9 per cent of the youth (in the 15-29 age group) have received formal vocational or technical training, while another 21.2 per cent received training through informal sources. Estimates from a 2025 study suggest that a 12-percentage-point increase in skilled workforce through investment in formal skilling could lead to more

90 Ibid note 88 above.

91 Annual PLFS report 2023-24.